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Orbital Mechanics

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Prof Rob Hewson (r.hewson@imperial.ac.uk)
Dr Davide Amato (d.amato@imperial.ac.uk)

Aeronautics Department, Imperial College London.

The N-body problem

We are interested in the motion of n particles or masses interacting with each other via mutual gravitation. We will denote the mass of each particle as m_i and the position and velocity of each particle as $\mathbf{r}_i$ and $\mathbf{v}_i$ respectively[1].

We can denote the distance between two bodies i and j as

$$r_{ij} := \|\mathbf{r}_i - \mathbf{r}_j\| = \sqrt{(\mathbf{r}_i - \mathbf{r}_j)^\top (\mathbf{r}_i - \mathbf{r}_j)}.$$

Newton's law of universal gravitation gives us the total force of attraction experienced by each of the bodies due to the others. The force experienced by body i due to the attraction of body j is Gm_im_j/r_{ij}^2. The constant G is the *universal gravitational constant*. In SI units, $G \simeq 6.67 \times 10^{-11}$ N $\cdot$ m^2 $\cdot$ kg^{-2}. The force on body i is directed radially towards particle j, i.e. in the direction of the unit vector $(\mathbf{r}_j - \mathbf{r}_i)/r_{ij}$.

In the first part of this course we will ignore any additional forces besides gravity that may act on the particles unless otherwise specified. Such forces can include atmospheric drag forces, solar pressure, use of thrusters on spacecraft, etc.

If we have a system of N particles and denote the total force acting on body i as $\mathbf{f}_i$, then the above gives us the total force applied to body i due to the other $N-1$ bodies:

Newton's Law of Universal Gravitation for N bodies *For a system of N particles each with mass m_i and position $\mathbf{r}_i$, the total force experienced by particle i is:*

$$\mathbf{f}_i = \sum_{j=1,\, i\neq j}^{N} G\frac{m_i m_j(\mathbf{r}_j - \mathbf{r}_i)}{r_{ij}^3}$$

Newton's second law of motion tells us that the acceleration of each particle is

$$\mathbf{f}_i = m_i\frac{d\mathbf{v}_i}{dt} = m_i\frac{d^2\mathbf{r}_i}{dt^2}.$$

Combining this with the law of gravitation of above, we get the following:

[1] In general these notes will follow the notation in the text by Battin (see suggested reading list), with vector quantities in bold, their associated magnitude in plain font, and unit vectors denoted as $\mathbf{i}$ with an appropriate subscript, i.e. for vector $\mathbf{r}$,

$$r := \|\mathbf{r}\|, \quad \hat{r}_r := \frac{\mathbf{r}}{r}.$$

Equations of motion for N bodies *For an N-body system, the equations of motion are:*

$$\frac{d^2\mathbf{r}_i}{dt^2} = \sum_{j=1,\, i\neq j}^{N} G\frac{m_j(\mathbf{r}_j - \mathbf{r}_i)}{r_{ij}^3}$$

Important: *Most of the physics used in this course is based on Newton's laws of motion and Newton's law of universal gravitation.* Almost everything *else in the course is based on mathematical manipulation of the above. If you are* extremely *good at maths, you can stop reading now!*

The above describes a set of second order, coupled, nonlinear differential equations for a set of N bodies. Except in certain special cases, there is no closed-form solution for these equations. However, if the problem is restricted to two bodies, we can solve the equations of motion explicitly.

The 2-body problem

For a system of two bodies, our equations of motion are:

$$m_1\frac{d^2\mathbf{r}_1}{dt^2} = G\frac{m_1 m_2(\mathbf{r}_2 - \mathbf{r}_1)}{r_{12}^3}$$
$$m_2\frac{d^2\mathbf{r}_2}{dt^2} = G\frac{m_1 m_2(\mathbf{r}_1 - \mathbf{r}_2)}{r_{21}^3}.$$

However, we generally are more interested in the *relative* motion of the two bodies. We can therefore define

$$\mathbf{r} := \mathbf{r}_2 - \mathbf{r}_1$$
$$\mathbf{v} := \mathbf{v}_2 - \mathbf{v}_1$$
$$r := \|\mathbf{r}\| = \|\mathbf{r}_2 - \mathbf{r}_1\|$$
$$\mu := G(m_1 + m_2)$$

It should be noting that when a satellite is orbiting a planet the mass of one body is significantly larger than the other. In this case it is reasonable to assume that $\mu = Gm_p$ (where m_p is the planet's mass, for Earth this value is approximately $\mu_E = 398\,600\,\mathrm{km}^3\,\mathrm{s}^{-2}$.

We can then take the difference between the two equations above, to arrive at the fundamental differential equation of orbital mechanics for the two-body problem:

$$\boxed{\frac{d^2\mathbf{r}}{dt^2} + \mu\frac{\mathbf{r}}{r^3} = 0.} \tag{1}$$

Note that we do *not* assume that either mass is fixed in space.

Integrals of Two-Body Motion

By taking various integrals of the two-body equations of relative motion, we can find several constants of motion for the system. The constants of motion will help us to characterize completely the solution of these equations.

The Angular Momentum Vector

We can take the cross product of relative position and acceleration and integrate to arrive at our first constant of orbital motion. First take the cross product

$$\frac{d}{dt}(\mathbf{r}\times\mathbf{v}) = \mathbf{r}\times\frac{d\mathbf{v}}{dt} = \mathbf{r}\times\left(-\frac{\mu}{r^3}\mathbf{r}\right) = 0,$$

then integrate once to get

$$\boxed{\mathbf{r}\times\mathbf{v} = \mathbf{h} = \text{constant.}}$$

The vector $\mathbf{h}$ is the (specific) *angular momentum* vector. The motion of the two bodies takes place in the plane normal to $\mathbf{h}$.

The Eccentricity Vector

We next take the cross product of $\mathbf{h}$ with $d\mathbf{v}/dt$ and apply various vector identities along with the fundamental equation of dynamics above, to obtain

$$\frac{d}{dt}\left[\mathbf{v}\times\mathbf{h} - \mu\frac{\mathbf{r}}{r}\right] = 0.$$

Verification of the above is left as a tutorial exercise. *(hint: use the identity* $\mathbf{A}\times(\mathbf{B}\times\mathbf{C}) = \mathbf{B}(\mathbf{A}\cdot\mathbf{C}) - \mathbf{C}(\mathbf{A}\cdot\mathbf{B})$ *and the quotient rule).* We then integrate once to obtain another constant, this time the vector *(Laplace vector)* $\mu\mathbf{e}$:

$$\boxed{\mathbf{v}\times\mathbf{h} - \mu\frac{\mathbf{r}}{r} = \mu\mathbf{e} = \text{constant.}} \tag{2}$$

The vector $\mathbf{e}$ is known more commonly as the *eccentricity* vector, and its magnitude e as the *eccentricity* of the orbit. The eccentricity vector points in the direction of *periapsis*, i.e.

the direction in which the two bodies make their closest approach. (An *apse* is a position in an orbit where the position and velocity vectors are orthogonal to each other).

This equation can then be rearranged to give the *Orbital Equation* by taking its dot product with the position vector $\mathbf{r}$:

$$\boxed{r = \frac{h^2}{\mu}\frac{1}{1 + e\cos\theta} \quad \text{or} \quad r = \frac{p}{1 + e\cos\theta}} \tag{3}$$

where p is the parameter of the orbit, defined by $p = \frac{h^2}{\mu}$.

This equation relates the relative distance between the bodies and the specific angular momentum, eccentricity and true anomaly. (You will show this in one of the tutorial exercises). The angle $\theta = 0$ corresponds to periapsis. It should be clear from the above that when $e < 0$, the orbit is bounded, and when $e \geq 1$ the orbit is unbounded. Using simple geometric arguments, it can be shown that there are three possible cases:

Eccentricity	Orbit type	Semi-major axis	Total energy
$e < 1$	Elliptical orbit	$a > 0$	negative
$e = 1$	Parabolic orbit	$a = \inf$	0
$e > 1$	Hyperbolic orbit	$a < 0$	positive

The Energy Integral

If we take the dot product of equation 1 with $\dot{\mathbf{r}}$ we can obtain:

$$\boxed{\mathcal{E} = \frac{1}{2}v^2 - \frac{\mu}{r} = \text{constant},} \tag{4}$$

where $\mathcal{E}$ is the total mechanical energy (per unit mass) of the system. The term $v^2/2$ is the *kinetic energy* (per unit mass), and the term $-\mu/r$ is the *potential energy*. We can rearrange this energy expression into the alternative form

$$\boxed{v^2 = \mu(\frac{2}{r} - \frac{1}{a}),} \tag{5}$$

which is referred to as the *energy integral* or the *vis-viva* integral. The derivation of this is a tutorial question.

Kepler's Laws of Planetary Motion

Kepler obtained his laws of planetary motion from observations of planetary motion. However, we are able to obtain them from Newton's laws of gravity and motion.

Kepler's First Law (1609)
Each of the planets orbits the sun in an ellipse with the sun at one focus.

The above is actually a restatement of the equation of orbit

$$r = \frac{p}{1 + e\cos\theta}.$$

Kepler's Second Law (1609)
A line joining a planet to the sun sweeps out equal areas in equal times.

This law can be verified by writing the position and velocity vectors in cylindrical coordinates

$$\mathbf{r} = r\hat{r}_r$$
$$\mathbf{v} = \frac{dr}{dt}\hat{r}_r + r\frac{d\theta}{dt}\hat{r}_\theta$$

so that

$$\mathbf{r} \times \mathbf{v} = r^2\frac{d\theta}{dt}\hat{r}_z = \mathbf{h}.$$

The angular momentum vector $\mathbf{h}$ is constant, and its magnitude

$$h = r^2\frac{d\theta}{dt} = 2\frac{dA}{dt}$$

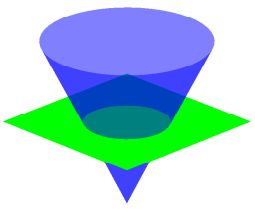
is proportional to the rate at which the radius vector sweeps out area.

Kepler's Third Law (1619)
The square of a planet's orbital period is proportional to the cube of its mean distance to the sun.

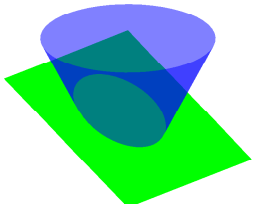
We will return to this later in the course.

Geometric Interpretations

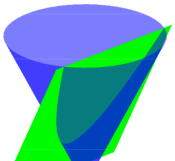
The three orbit types (elliptical, parabolic and hyperbolic) are referred to as *conic sections*, because they match the shape one obtains by slicing through a right circular cone.

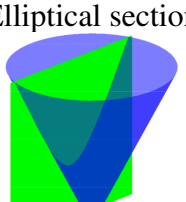


Circular section



Elliptical section





Orbital Velocity and Transfers

The angular velocity of the orbiting mass can be described by the rate of change of the true anomaly, $\dot{\theta}$, this results in the velocity perpendicular to the displacement vector, $\mathbf{r}$, being given by,

$$v_\perp = r\dot{\theta} \quad \text{or} \quad \dot{\theta} = \frac{v_\perp}{r}.$$

The specific angular momentum can be given by multiplying the perpendicular velocity by the magnitude of the displacement ($h = rv_\perp$),

$$h = r^2\dot{\theta}$$

We can determine $v_\perp$ from $h = rv_\perp$ and the orbital equation (equation (3)).

$$\boxed{v_\perp = \frac{\mu}{h}\left(1 + e\cos\theta\right)},$$

Differentiating equation (3) with respect to time gives the radial velocity (v_r),

$$\dot{r} = v_r = \frac{h^2}{\mu}\left[-\frac{e\left(-\dot{\theta}\sin\theta\right)}{\left(1+e\cos\theta\right)^2}\right] = \frac{h^2}{\mu}\frac{e\sin\theta}{\left(1+e\cos\theta\right)^2}\frac{h}{r^2}$$

$$= \boxed{\frac{\mu}{h}e\sin\theta}$$

The minimum radius, known as the periapsis, r, given by equation (3) is obtained when $\theta = 0$ (unless $e = 0$ in which case the radius is constant- see the next section on circular orbits), in which case the minimum is given by,

$$r_p = \frac{h^2}{\mu}\frac{1}{1+e}$$

when $\theta = 0$, $v_\perp = 0$. The flight angle γ is the angle between the perpendicular to the position vector, r_p, (also known as the local horizon) and the flight direction. This is given by,

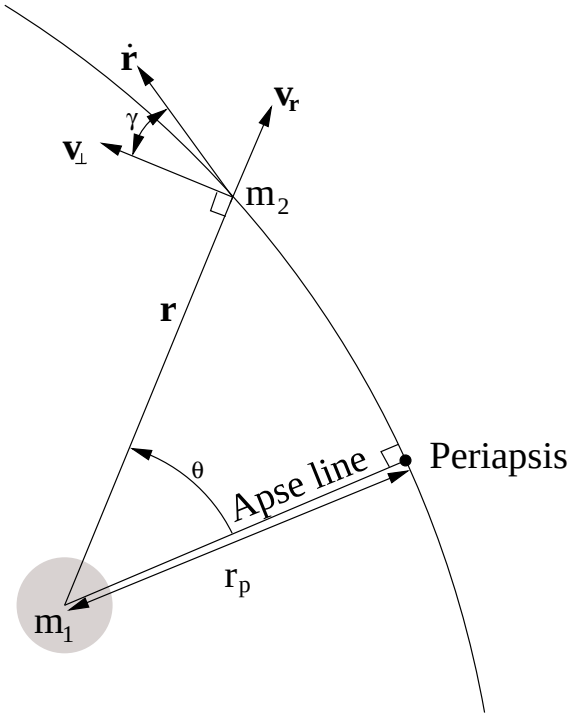
$$\boxed{\tan\gamma = \frac{v_r}{v_\perp}}$$

or

$$\boxed{\tan\gamma = \frac{e\sin\theta}{1+e\cos\theta}}.$$

As $\cos\theta = \cos\left(-\theta\right)$ the trajectory is symmetric about the $\theta = 0, \pi$ line, also known as the *apse line*. The line between the two positions when $\theta = \pm\frac{\pi}{2}$ is known as the latus rectum, with the line between the central attracting body and the orbiting satellite known as the semi-latus rectum, or also known as the *parameter of the orbit*, p, which can be obtained from equation (3),

$$p = \frac{h^2}{\mu}.$$



Position and velocity of mass, m_2, moving round mass m_1

Circular Orbits

Setting the eccentricity to zero in the orbital equation results in,

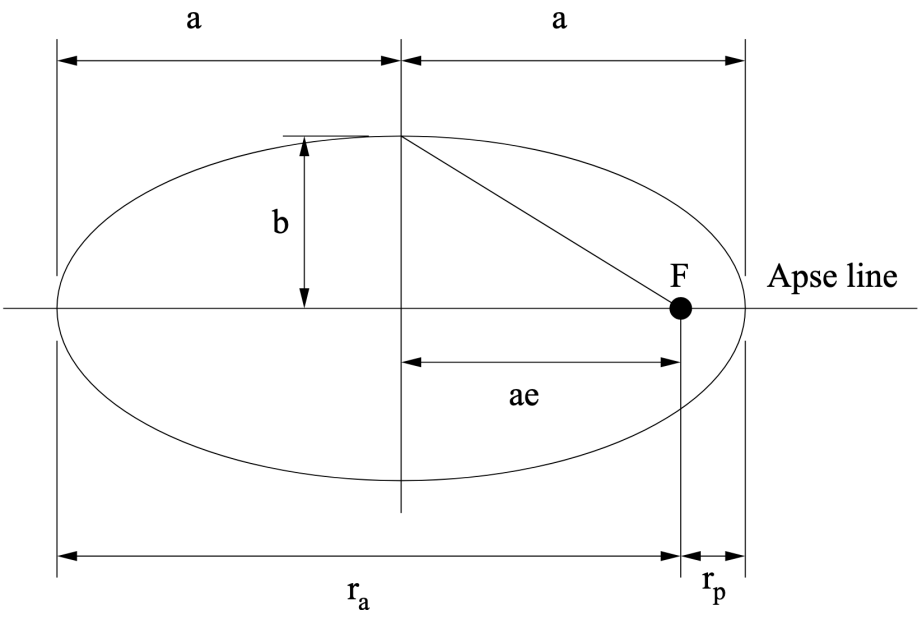
$$r = \frac{h^2}{\mu},$$

in this case the radius is constant and describes the circular orbit. As the change in radius is zero ($\dot{r} = 0$) the angular momentum is simply $h = rv$, which can be substituted into the equation above to give the velocity and period of the orbit,

$$v_{\text{circular}} = \sqrt{\frac{\mu}{r}} \quad \text{and} \quad P = \frac{2\pi}{\sqrt{\mu}} r^{\frac{3}{2}}. \tag{6}$$

Elliptical Orbits

Elliptical orbits are described by the orbital equation for $0 < e < 1$. In this case the orbital equation varies with the true anomaly (θ) but the denominator remains positive, ensuring that the position vector remains bounded. It can be seen by inspecting equation (3) that the position vector is at its smallest when $\theta = 0$ (periapsis, r_p) and at its largest when $\theta = 180$ (apoapsis, r_a). When $0 < e < 1$ equation (3) describes an ellipse.



Elliptical Orbit

If we give the major axis of the ellipse length $2a$ it is clear that $2a = r_a + r_p$, which when the orbital equations for r_a and r_p are substituted results in,

$$a = \frac{h^2}{\mu}\frac{1}{1-e^2}, (a \text{ is the semi-major axis of the ellipse.}) \tag{7}$$

which can be rearranged to give,

$$\frac{h^2}{\mu} = a\left(1-e^2\right), \tag{8}$$

which can then be substituted into the orbital equation to give an alternative form of the orbital equation,

$$r = a\frac{1-e^2}{1+e\cos\theta}.$$

Using Pythagoras' theorem we can determine the semi-minor axis,

$$b = a\sqrt{1-e^2} \tag{9}$$

Which shows that the orbital energy is independent of the eccentricity and depends only on the semi-major axis. We may also determine the time taken for an orbit, this is done using Kepler's second law that state's that the area swept out by an orbiting body increases constantly with time ($\frac{dA}{dt} = \frac{h}{2}$). The area of one elliptical orbit is given by $A = \pi ab$. Therefore $\pi ab = \frac{h}{2}P$ or,

$$P = \frac{2\pi ab}{h}.$$

Substituting a and b from equations (9) and (7) gives the orbit period as,

$$P = \frac{2\pi}{\mu^2}\left(\frac{h}{\sqrt{1-e^2}}\right)^3,$$

we can substitute equation (8) into this equation to give the period as a function of only the semi-major axis,

$$\boxed{P = \frac{2\pi}{\sqrt{\mu}}a^{\frac{3}{2}}}.$$

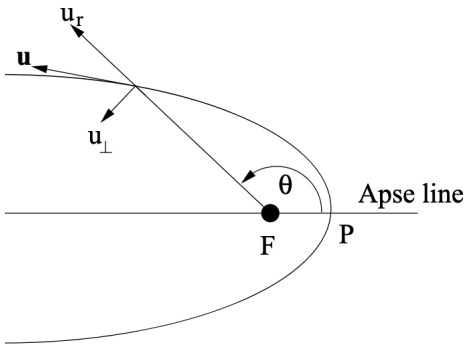
Note that this expression is identical to the equation describing the period of a circular orbit with the radius replaced with the semi-major axis, this also shows that like the elliptical orbit's energy the period is a function of the semi-major axis only. This relationship is known as Kepler's third law which states that the period of an orbit is proportional to the three-halves power of the semi-major axis. A final observation we can make of elliptic orbits is the relationship between e and r_a and r_p,

$$\boxed{\frac{r_p}{r_a} = \frac{1-e}{1+e} \quad \text{or} \quad e = \frac{r_a - r_p}{r_a + r_p}}.$$

It should be noted that $r_a - r_p$ is the distance between foci of the ellipse and $r_a + r_p$ is the length of the major axis, therefore,

$$e = \frac{\text{distance between foci}}{\text{length of major axis}}.$$

Parabolic Trajectories



Parabolic Trajectory

If $e = 1$ then the orbit equation becomes,

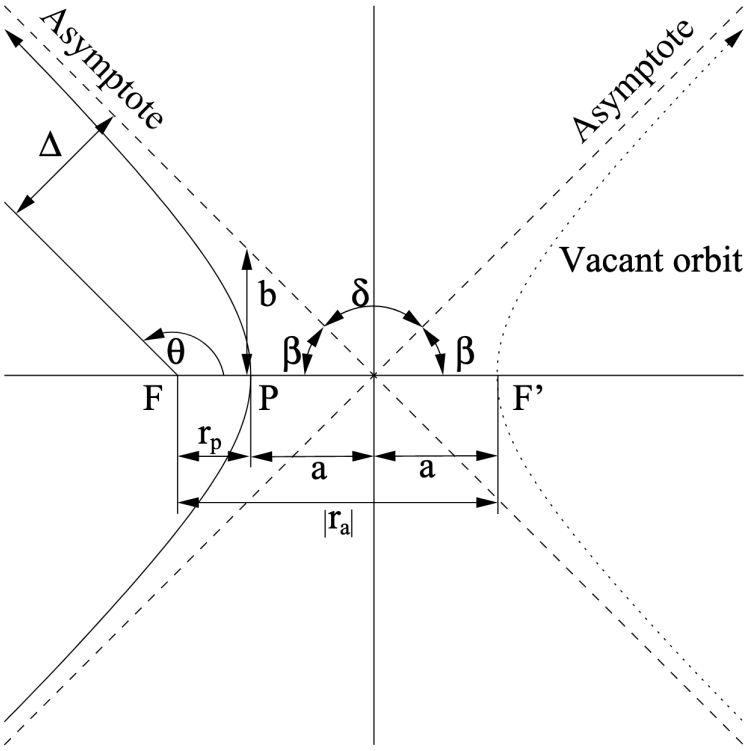
$$r = \frac{h^2}{\mu}\frac{1}{1+\cos\theta},$$

and it can be seen that as the true anomaly $\theta \to 180^\circ$ then $r \to \infty$. From equation (4) it is clear that the energy of a parabolic trajectory is zero, and so the velocity is given by,

$$v_{\text{esc}} = \sqrt{\frac{2\mu}{r}}$$

From this we see that as the body coasts to infinity it's velocity will tend to zero, and it will not return. It is for this reason that parabolic velocities are termed escape trajectories. If we consider a satellite in a circular orbit, the velocity v_0 is given by equation (6), the relationship between circular orbit velocity and escape velocity is therefore $v_{\text{esc}} = \sqrt{2}v_0$.

Hyperbolic Trajectories



Hyperbolic Trajectory

When $e > 1$ equation (3) describes a hyperbolic flight path. As with the parabolic trajectory the r tends to infinity for certain θ. Rather than only occurring as $\theta \to 180^\circ$ for hyperbolic trajectories the value of θ for which r tends to infinity is given by,

$$\theta_\infty = \cos^{-1}\left(\frac{-1}{e}\right),$$

θ_∞ is known as the true anomaly of asymptote, and must always lie between 90° and 180°. Using trigonometric identities we can write the equation above as,

$$\sin\theta_\infty = \frac{\sqrt{e^2-1}}{e}.$$

When $\theta_\infty < \theta < (360^\circ - \theta_\infty)$ a vacant orbit is traced out, this orbit is impossible as it would require a repulsive gravitational force. Considering the vacant orbit is useful in analysing the hyperbolic trajectory. The periapsis lies on the apse line on the physical hyperbola where as the apoapsis lies on the apse line of the vacant hyperbolic trajectory. The centre of the hyperbola lies half-way between the periapsis and apoapsis.

The asymptotes of the hyperbola are straight lines which the hyperbolic curves tend towards as they come from and approach infinity. The turn angle δ is the angle through which the velocity vector is rotated as it passes the attracting body m_1.

$$\delta = 2\sin^{-1}\left(\frac{1}{e}\right)$$

As with the elliptical orbits and parabolic trajectories the distance from the focus to the periapsis is given by setting $\theta = 0$ in the orbit equation,

$$r_p = \frac{h^2}{\mu}\frac{1}{1+e}.$$

By setting $\theta = 0^\circ$ and $\theta = 180^\circ$ the orbital equation gives the distance from the focus to the periapsis (r_p) and apoapsis (r_a) respectively,

$$r_p = \frac{h^2}{\mu}\frac{1}{1+e} \quad \text{and} \quad r_a = \frac{h^2}{\mu}\frac{1}{1-e}.$$

as $e > 1$ then r_a is negative, leading to the apoapsis being in the opposite direction to the $\theta = 180^\circ$ direction. As the distance between the periapsis and apoapsis ($2a$) is given by $2a = |r_a| - r_p = -r_a - r_p$, which when r_a and r_p are substituted gives,

$$2a = -\frac{h^2}{\mu}\left(\frac{1}{1-e} + \frac{1}{1+e}\right),$$

which can be rearranged to give,

$$a = \frac{h^2}{\mu}\frac{1}{e^2-1}.$$

Which in turn allows the orbital equation to be written as,

$$r = a\frac{e^2-1}{1+e\cos\theta}.$$

It therefore follows that,

$$r_p = a\,(e-1) \quad \text{and} \quad r_a = -a\,(e+1)\,.$$

For a hyperbolic trajectory the semi-minor axis, b, is the distance from the periapsis to an asymptote measured perpendicular to the apse line.

$$b = a\tan\beta = a\frac{\sin\beta}{\cos\beta} = a\frac{\sin(180-\theta_\infty)}{\cos(180-\theta_\infty)} = a\frac{\frac{\sqrt{e^2-1}}{e}}{-\left(-\frac{1}{e}\right)} = a\sqrt{e^2-1}$$

The angle made between the asymptote and a parallel line running through the focus is called the aiming radius and is given by,

$$\Delta = a\sqrt{e^2 - 1} \tag{10}$$

which is the same as the semi-minor axis of the orbit, therefore the aiming radius is equal to the semi-minor axis of the hyperbolic trajectory. If we now consider the energy of the hyperbolic trajectory, from equation (4) we can derive the equation for the energy of a hyperbolic trajectory as,

$$\varepsilon = \frac{\mu}{2a},$$

equation (4) then becomes,

$$\frac{u^2}{2} - \frac{\mu}{r} = \frac{\mu}{2a}. \tag{11}$$

the velocity at which the body arrives at infinity is therefore,

$$v_\infty = \sqrt{\frac{\mu}{a}}$$

Which means we can write equation (11) as,

$$\frac{v^2}{2} - \frac{\mu}{r} = \frac{v_\infty^2}{2}.$$

The escape velocity has been previously defined as $v_{\text{esc}} = \sqrt{\frac{2\mu}{r}}$, we are therefore able to write the velocity for the hyperbolic trajectory,

$$v^2 = v_{\text{esc}}^2 + v_\infty^2.$$

The hyperbolic excess speed, v_∞, represents the excess kinetic energy above that required to simply escape the attraction of mass m_1.

Elliptic Orbits and Kepler's Equation

So far we have identified 6 constants of integration for the 2nd order differential equation describing the motion of 2-bodies, i.e. we have found the constant vectors $\mathbf{e}$ and $\mathbf{h}$. However, these two vectors are not independent (i.e. $\mathbf{h} \cdot \mathbf{e} = 0$), so there is still one constant missing.

Time of pericenter passage

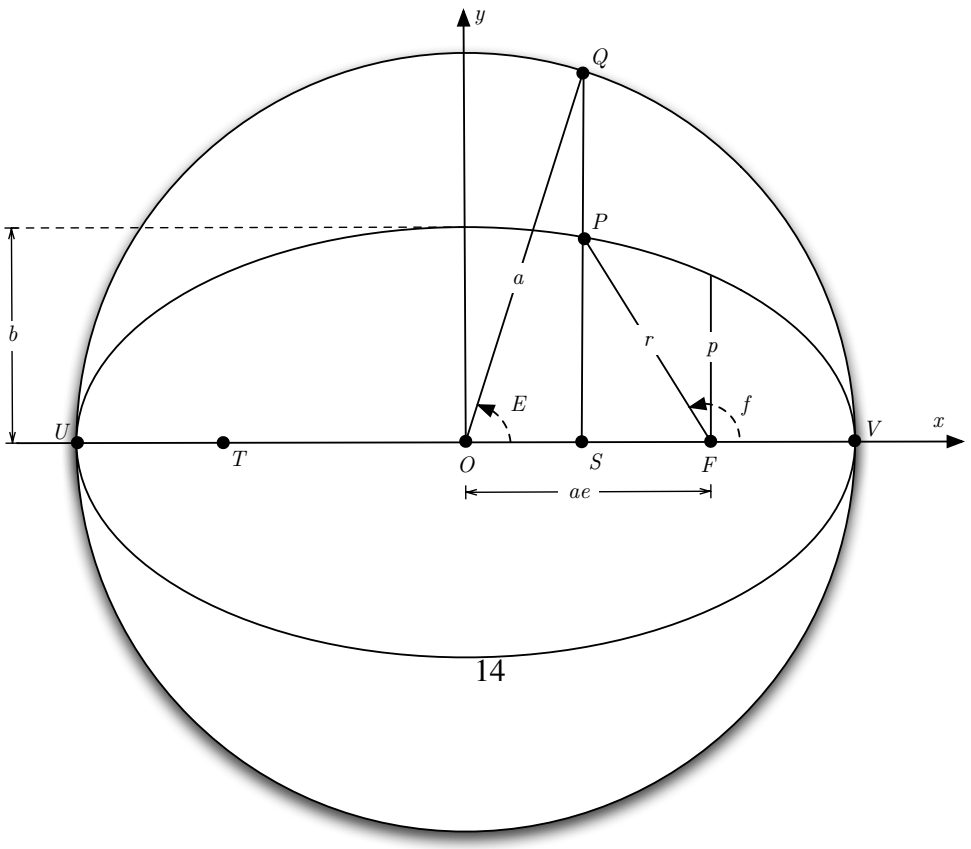
The missing constant is the one relating *time* to *orbital position*. Using the equation of orbit from the previous lecture, we can write

$$\boxed{\sqrt{\frac{\mu}{p^3}}dt = \frac{d\theta}{(1+e\cos\theta)^2}.} \tag{12}$$

If we integrate this equation once (not easy!), we get a relationship between the true anomaly f and the time t, plus our final constant of integration. The usual choice for this constant is the *time of pericenter passage*, denoted τ, which gives the time of closest approach of the two bodies.

The eccentric anomaly

In order to find a solution to the above differential equation, we will first require a few new definitions relating to the geometry of our orbit. In particular, we will define a new angle E, known as the *eccentric anomaly*. The relation of this angle to the true anomaly θ is shown in the figure below:



In this figure, we can see the following features of the orbit:

a	length of the semimajor axis (same as the distance OV)
b	length of the semiminor axis
p	the parameter
F	the focus of the orbit
T	the vacant focus
O	the center of the orbit
U	apoapsis
V	periapsis
f	the true anomaly, θ
E	the eccentric anomaly

The outer circle in the figure (whose radius matches the semimajor axis of the orbital ellipse), is known as the *auxiliary circle*, and is a useful device for determining orbital position as a function of time.

We first find a relationship between the true and eccentric anomalies. From the figure,

$$\cos E = \frac{ae + r\cos\theta}{a}.$$

From the equation of orbit, we also know that

$$r = \frac{a(1-e^2)}{1+e\cos\theta}.$$

Combining the two, we find the relationship between E and θ:

$$\boxed{\cos E = \frac{e+\cos\theta}{1+e\cos\theta}, \qquad \cos\theta = \frac{\cos E - e}{1-e\cos E}}$$

However, as for any value of $\cos E$ between -1 and 1 there are two possible values of E between 0° and 360°. To resolve this quadrant ambiguity we use the trigonometric identity and some manipulation of the equation for an ellipse,

$$\tan^2\frac{E}{2} = \frac{1-\cos E}{1+\cos E} \tag{13}$$

This leads to,

$$\tan^2\frac{E}{2} = \frac{1-e}{1+e}\times\frac{1-\cos\theta}{1+\cos\theta} = \frac{1-e}{1+e}\tan^2\frac{\theta}{2}. \tag{14}$$

giving the two important results,

$$\tan\frac{E}{2} = \sqrt{\frac{1-e}{1+e}}\tan\frac{\theta}{2} \quad \text{and} \quad E = 2\tan^{-1}\left(\sqrt{\frac{1-e}{1+e}}\tan\frac{\theta}{2}\right) \tag{15}$$

Kepler's Equation

The reason for introducing the eccentric anomaly is that we can relate it to an explicit function of time. First note that the ratio of y positions of points on the ellipse to points on the auxiliary circle (e.g. the ratio of the lengths SP/SQ) is always b/a.

Recall from Kepler's second law that equal areas of the ellipse are swept out in equal times. We would like to find the area swept out by the true anomaly θ, i.e. the area of the sector of the ellipsoid bounded by the points FPV, which we will denote Δ.

If we denote the area of the triangle SPF as Γ, then simple geometric arguments show that

$$\Gamma = \frac{ab}{2}\left(e\sin E - \cos E \sin E\right)$$

We can then express the sum $\Delta + \Gamma$ as a fraction of the area of the auxiliary circle swept out by E minus the area of triangle OSQ.

$$\frac{b}{a}\cdot\left[\frac{E}{2\pi}\cdot\pi a^2 - \frac{a^2}{2}\sin E\cos E\right] = \Delta + \Gamma,$$

so that

$$\Delta = \frac{ba}{2}(E - e\sin E).$$

From Kepler's second law (equal areas are swept out in equal times), we can conclude that

$$\frac{\Delta}{\pi ab} = \frac{(t-\tau)}{P},$$

where $P = 2\pi\sqrt{a^3/\mu}$ is the orbital period and τ is the time at periapsis. The resulting expression is *Kepler's equation:*

$$\boxed{M = \sqrt{\frac{\mu}{a^3}}(t-\tau) = E - e\sin E.} \tag{16}$$

The term M is the *mean anomaly*, and can be thought of as the position of a point that travels the auxiliary circle at constant speed and with period P. The rate of change of the mean anomaly is the *mean motion*

$$n = \sqrt{\frac{\mu}{a^3}}. \tag{17}$$

Kepler's Equation *relates the mean anomaly M to the eccentric anomaly E:*

$$M = E - e\sin E.$$

Solving Kepler's Equation

The advantage of the above equation is that it is very easy to find a value for M given the current time and the time at periapsis. If we can solve Kepler's equation for E (and consequently f) given M, then we can determine the position of an orbiting body as a function of time. Unfortunately, solving Kepler's equation is not straightforward.

There are a huge number of methods for solving this equation (see, for example, all of chapter 5 of the book by Battin). The most straightforward of these (but certainly not the best) is the method of successive substitution. Starting from $E_0 = 0$, assign:

$$E_k \leftarrow M + e \sin E_{k-1},$$

which is guaranteed to converge (cf. Battin §5.1). An efficient way to solve Kepler's equation (along with many others) is the *Newton-Raphson method*. We rearrange the terms in equation (16) to define

$$f(E) = E - e \sin E - M.$$

A solution of Kepler's equation is a root of $f(E)$, that is a value of E such that $f(E) = 0$. We start by taking an initial guess for the solution E_0. To refine our guess, we approximate the function $f(E)$ with its tangent at E_0 and take as our improved guess E_1 the intercept of the tangent with the abscissae. We then repeat this process until convergence. The slope of the tangent for the ith approximation of the root is

$$f'(E_i) = \frac{0 - f(E_i)}{E_{i+1} - E_i},$$

which, upon rearranging, gives the formula for the ith Newton-Raphson iteration

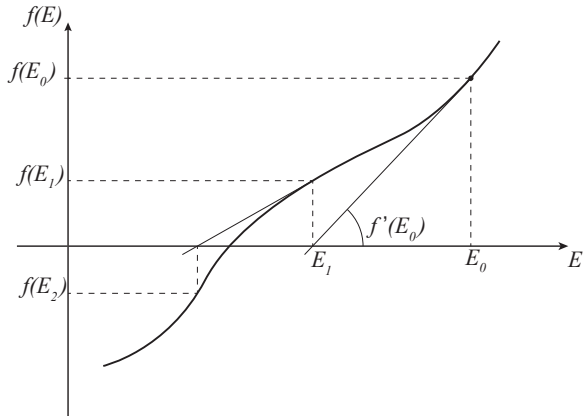
$$\begin{aligned} E_{i+1} &= E_i - \frac{f(E_i)}{f'(E_i)}, \quad i = 0, 1, \ldots \\ f(E_i) &= E_i - e \sin E_i - M \\ f'(E_i) &= 1 - e \cos E_i. \end{aligned}$$

The process is illustrated graphically in the figure below. A reasonable choice for the initial guess is $E_0 = M$, although more refined guesses have been devised to improve convergence.

Open trajectories

Kepler's equation has different forms for parabolic and hyperbolic trajectories. For parabolic trajectories, integrating equation (12) $e = 1$ leads to *Barker's equation*,

$$\boxed{M_{\mathrm{p}} = \frac{\mu^2 (t - \tau)}{h^3} = \frac{1}{2} \tan \frac{\theta}{2} + \frac{1}{6} \tan^3 \frac{\theta}{2}},$$



Newton-Raphson method starting from E_0 as initial guess.

where M_{p} is a quantity that is linear in time and thus is the "mean anomaly" for a parabolic trajectory. It is possible to solve Barker's equation algebraically, unlike Kepler's equation for elliptical and hyperbolic trajectories.

For hyperbolic orbits, we can define a quantity

$$F = \ln \frac{\sqrt{e+1} + \sqrt{e-1}\tan(\theta/2)}{\sqrt{e+1} - \sqrt{e-1}\tan(\theta/2)} \tag{18}$$

that is analogous to the eccentric anomaly for elliptical orbits. To transform between F and θ, we can use the equation

$$\tan\frac{\theta}{2} = \sqrt{\frac{e+1}{e-1}}\tanh\frac{F}{2},$$

which is easier to handle with respect to equation (18). Kepler's equation for hyperbolic trajectories takes the form

$$\boxed{M_{\mathrm{h}} = \frac{\mu^2}{h^3}\sqrt{(e^2-1)^3}(t-\tau) = e\sinh F - F}$$

As with the elliptical case, solving for time given the anomaly F is straightforward, but solving for the anomaly given the time requires finding a root of

$$f(F) = e\sinh F - F,$$

with derivative

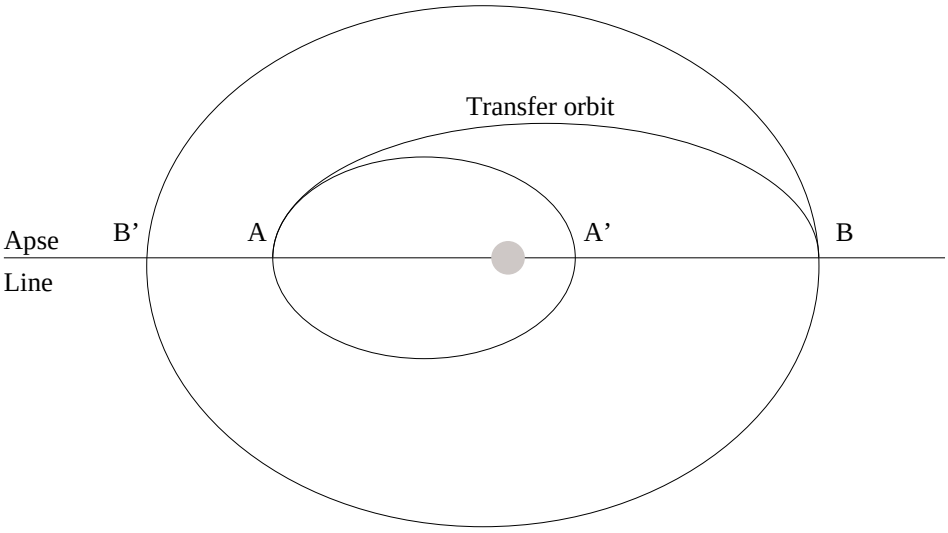
$$f'(F) = e\cosh F - 1.$$

We can use these expressions to solve for the root using the Newton-Rhapson method.

Orbital Manoeuvres

Hohmann Transfer

The Hohmann transfer is the most energy efficient two-impulse manoeuvre for transferring between two co-planar circular orbits sharing a common focus.



Hohmann Transfer Orbit

Two impulses are used to produce two velocity increments (at A and B or A' and B'), the first is used when the spacecraft is at the periapsis or apoapsis to transfer it to a new orbit that has the periapsis or apoapsis at the same location as the desired orbit, which is then reached by means of the second velocity increment which puts the spacecraft in the desired orbit. This applies whether moving from a lower to higher orbit or from a higher orbit to lower orbit, as in this case the impulses are of the same magnitude as from a lower to higher orbit but in the opposite direction. Therefore the total of the velocity increments and the total propellant expenditure is the same regardless of whether the spacecraft is going from a low to high orbit or vice-versa.

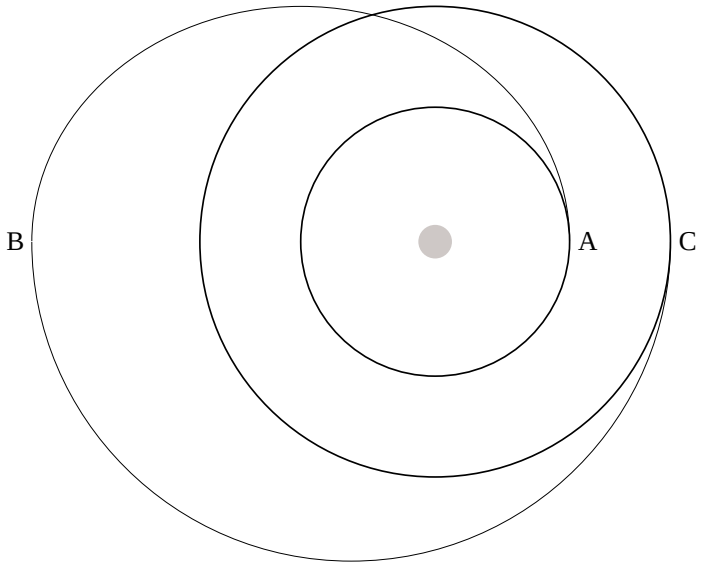
To determine the velocity increment required for the transfer we calculate the velocity at each transfer point for the orbits and the transfer orbits, the velocity increments are then sum of the differences in velocities at each transfer point between the orbits and the transfer orbit. As there are two different positions at which the transfer can occur (marked A and A', and B and B' on the diagram), generally the most efficient if the transfer is the impulse at the lower orbit occurs at the periapsis. If the inner orbit is circular then the impulse should occur at the apoapsis of the outer orbit.

Bi-elliptic Hohmann Transfer

Rather than using a single semi-elliptical orbit to transfer between two orbits the bi-elliptic Hohmann orbit uses two semi-elliptical orbits to transfer from one orbit to another with impulses occurring at A, B and C. For the bi-elliptic orbit to be more efficient than the Hohmann transfer,

$$\Delta v_{\text{total, bi-elliptical}} < \Delta v_{\text{total, Hohmann}} \tag{19}$$

which only occurs when the outer orbit is greater than around 12 times that of the inner orbit for circular orbits.



Bi - Elliptic Hohmann Transfer Orbit

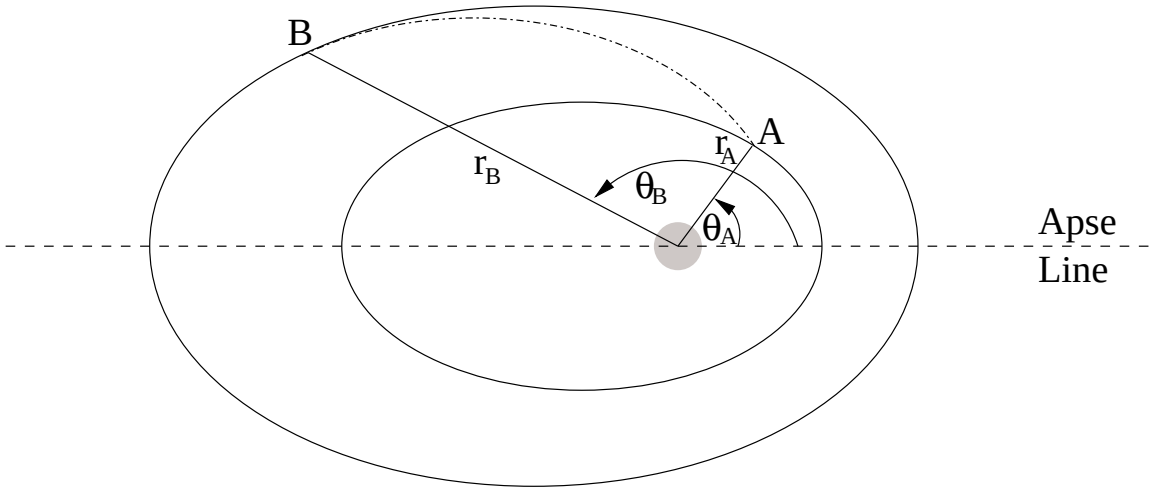
The bi-elliptic transfer orbit tends to be more efficient than the Hohmann transfer orbit for small changes in orbital radii. However, the longer flight time of the bi-elliptic transfer orbit may offset any small gains in efficiency.

Phasing Manoeuvre

A phasing manoeuvre involves a two-impulse Hohmann manoeuvre from and then back to the same orbit. The period of the transfer ellipse is chosen to return the spacecraft to the original orbit at the required time. The firing of the first impulse is a little counter-intuitive. In order to increase the semi-major axis of the orbit and therefore increase the period/slow down the angular velocity the speed of the spacecraft is increased. The opposite is true if the angular velocity is to be increased. The spacecraft thrusters are fired forward to reduce the semi-major axis and increase the angular velocity, allowing the spacecraft to catch up and rendez-vous, whether it be with another spacecraft or with a prescribed position in space.

Non-Hohmann Transfers with a Common Apse Line

This transfer is from one orbit to another with a common apse line, but with the impulses taking place at arbitrary points on the starting and final orbit. For this to be achieved an elliptical orbit must link the starting and final orbits and must also have a common apse line. As all three orbits have a common apse line the true anomalies θ_A and θ_B of the initial position on the starting orbit and the final position on the final orbit respectively are also the true anomalies at which the impulses are fired in the transfer orbit.



Common Apse Line Transfer

Care must be taken in calculating the Δv term as it is the magnitude of the change in velocity between orbits and not the change in magnitudes of the two orbits, i.e.

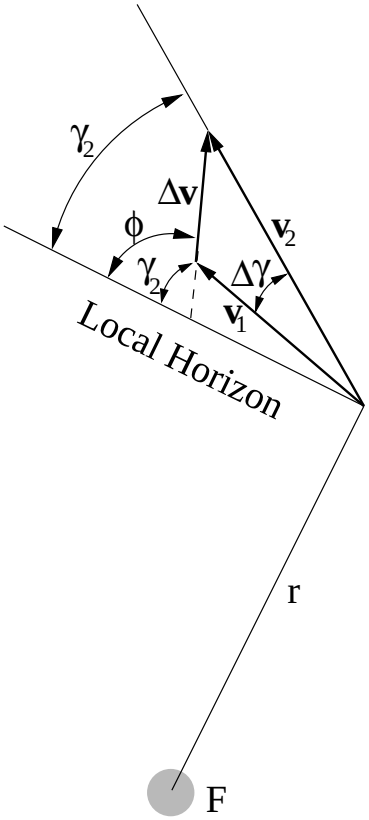
$$\Delta v = |\mathbf{v_2} - \mathbf{v_1}|$$

NOT $\Delta v = |\mathbf{v_2}| - |\mathbf{v_1}|$ (unless the two $\mathbf{v_1}$ and $\mathbf{v_2}$ are parallel as is the case for the Hohmann transfer). From the law of cosines, the magnitude of the vector opposite angle $\Delta\gamma$ is,

$$\Delta v = \sqrt{v_1^2 + v_2^2 - 2v_1 v_2 \cos\Delta\gamma}, \tag{20}$$

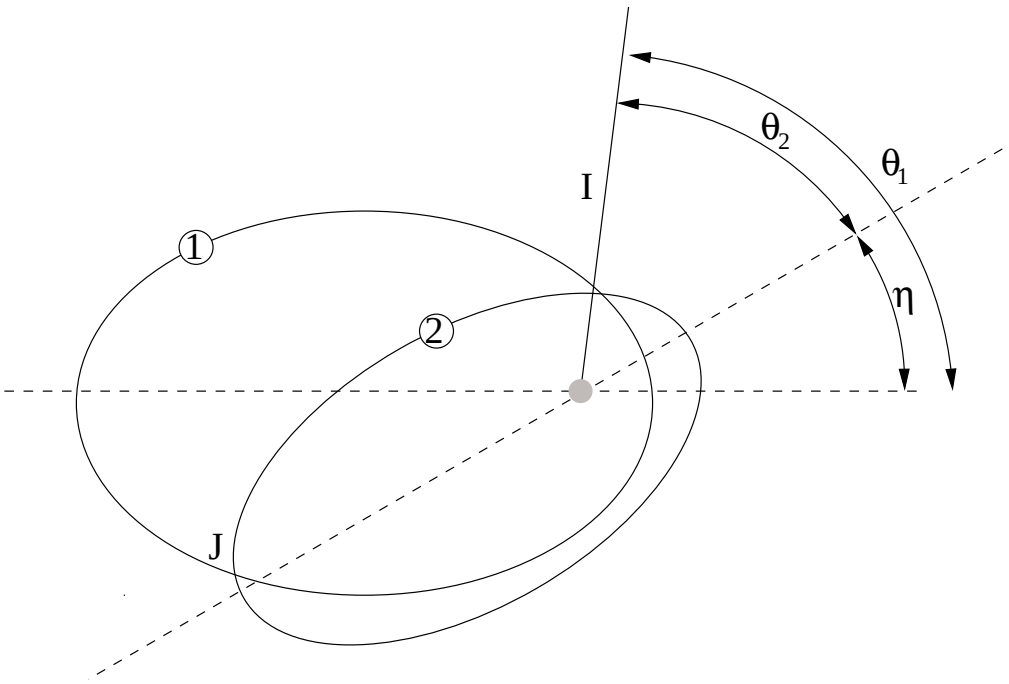
where $v = |\mathbf{v}|$ and $\Delta\gamma = \gamma_2 - \gamma_1$ which is the change in flight angle relative to the local horizon. The direction of $\Delta\mathbf{v}$ gives the alignment of the thruster that produces the impulse. This can be given in terms of the local horizon,

$$\tan\phi = \frac{\Delta v_r}{\Delta v_\perp}.$$



Change in flight angle and velocity at intersection of two orbits

Apse Line Rotation



Apse Line Rotation

The orbital transfers above describe transfers between orbits with common apse lines. We will now consider the transfer between orbits with different apse lines (from orbit 1 to 2 occurring at point I or J). This transfer can occur when the two orbits cross. The rotation of the apse line by η is the difference between the true anomalies of the two orbits.

$$\eta = \theta_1 - \theta_2 \tag{21}$$

To be able to compute the transfer from one orbit to another we need to determine the true anomalies on both orbits at which the transfer occurs. Its is then a case of determining the Δv required to change the orbit of the spacecraft. To move from one orbit to another occurs at the point of intersection, at which the radius of the two orbits is the same.

$$r_1 = r_2 \tag{22}$$

$$\frac{h_1^2}{\mu}\frac{1}{1+e_1\cos\theta_1} = \frac{h_2^2}{\mu}\frac{1}{1+e_2\cos\theta_2} \tag{23}$$

Which can be rearranged to give,

$$e_1 h_2^2 \cos\theta_1 - e_2 h_1^2 \cos\theta_2 = h_1^2 - h_2^2. \tag{24}$$

As $\theta_2 = \theta_1 - \eta$ and from the trig identity, $\cos(\theta_1 - \eta) = \cos\theta_1\cos\eta + \sin\theta_1\sin\eta$

$$a\cos\theta_1 + b\sin\theta_1 = c, \tag{25}$$

where,

$$a = e_1 h_2^2 - e_2 h_1^2 \cos\eta \quad b = -e_2 h_1^2 \sin\eta \quad c = h_1^2 - h_2^2. \tag{26}$$

Inspecting equation (25) shows that there are two roots, which correspond the the two points where the orbits cross. These are given by,

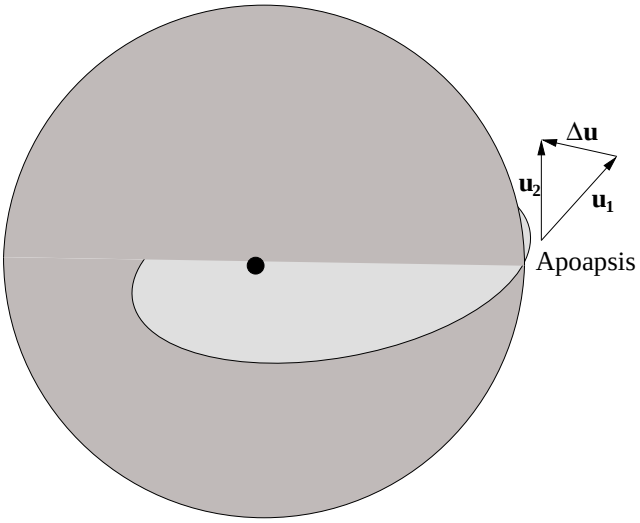
$$\theta_1 = \phi \pm \cos^{-1}\left(\frac{c}{a}\cos\phi\right), \tag{27}$$

where,

$$\phi = \tan^{-1}\frac{b}{a} \tag{28}$$

Plane Change Manoeuvres

So far we have only considered changes in orbits that have all taken place in the same plane. However a change in plane may be required. Consider two orbits in different planes but with a common focus.



Orbital plane change

To change the orbit plane a common point on both the first and second orbits is required. Further orbital changes may then be required once the plane change has been made. The change in velocity required to transfer the spacecraft from one orbit to another can be divided into the perpendicular and radial velocity directions. The radial velocity unit vector $\hat{\mathbf{r}}_\mathbf{r}$ is the same for both orbits at the point of intersection as they both have the same focus.

$$\Delta\mathbf{v} = \mathbf{v_2} - \mathbf{v_1} \tag{29}$$

$$= (v_{r2} - v_{r1})\,\hat{\mathbf{v}}_\mathbf{r} + v_{\perp 2}\hat{\mathbf{v}}_\mathbf{\perp 2} - v_{\perp 1}\hat{\mathbf{v}}_\mathbf{\perp 1} \tag{30}$$

We can find the magnitude of Δv by using the dot product, $\Delta v^2 = \Delta\mathbf{v}\cdot\Delta\mathbf{v}$, that the dot product of parallel unit vectors is one and perpendicular vectors is zero and the angle

between $\hat{\mathbf{v}}_{\perp\mathbf{1}}$ and $\hat{\mathbf{v}}_{\perp\mathbf{2}}$ is $\cos\delta$.

$$\begin{aligned}\Delta v^2 &= \left[(v_{r2}-v_{r1})\,\hat{\mathbf{v}}_{\mathbf{r}} + v_{\perp 2}\hat{\mathbf{v}}_{\perp\mathbf{2}} - v_{\perp 1}\hat{\mathbf{v}}_{\perp\mathbf{1}}\right]\cdot\left[(v_{r2}-v_{r1})\,\hat{\mathbf{v}}_{\mathbf{r}} + v_{\perp 2}\hat{\mathbf{v}}_{\perp\mathbf{2}} - v_{\perp 1}\hat{\mathbf{v}}_{\perp\mathbf{1}}\right] &(31)\\ &= (v_{r2}-v_{r1})^2 + v_{\perp 1}^2 + v_{\perp 2}^2 - 2v_{\perp 1}v_{\perp 2}\left(\hat{\mathbf{v}}_{\perp\mathbf{1}}\cdot\hat{\mathbf{v}}_{\perp\mathbf{2}}\right) &(32)\\ &= (v_{r2}-v_{r1})^2 + v_{\perp 1}^2 + v_{\perp 2}^2 - 2v_{\perp 1}v_{\perp 2}\cos\delta. &(33)\end{aligned}$$

From the following definitions of the flight path angle,

$$\begin{aligned}v_{r1} &= v_1\sin\gamma_1 &(34)\\ v_{\perp 1} &= v_1\cos\gamma_1 &(35)\\ v_{r2} &= v_2\sin\gamma_2 &(36)\\ v_{\perp 2} &= v_2\cos\gamma_2 &(37)\end{aligned}$$

leads to,

$$\Delta v = \sqrt{v_1^2 + v_2^2 - 2v_1v_2\left(\cos\Delta\gamma - \cos\gamma_2\cos\gamma_1\left(1-\cos\delta\right)\right)} \tag{38}$$

When $\delta = 0$ as is the case for in plane orbital transfers this equation (38) becomes equation (20). To keep Δu a minimum the radial velocity should be zero and $u_\perp$ should be smallest - at the apoapse this reduces equation (33) to:

$$\Delta v = \sqrt{v_1^2 + v_2^2 - 2v_1v_2\cos\delta} \tag{39}$$

Common Types of Orbit

There are a number of common types of orbits, the major types are given below.

Low Earth Orbit

Spacecraft in low earth orbit are close to the earth and fast moving. They are therefore ideal for earth observations. In order to cover the entire earth a low earth orbit can be placed in a polar orbit. Landsat 7 has an inclination of 98° and is theoretically able to observe the entire earth. A particularly important type of Low Earth Orbit is a *Sun-synchronous orbit*. Sun-synchronous orbits exploit the effect of Earth oblateness perturbations on Keplerian motion to pass over the different areas of the Earth always at the same time of the day. This is particularly advantageous for Earth observation missions, because having the same lighting conditions considerably simplifies the design of the solar power system and of optical observation payloads. Landsat 9 is one example of a spacecraft placed on such an orbit, at 98.2° inclination and at an altitude of $705\,\mathrm{km}$. We will study the mathematical conditions imposed to achieve Sun-Synchronous Orbits in the lecture on Orbit Perturbations.

Geosynchronous Orbits

A geosynchronous orbit is an orbit which has an orbital period close to that of the earths rotation. A geostationary orbit is a special case of the geosynchronous orbit where inclination = 0 and the period is equal to the rotation period of the earth (approx 1436 minutes), corresponding to a circular orbit of approx. 35,700km altitude. A satellite in this orbit appears essentially stationary in the sky, which is why this orbit is used extensively for telecommunications and weather satellites. In reality lunar and solar gravitational influences perturb the satellites orbit, so that through the day the satellites position shifts slightly.

Compared with the LEO orbit of Landsat, a much larger portion of the earth's surface is visible from the TDRS-7 spacecraft. The zone of visibility of the spacecraft has been highlighted by a cone. Approximately 40% of the earths surface can be viewed at any one time from geostationary altitude. Additionally, the spacecraft orbit is sunlight apart from a small zone which passes into the earths shadow. Actually, geostationary satellites only experience eclipses at two periods of the year - for a few weeks at a time at the spring and autumn equinoxes. The reason for this is simple. The earths rotation axis is inclined with respect to the ecliptic, hence the earth's shadow cone misses the plane of a zero inclination geostationary orbit apart from the times when the suns declination is close to zero. This occurs twice a year, once at the spring equinox and once at the autumn equinox.

Geosynchronous Transfer Orbits

There are many rocket boosters which are observable in transfer orbits. These are the orbits used to transfer the satellite from an initial low earth orbit to the final orbit. The orbit used for transfer to geostationary orbit is named appropriately enough a "geostationary transfer orbit" (GTO). A standard GTO is an orbit which requires the minimum energy to reach geostationary altitude (A Hohmann transfer ellipse).

In recent years modified versions of the GTO orbit have been used. A super-synchronous orbit is one where the apogee is significantly greater than geosynchronous altitude. Why send the payload higher than the target orbit altitude? The reason is because the payload still has to adjust its inclination from the launch inclination (anything from 5 to 51) to 0. This maneuver is very expensive in terms of energy, much more so than an in plane change of orbital altitude. The energy required to do this maneuver decreases with orbital altitude, so it requires less fuel to perform this plane change at high altitude (e.g.. 60,000 or 70,000 km) and then descend to a geostationary orbit rather than do the plane change at geostationary height. Many super-synchronous transfer orbits also have very low perigee altitudes, to accelerate their decay and reduce the amount of debris in orbit. Sub-synchronous transfer orbits also exist where the rocket only sends the payload part of the way to geostationary height, the payload then uses its own propulsion system to reach the final orbit.

Molniya Orbits

Many Russian cities are at high northern latitudes where it is impractical to use geostationary satellites for telecommunications since the satellite would appear either low on the horizon on not visible at all. To overcome this problem Molniya ("lightning") satellites are used for communications in these regions. The orbit used by these satellites is a 12h, high inclination elliptical orbits. This is a 1470 x 38900 km, 63.4 inclination orbit. Again this orbit has special features which make it well suited for telecommunications. First the period is 12h, so there are 2 orbits per day.

As a result the ground track of the orbit repeats at the same time of day each day. Since the orbit is elliptical, the satellite spends most of it's time near apogee (where its velocity is slowest), so for 11h of each orbit the satellite is above the horizon for high northern latitudes. Additionally, for several hours per day the satellite moves only very slowly across the sky (as can be seen from the ground track), making it easy to follow with a communications antenna.

Mid-Earth Orbit

Mid earth orbit (MEO) is a term used to describe 12h period, medium inclination orbits generally used for Global Positioning Satellites. With a constellation of 24 appropriately spaced satellites in approx 20,000 km near circular orbits it is possible to ensure that at least 4 satellites are visible from any one location at any time to ensure reliable navigation using signals from these GPS satellites. The orbit is resonant with the earths rotation period (2 orbits per day) so the orbit track repeats itself each day. Interplanetary Strategies In the previous section we considered the orbital motion of a two body system. In this section we will consider the motion of a spacecraft between planets from launch, within the earths gravitational influence to the heliocentric (sun centered) orbits and planetary flybys and orbits. The analysis is undertaken by considering the radius influence of the planets gravity to that of the sun.

Earth-based Reference Frames

Earth-centred inertial (ECI) reference frame

In order to characterize the position and velocity of a satellite relative to the Earth (or any other central body), we require an inertial frame of reference. Defining such a frame of reference (e.g. for satellites orbiting Earth) is not straightforward, because the orbit plane and spin axis of the earth are not fixed, to say nothing of the fact that the Earth orbits the Sun and thus experiences an acceleration. Two of the most important sources of difficulty are:

- **Precession of the Equinoxes**: The motion of the Earth's spin axis. The spin axis is inclined relative to the Earth's orbit around the sun at an angle of $\approx 23.4^\circ$ (which is time-varying), and this vector traces a cone in space with a period of $\approx 26,000$ years.
- **Precession of the Ecliptic**: The *ecliptic plane* is the plane of the Earth's rotation about the sun (and is the plane in which all of constellations of the zodiac are found). The ecliptic plane is not fixed in space, because the angular momentum vector of the Earth's orbit drifts with a period of $\approx 100,000$ years.

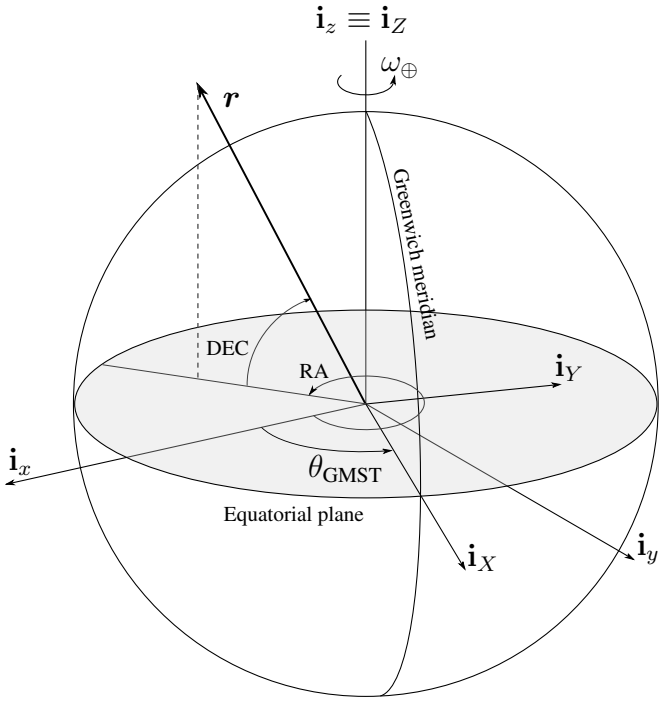
The *vernal equinox* is the point in the Earth's orbit where the plane of the equator intersects the ecliptic (i.e. where the Earth is not tilted towards or away from the sun). The vernal equinox is time varying due to the precession of the equinoxes. It is also called the *first point of Aries*, although this point no longer corresponds exactly to the vernal equinox.

The *invariable plane* is the plane defined by the total angular momentum of the solar system, and is dominated by the angular momentum of the planet Jupiter. The earth's orbit is inclined to the invariable plane by $\approx 1^\circ$.

Given the above, it is important to be very careful when defining coordinate systems for use in orbital mechanics. One of the most important frames in orbital mechanics is the **J2000** Earth-centred inertial (ECI) frame, which has its origin in the barycentre of the Earth and uses:

$\mathbf{i}_x$	unit vector in the direction of the mean vernal equinox on 1 January 2000 at 12:00:00.00. (Sometimes denoted as Υ in reference to the constellation Aries).
$\mathbf{i}_z$	unit vector in the direction of the Earth's mean rotation axis on 1 January 2000 at 12:00:00.00.
$\mathbf{i}_y$	completes the set, i.e. $\mathbf{i}_y = \mathbf{i}_z \times \mathbf{i}_x$.

Note that the fundamental plane (x, y) of this reference frame intersects the Earth at the equator. The *J* in J2000 stands for *Julian Date*, which is the number of days elapsed since January 1, 4713 BC at noon, so that the directions of the unit vectors in the J2000 system are relative to Julian Date 2451545.0. It is customary for inertial reference systems to be



ECI and ECEF frames.

updated every 50 years – starting in 2025 most calculations will be made with respect to the J2050 coordinate system. The position and velocity (or orbital elements) of a spacecraft, along with its equations of motion, are usually expressed in this frame.

In many cases, we can neglect the fact that the origin of the ECI frame (which, more precisely, is "pseudo-inertial") is subject to accelerations due to the Sun and Moon, and consider the ECI frame as a purely inertial frame. This will become more apparent once we start looking into perturbations.

Right Ascension and Declination

Celestial objects, such as stars, planets, and spacecraft, can be represented as points on an imaginary *celestial sphere* of arbitrary radius centred on the Earth. Objects are located on the celestial sphere through two angles:

- the *Right Ascension* (RA), that is the angle between the vernal equinox direction and the projection of the position vector of the object on the equatorial plane;
- the *Declination* (DEC), that is the angle between the equatorial plane and the position vector.

Any point on the celestial sphere can be identified with RA $\in [0°, 360°)$, DEC $\in [-90°, 90°]$.

The position vector of an object in the ECI frame is

$$\boldsymbol{r} = x\mathbf{i}_x + y\mathbf{i}_y + z\mathbf{i}_z. \tag{40}$$

To calculate RA and DEC from $\boldsymbol{r}$, we first need to compute the position vector magnitude,

$$r = \sqrt{x^2 + y^2 + z^2}, \tag{41}$$

and then we express the position vector in terms of its direction cosines $l = x/r, m = y/r, n = z/r$ as

$$\boldsymbol{r} = r\left(l\mathbf{i}_x + m\mathbf{i}_y + n\mathbf{i}_z\right). \tag{42}$$

The direction cosines are related to the right ascension and declination through:

$$\begin{aligned} l &= \cos(\text{DEC})\cos(\text{RA}) \\ m &= \cos(\text{DEC})\sin(\text{RA}) \\ n &= \sin(\text{DEC}) \end{aligned}. \tag{43}$$

To compute the right ascension, we start from

$$\tan \text{RA} = \frac{m}{l}. \tag{44}$$

Because the codomain of the arctangent function is $[-90°, 90°]$, but RA $\in [0°, 360°)$, we first need to determine the quadrant of the projection of the position vector on the equatorial plane, $\boldsymbol{r}_{\text{eq}} = r\left(l\mathbf{i}_x + m\mathbf{i}_y\right)$. If $l > 0$, $\boldsymbol{r}_{\text{eq}}$ is in either the first or fourth quadrants; conversely, if $l < 0$ then $\boldsymbol{r}_{\text{eq}}$ is in either the second or third quadrants. The right ascension can then be calculated as

$$\text{RA} = \begin{cases} \arctan\left(\frac{m}{n}\right) & \text{if } m > 0, l > 0 \\ 180° + \arctan\left(\frac{m}{n}\right) & \text{if } l < 0 \\ 360° + \arctan\left(\frac{m}{n}\right) & \text{if } m < 0, l > 0 \end{cases}. \tag{45}$$

For the declination, the task is easier:

$$\sin \text{DEC} = n. \tag{46}$$

The codomain of the arcsine function is $[-90°, 90°]$, which is the same range as that of the declination. Therefore, the declination can be calculated as

$$\text{DEC} = \arcsin n. \tag{47}$$

If DEC $= \pm 90°$, the object is at a pole, and any value of RA will describe the same point in space.

Note that if we know $\boldsymbol{r}$ we can compute RA and DEC, but the converse is not true. To compute the position vector from RA and DEC, we also need to know the position magnitude r.

Earth-centred, Earth-fixed (ECEF) frame

The Earth-centred, Earth-fixed (ECEF) frame shares its origin (the Earth's barycentre) and fundamental plane (the equatorial plane) with the ECI frame. However, the unit vectors of the ECEF frame are:

$\mathbf{i}_X$	vector in the direction of the intersection of the Greenwich meridian with the equatorial plane.
$\mathbf{i}_Z$	vector in the direction of the Earth's rotation axis, coincides with $\mathbf{i}_z$.
$\mathbf{i}_Y$	completes the set, i.e. $\mathbf{i}_y = \mathbf{i}_z \times \mathbf{i}_x$.

Therefore, the vectors $\mathbf{i}_X, \mathbf{i}_Y$ are attached to the Earth's surface and rotate once per day. The angle between $\mathbf{i}_X$ and $\mathbf{i}_x$ is the *Greenwich mean sidereal time* θ_{GMST}, and can be measured in either hours or degrees. It might be surprising to see an angle expressed in units of time! The reason is that θ_{GMST} expresses the local time at the Greenwich meridian. We can easily switch between hours to degrees by considering that

$$1\,\text{h} = 15^\circ.$$

In the simplest approximation, this angle is

$$\theta_{\text{GMST}} = \omega_\oplus (t - t_0). \tag{48}$$

The Earth completes one full revolution every sidereal day, therefore its angular velocity is

$$\omega_\oplus = \frac{360^\circ}{\text{sidereal day}} = \frac{360^\circ}{1 \text{ sidereal day}} = \frac{360^\circ}{23^\text{h}\,56^\text{m}\,4.090524^\text{s}}.$$

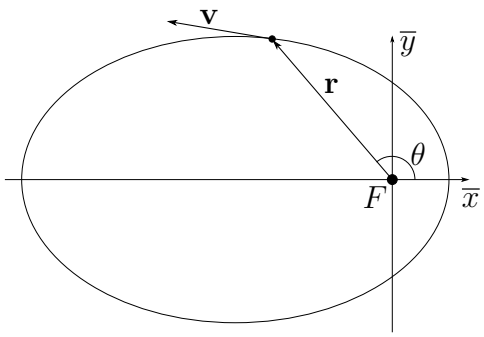
Thus, in this first approximation we can transform coordinates from the ECI coordinates (x, y, z) to the ECEF coordinates (X, Y, Z) through the following rotation matrix:

$$\begin{Bmatrix} X \\ Y \\ Z \end{Bmatrix} = \mathbf{R}_3 \left[\theta_{\text{GMST}}\right] = \begin{bmatrix} \cos\theta_{\text{GMST}} & \sin\theta_{\text{GMST}} & 0 \\ -\sin\theta_{\text{GMST}} & \cos\theta_{\text{GMST}} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{Bmatrix} x \\ y \\ z \end{Bmatrix}.$$

Perifocal frame

The perifocal frame is centred in the focus of the orbit (i.e., the Earth), and has unit vectors:

$\mathbf{i}_e = \mathbf{e}/e$	vector directed along the eccentricity vector, pointing towards periapsis.
$\mathbf{i}_p$	completes the set, $\mathbf{i}_p = \mathbf{i}_h \times \mathbf{i}_e$.
$\mathbf{i}_h = \mathbf{h}/h$	vector directed along the angular momentum vector, out of the orbital plane.



Perifocal frame.

The plane $(\overline{x}, \overline{y})$ is the orbital plane.

The kinematics of a satellite can be elegantly described in the perifocal frame, and the frame will be useful when introducing the orbital elements. The position and velocity vectors are expressed in perifocal coordinates as

$$\mathbf{r} = \overline{x}\,\mathbf{i}_e + \overline{y}\,\mathbf{i}_p \tag{49}$$

$$\mathbf{v} = \dot{\overline{x}}\,\mathbf{i}_e + \dot{\overline{y}}\,\mathbf{i}_p. \tag{50}$$

Note that

$$\overline{x} = r\cos\theta, \quad \dot{\overline{x}} = \dot{r}\cos\theta - r\dot{\theta}\sin\theta$$
$$\overline{y} = r\sin\theta, \quad \dot{\overline{y}} = \dot{r}\sin\theta + r\dot{\theta}\cos\theta.$$

By substituting the orbit equation (3) and considering that $\dot{\theta} = h/r^2$,

$$\dot{r} = \frac{\mu}{h} e \sin\theta, \tag{51}$$

we have that the position and velocity vectors in the perifocal frame are written as

$$\mathbf{r} = \frac{h^2}{\mu}\frac{1}{1 + e\cos\theta}\left(\cos\theta\,\mathbf{i}_e + \sin\theta\,\mathbf{i}_p\right)$$
$$\mathbf{v} = \frac{\mu}{h}\left[-\sin\theta\,\mathbf{i}_e + (e + \cos\theta)\;\mathbf{i}_p\right].$$

State Vector and Orbital Elements

In order to describe the current and future locations of a satellite in three-dimensional space we need at least six independent scalar quantities. This reflects the fact that we derived six constants of integration from the two-body equations of motion. The vector

collecting these quantities is the *state vector* of the satellite. So far, we have considered these six quantities to be the Cartesian (i.e., rectangular) coordinates of the position and velocity vectors along the inertial reference frame; therefore, the *state vector* in Cartesian coordinates is $\mathbf{x} = [\mathbf{r}, \mathbf{v}]$.

However, this is not the only state vector choice. In fact, we can describe the state vector of a satellite through a set of six *orbital elements*, which gives us a clear picture of the size, shape, and orientation of the orbit in three-dimensional space. The set of classical orbital elements is the following:

a	the semimajor axis
e	the eccentricity
i	the inclination
Ω	the right ascension of the ascending node (RAAN)
ω	the argument of periapsis
θ	the true anomaly

The first two values above describe the shape of the orbit (a, e). The angles (i, Ω, ω) describe the orientation of the orbit plane relative to the inertial frame of reference, and the angle θ describes the location of the satellite within the plane of the orbit. The *ascending node* (denoted ☊) is the intersection of the orbital plane with the reference plane, on the ascending side of the orbit. Equivalent sets of orbital elements might swap the semi-major axis a for the semi-latus rectum p or the specific angular momentum magnitude h, or might include the mean anomaly M or time of periapsis passage t_p instead of the true anomaly θ.

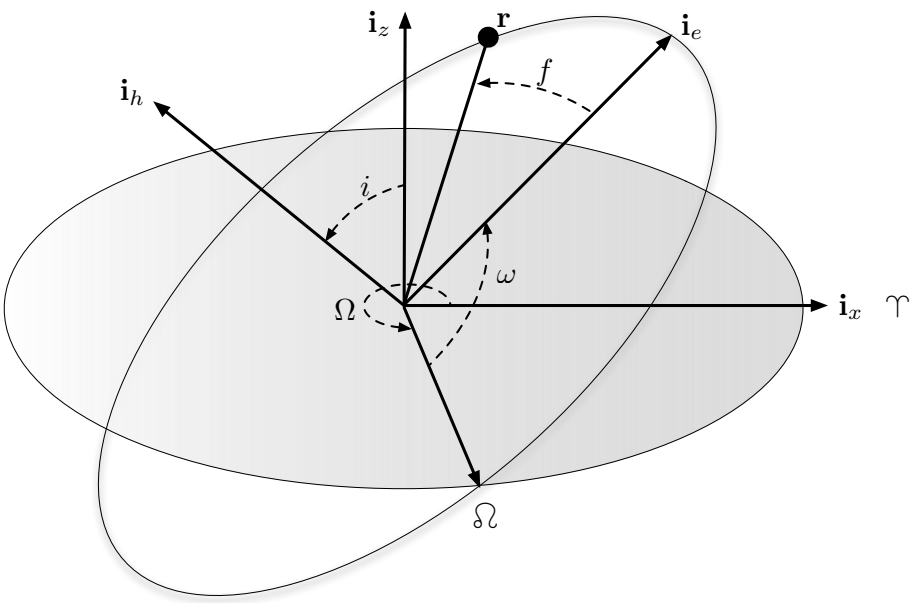
The transformation from orbital elements to position and velocity works as follows. In the previous section, we derived the position and velocity of a satellite in the perifocal frame as

$$\mathbf{r}_\mathrm{p} = \frac{h^2}{\mu}\frac{1}{1+e\cos\theta}\left(\cos\theta\,\mathbf{i}_e + \sin\theta\,\mathbf{i}_p\right)$$
$$\mathbf{v}_\mathrm{p} = \frac{\mu}{h}\left[-\sin\theta\,\mathbf{i}_e + (e+\cos\theta)\;\mathbf{i}_p\right],$$

where the subscript $_\mathrm{p}$ denotes perifocal coordinates. To obtain the position and velocity in the inertial frame, $\mathbf{r}_\mathrm{I}, \mathbf{v}_\mathrm{I}$, we need to rotate the perifocal frame into the inertial frame. The rotation is given by

$$\mathbf{r}_\mathrm{I} = \mathbf{A}\mathbf{r}_\mathrm{p}$$
$$\mathbf{v}_\mathrm{I} = \mathbf{A}\mathbf{v}_\mathrm{p}$$

where each column of the direction cosine matrix $\mathbf{A}$ is formed by the projection (given by the dot product) of the unit vectors of the inertial reference frame $\mathbf{i}_x, \mathbf{i}_y, \mathbf{i}_z$ on those of the



The classical orbital elements.

perifocal frame:

$$\mathbf{A} = \begin{bmatrix} \mathbf{i}_x \cdot \mathbf{i}_e & \mathbf{i}_x \cdot \mathbf{i}_p & \mathbf{i}_x \cdot \mathbf{i}_h \\ \mathbf{i}_y \cdot \mathbf{i}_e & \mathbf{i}_y \cdot \mathbf{i}_p & \mathbf{i}_y \cdot \mathbf{i}_h \\ \mathbf{i}_z \cdot \mathbf{i}_e & \mathbf{i}_z \cdot \mathbf{i}_p & \mathbf{i}_z \cdot \mathbf{i}_h \end{bmatrix}$$

$$= \begin{bmatrix} \cos\Omega\cos\omega - \sin\Omega\sin\omega\cos i & -\cos\Omega\sin\omega - \sin\Omega\cos\omega\cos i & \sin\Omega\sin i \\ \sin\Omega\cos\omega + \cos\Omega\sin\omega\cos i & -\sin\Omega\sin\omega + \cos\Omega\cos\omega\cos i & -\cos\Omega\sin i \\ \sin\omega\sin i & \cos\omega\sin i & \cos i \end{bmatrix}.$$

This can also be derived by considering the orthogonal rotation matrices associated with rotations Ω, i and then ω (in that order).

$$\mathbf{R}_3\left[\Omega\right] = \begin{bmatrix} \cos\Omega & \sin\Omega & 0 \\ -\sin\Omega & \cos\Omega & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{R}_1\left[i\right] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos i & \sin i \\ 0 & -\sin i & \cos i \end{bmatrix}$$

$$\mathbf{R}_3\left[\omega\right] = \begin{bmatrix} \cos\omega & \sin\omega & 0 \\ -\sin\omega & \cos\omega & 0 \\ 0 & 0 & 1 \end{bmatrix}$$
$$\mathbf{A} = \left[\mathbf{R}_3\left(\omega\right)\mathbf{R}_1\left(i\right)\mathbf{R}_3\left(\Omega\right)\right]^T.$$

Computing the orbital elements corresponding to the Cartesian state in the inertial frame $\mathbf{x} = [\mathbf{r}, \mathbf{v}] = [r_x, r_y, r_z, v_x, v_y, v_z]$ is a bit more involved. First, compute the radius and velocity vector magnitudes,

$$r = \sqrt{r_x^2 + r_y^2 + r_z^2}$$
$$v = \sqrt{v_x^2 + v_y^2 + v_z^2}.$$

Depending on the particular set of orbital elements used, the first element can be regarded as the specific angular momentum magnitude $\mathbf{h}$, the semi-latus rectum p, or the semi-major axis a. Compute the angular momentum vector along with its magnitude as

$$\mathbf{h} = \mathbf{r} \times \mathbf{v} = [r_y v_z - r_z v_y, r_z v_x - r_x v_z, r_x v_y - r_y v_x] = [h_x, h_y, h_z]$$
$$h = \sqrt{h_x^2 + h_y^2 + h_z^2}.$$

The semi-major axis can be computed by rearranging the terms in the vis-viva equation (5),

$$a = \frac{1}{\frac{2}{r} - \frac{v^2}{\mu}}.$$

For parabolic trajectories $a \to \infty$, and for hyperbolic trajectories $a < 0$. Often, a better choice for representing all orbit types is the semi-latus rectum is the first element,

$$p = \frac{h^2}{\mu},$$

which is positive and well-defined in all cases.

The inclination is the angle between the angular momentum vector and the z-axis of the inertial frame, and ranges from $0°$ (for equatorial, *prograde* orbits) to $180°$ (for equatorial, *retrograde* orbits). This corresponds to the range of the arccosine function, therefore the inclination is simply

$$i = \cos^{-1}\left(\frac{h_z}{h}\right).$$

Next, we compute the nodal vector $\mathbf{N}$ along with its magnitude N

$$\mathbf{N} = \mathbf{i}_z \times \mathbf{h},$$
$$N = \sqrt{h_x^2 + h_y^2}.$$

The nodal vector $\boldsymbol{N}$ is in the equatorial plane and points towards the direction of the ascending node, ☊. The RAAN is the angle between the nodal vector and the x-axis of the ECI frame, and can range from $0°$ to $360°$. The range of the arccosine is $[0°, 180°]$, therefore, we must the resolve a quadrant ambiguity. To do so, note that if $N_y > 0$ the RAAN is in the first or second quadrants, vice versa if $N_y < 0$ the RAAN is in the third or fourth quadrants. Therefore

$$\Omega = \begin{cases} \cos^{-1}\left(\frac{N_x}{N}\right), \text{if } N_y \geq 0 \\ 360° - \cos^{-1}\left(\frac{N_x}{N}\right), \quad \text{if } N_y < 0 \end{cases}.$$

The eccentricity vector is computed by rearranging equation (2),

$$\mathbf{e} = \frac{\mathbf{v} \times \mathbf{h}}{\mu} - \frac{\mathbf{r}}{r},$$

and the eccentricity is the magnitude of the eccentricity vector,

$$e = \sqrt{\mathbf{e} \cdot \mathbf{e}} = \sqrt{1 + \frac{h^2}{\mu^2}\left(\frac{2\mu}{r} - v^2\right)}, \tag{52}$$

where the derivation of the above relation is left as an exercise. Equation (52) shows that there are two kinds of orbits for which $e = 0$: rectilinear orbits for which $h = 0$, and parabolic orbits for which the orbital energy is zero and thus $v^2 = \frac{2\mu}{r}$; either of them is rarely encountered in practice. The argument of periapsis is the angle between the eccentricity vector $\mathbf{e}$, which points in the direction of periapsis, and the nodal vector $\boldsymbol{N}$. Analogously to the RAAN, it ranges from $0°$ to $360°$, therefore we must resolve a quadrant ambiguity by looking at the sign of the component of the eccentricity vector along the z-axis of the inertial frame. If $e_z \geq 0$, then the periapsis is above the equatorial plane ($0° \leq \omega \leq 180°$); otherwise, if $e_z < 0$ it is below the equatorial plane ($180° < \omega < 360°$). Thus

$$\omega = \begin{cases} \cos^{-1}\left(\frac{\mathbf{N}\cdot\mathbf{e}}{Ne}\right), \quad \text{if } e_z \geq 0 \\ 360° - \cos^{-1}\left(\frac{\mathbf{N}\cdot\mathbf{e}}{Ne}\right), \quad \text{if } e_z < 0 \end{cases}.$$

Finally, we compute the true anomaly, that is the angle between the radius and the eccentricity vectors. As with the RAAN and argument of periapsis, we need to resolve a quadrant ambiguity. This is done by recognising that the radial velocity $(\mathbf{r}/r) \cdot \mathbf{v}$ is positive if the satellite is flying away from periapsis, and negative otherwise,

$$\theta = \begin{cases} \cos^{-1}\left(\frac{\mathbf{e}\cdot\mathbf{r}}{er}\right), \quad \text{if } \mathbf{r}\cdot\mathbf{v} \geq 0 \\ 360° - \cos^{-1}\left(\frac{\mathbf{e}\cdot\mathbf{r}}{er}\right), \quad \text{if } \mathbf{r}\cdot\mathbf{v} < 0 \end{cases}.$$

If the mean anomaly rather than the true anomaly is wanted, we first need to compute the eccentric anomaly through

$$E = 2\tan^{-1}\left(\sqrt{\frac{1-e}{1+e}}\tan\frac{\theta}{2}\right). \tag{15}$$

Then, the mean anomaly is obtained by solving Kepler's equation,

$$M = E - e\sin(E). \tag{53}$$

Propagating orbits

Let's assume that we know the position $\mathbf{r}_0$ and velocity $\mathbf{v}_0$ of a satellite at an initial time t_0. How do we obtain the position and velocity at any time $t \neq t_0$? We have two ways of achieving this. The first exploits the fact that the orbital elements are constants of the two-body problem, thus we can follow the steps:

1. Transform the state vector at time t_0 from Cartesian coordinates $\mathbf{x}(t_0) = [\mathbf{r}_0, \mathbf{v}_0]$ to orbital elements $\mathbf{y}(t_0) = [h_0, e_0, i_0, \Omega_0, \omega_0, M_0]$ using the procedures in the previous section.
2. At any time, all orbital elements are constant except for the mean anomaly, which locates the satellite along its orbit. The new value of the mean anomaly is $M(t) = M_0 + n(t - t_0)$, where n is the mean motion defined in equation (17).
3. Compute the eccentric anomaly $E(t)$ by solving Kepler's equation.
4. Obtain the true anomaly by inverting equation (15), giving

$$\theta(t) = 2\tan^{-1}\left(\sqrt{\frac{1+e}{1-e}}\tan\frac{E(t)}{2}\right)$$

5. Transform the state vector in orbital elements $\mathbf{y}(t) = [h_0, e_0, i_0, \Omega_0, \omega_0, \theta(t)]$ to Cartesian coordinates $\mathbf{r}(t), \mathbf{v}(t)$.

For hyperbolic and parabolic orbits, we have to solve the corresponding versions of Kepler's equation. This method may be cumbersome to implement if all that is desired is propagating position and velocity. Another, more elegant way of propagating orbits in time relies on the so-called *Lagrange coefficients*, which are also useful for other applications in orbit determination.

The Lagrange coefficients

According to equations (49), (50), the position and velocity vectors at a time t are expressed in the perifocal frame as

$$\mathbf{r}(t) = \overline{x}(t)\,\mathbf{i}_e + \overline{y}(t)\,\mathbf{i}_p \tag{49}$$

$$\mathbf{v}(t) = \dot{\overline{x}}(t)\,\mathbf{i}_e + \dot{\overline{y}}(t)\,\mathbf{i}_p. \tag{50}$$

At t_0, the initial position and velocity in the perifocal frame are

$$\mathbf{r}(t_0) = \overline{x}_0\,\mathbf{i}_e + \overline{y}_0\,\mathbf{i}_p \tag{49}$$
$$\mathbf{v}(t_0) = \dot{\overline{x}}_0\,\mathbf{i}_e + \dot{\overline{y}}_0\,\mathbf{i}_p. \tag{50}$$

We can solve for the unit vectors of the perifocal frame $\mathbf{i}_e, \mathbf{i}_p$ as a function of the coordinates $\overline{x}, \overline{x}_0, \overline{y}, \overline{y}_0, \dot{\overline{x}}, \dot{\overline{x}}_0, \dot{\overline{y}}, \dot{\overline{y}}_0$ and plug the resulting expressions into equations (49), (50). The position and velocity are rewritten as

$$\mathbf{r} = f\mathbf{r}_0 + g\mathbf{v}_0 \tag{54}$$
$$\mathbf{v} = \dot{f}\mathbf{r}_0 + \dot{g}\mathbf{v}_0 \tag{55}$$

where

$$f = \frac{\overline{x}\,\dot{\overline{y}}_0 - \overline{y}\,\dot{\overline{x}}_0}{h} \quad g = \frac{-\overline{x}\,\overline{y}_0 + \overline{y}\,\overline{x}_0}{h}$$
$$\dot{f} = \frac{-\dot{\overline{x}}\,\overline{y}_0 + \dot{\overline{y}}\,\overline{x}_0}{h} \quad \dot{g} = \frac{-\dot{\overline{x}}\,\overline{y}_0 + \dot{\overline{y}}\,\overline{x}_0}{h}\ .$$

These are called the *Lagrange coefficients*. Conservation of angular momentum results in the relation

$$f\dot{g} - g\dot{f} = 1,$$

whose derivation is left as an exercise (hint: use equations (54), (55) to compute the angular momentum vector). Therefore, if any three of the four coefficients are known, the fourth can be derived through the above relation.

The Lagrange coefficients can be rewritten as a function of the initial and final radii r_0, r, the angular momentum h, and the difference in true anomalies $\Delta\theta = \theta(t) - \theta_0$ as follows:

$$f = 1 - \frac{\mu r}{h^2}\left(1 - \cos\Delta\theta\right) \qquad g = \frac{r r_0}{h}\sin\Delta\theta$$
$$\dot{f} = \frac{\mu}{h}\frac{1-\cos\Delta\theta}{\sin\Delta\theta}\left[\frac{\mu}{h^2}\left(1-\cos\Delta\theta\right) - \frac{1}{r_0} - \frac{1}{r}\right] \quad \dot{g} = 1 - \frac{\mu r_0}{h^2}\left(1-\cos\Delta\theta\right)\ . \tag{56}$$

These equations, along with equations (54), (55), easily allow us to propagate the position and velocity through a true anomaly interval $\Delta\theta$. To perform orbit propagation with Lagrange coefficients through a time interval Δt, we need to compute the corresponding true anomaly interval $\Delta\theta$. We will leave the derivation of $\Delta\theta$ from Δt knowing $\mathbf{r}_0, \mathbf{v}_0, t_0, t$ as an exercise (hint: we only need to compute the angular momentum and eccentricity).

Preliminary Orbit Determination

So far, we have looked at how satellites move in space. However, in order to carry out a successful space mission, we need to know where a specific satellite is (that is, what are its current position and velocity $\mathbf{r}(t), \mathbf{v}(t)$) and where it will be in the future from *observations* taken from ground stations. Computing orbits from observations is called *orbit determination*.

We will now look at the methods used to compute orbits from observations. We will focus on *Gibbs's method*, which allows us to compute an orbit from three geocentric position vectors, and on solving *Lambert's problem* to compute an orbit connecting two geocentric position vectors given the time between them. If, instead of vectors, we have at least three sets of angular measurements (right ascension and declination) and therefore the range to the object is unknown, we can use *Gauss's method* of orbit determination. We will not cover Gauss's method in this module, but be aware of its existence!

All of these methods rely on the assumption that the motion of the satellite is purely Keplerian, i.e., that there are no perturbing accelerations acting on the satellite except the main gravitational acceleration from the primary body. In the real world, satellites are subject to additional forces arising from the non-uniform mass distribution of the Earth, aerodynamic forces, radiation pressure, and several additional causes. Real satellites observations are also never perfect and they are affected by sensor noise. Because the methods that we are going to explore do not take into account perturbations and sensor noise, they are considered *preliminary* orbit determination methods, which are used to generate "first guesses" for the satellite's orbit that can then be refined using more advanced methods. Operational orbit determination is done by using statistical estimation methods based on probability and dynamical systems theory, which we will not cover in this course.

Gibbs' method

In Gibbs' method, we start from three coplanar position vectors $\mathbf{r}_1 = \mathbf{r}(t_1), \mathbf{r}_2 = \mathbf{r}(t_2), \mathbf{r}_3 = \mathbf{r}(t_3)$ corresponding to times $t_1 < t_2 < t_3$. Our goal is to derive the set of six orbital elements describing the orbit $a, e, i, \Omega, \omega, M$. We do this by finding a formula to compute the velocity at any of the times t_1, t_2, t_3, and, once we have the velocity, we can transform $\mathbf{r}, \mathbf{v}$ into orbital elements following the procedure shown in the previous lecture.

We start by using the definition of the eccentricity vector $\mathbf{e}$, equation (2), to arrive at a new expression for the velocity vector $\mathbf{v}$. To solve for $\mathbf{v}$, we cross both sides of equation (2) by $\mathbf{h}$, use the *bac-cab* vector identity

$$\boxed{\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}\,(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}\,(\mathbf{A} \cdot \mathbf{B})} \tag{57}$$

and rearrange the terms to arrive at the following expression for the velocity:

$$\mathbf{v} = \frac{\mu}{h^2}\left(\frac{\mathbf{h}\times\mathbf{r}}{r} + \mathbf{h}\times\mathbf{e}\right).$$

Recalling the definition of the perifocal frame we see that, because $\mathbf{h} = h\mathbf{i}_h, \mathbf{e} = e\mathbf{i}_e, \mathbf{i}_p = \mathbf{i}_h \times \mathbf{i}_e$,

$$\mathbf{v} = \frac{\mu}{h}\left(\frac{\mathbf{i}_h\times\mathbf{r}}{r} + e\mathbf{i}_p\right). \tag{58}$$

Therefore, to find the velocity corresponding to any of the position vectors $\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3$, we just need to know the angular momentum magnitude h and the unit vectors of the perifocal frame $\mathbf{i}_h$, $\mathbf{i}_p$.

To compute the first two quantities, we define the two auxiliary vectors

$$\mathbf{N} := r_1(\mathbf{r}_2\times\mathbf{r}_3) + r_2(\mathbf{r}_3\times\mathbf{r}_1) + r_3(\mathbf{r}_1\times\mathbf{r}_2) \tag{59}$$

$$\mathbf{D} := \mathbf{r}_1\times\mathbf{r}_2 + \mathbf{r}_2\times\mathbf{r}_3 + \mathbf{r}_3\times\mathbf{r}_1. \tag{60}$$

By using the orbit equation, along with the fact that the three position vectors are coplanar, it can be shown that

$$h = \sqrt{\mu\frac{N}{D}} \tag{61}$$

$$\mathbf{i}_h = \frac{\mathbf{D}}{D}, \tag{62}$$

where the last equation can be easily understood by realizing that the vector resulting from the three cross products in the right-hand side of equation (60) are orthogonal to the orbital plane.

The only quantity left to compute in order to evaluate equation (58) is the unit vector

$$\mathbf{i}_p = \mathbf{i}_h\times\hat{\mathbf{p}} = \frac{1}{De}\mathbf{D}\times\mathbf{e} =$$
$$\frac{1}{De}\left[(\mathbf{r}_1\times\mathbf{r}_2)\times\mathbf{e} + (\mathbf{r}_2\times\mathbf{r}_3)\times\mathbf{e} + (\mathbf{r}_3\times\mathbf{r}_1)\times\mathbf{e}\right].$$

We use the orbit equation (3) and the *bac-cab* identity, equation (57), to arrive at

$$\mathbf{i}_p = \frac{1}{De}\mathbf{S}, \tag{63}$$

where the auxiliary vector $\mathbf{S}$ is

$$\mathbf{S} = \mathbf{r}_1(r_2 - r_3) + \mathbf{r}_2(r_3 - r_1) + \mathbf{r}_3(r_1 - r_2).$$

Once $h, \mathbf{i}_h, \mathbf{i}_p$ are computed through equations (61 – 63), we plug them into equation (58), along with any of the three position vectors, to obtain the corresponding velocity vector $\mathbf{v}$.

At this point, we can to transform the position and velocity vectors into orbital elements. Because each of the position vectors $\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3$ relate to the same Keplerian orbit, the orbital elements resulting from computing any of the states $(\mathbf{r}_1, \mathbf{v}_1), (\mathbf{r}_2, \mathbf{v}_2), (\mathbf{r}_3, \mathbf{v}_3)$ will result in the same orbital elements $(a, e, i, \Omega, \omega)$, whereas the true anomaly θ will increase with time: $\theta(t_3) > \theta(t_2) > \theta(t_1)$.

Lambert's problem

Consider now the case in which we know that an object passes through two position vectors $\mathbf{r}_1 = \mathbf{r}(t_1), \mathbf{r}_2 = \mathbf{r}(t_2)$, and we want to know the orbit (or orbits) that connect them. This is *Lambert's problem*, and consists of solving the two-point boundary value problem

$$\begin{aligned} \ddot{\mathbf{r}} &= -\frac{\mu}{r^3}\mathbf{r} \\ \mathbf{r}(t_1) &= \mathbf{r}_1 \\ \mathbf{r}(t_2) &= \mathbf{r}_2. \end{aligned}$$

The time interval $t_2 - t_1 = \Delta t$ is called the *transfer time*. Although here we consider Lambert's problem as an algorithm for orbit determination, it is equally relevant for the case in which we want to manoeuvre a satellite from one position to another in a given time (this is called *interception*). In fact, Lambert's problem is often used in the preliminary design of interplanetary missions.

We will explore some geometric properties of Lambert's problem that will be very helpful to understand its solutions. Only the case of elliptical transfer orbits is considered here as it is the most relevant for real-world applications; however, similar treatments for hyperbolic transfer orbits can be found in Battin's book. A lot of solution algorithms for Lambert's problem exist in the literature, and this is still an active area of research.

Geometric properties of Lambert's problem

The figure below shows the basic geometry of Lambert's problem, where P_1 and P_2 are our departure and arrival positions, respectively. For the remainder of this section and without loss of generality, we assume that the arrival radius is larger than the departure radius, $r_2 = \|\mathbf{r}_2\| \geq r_1 = \|\mathbf{r}_1\|$. Because gravity is a conservative force, if we need the solution for the case $r_2 < r_1$ we can simply change the sign of the initial velocity for the case $r_2 \geq r_1$. The departure and arrival position radii are separated by the *transfer angle* $\Delta\theta$, and the distance between the two vectors is the *chord length* $c = \|(\mathbf{r}_2 - \mathbf{r}_1)\|$. The triangle $P_1 F P_2$ is called the *space triangle*.

The transfer orbit is an elliptical orbit around the Earth (or any point mass), which occupies the focus F, and has vacant focus F^*. By definition, the sum of the distances from a point

on the ellipse to the foci are equal to the major axis of the ellipse; therefore the distances of P_1, P_2 from the vacant focus are

$$P_1F^* = 2a - r_1$$
$$P_2F^* = 2a - r_2.$$

How many transfer orbits are possible between P_1 and P_2? It depends on the value of the transfer orbit semi-major axis a. For positive semi-major axis values $a < a_{\rm m}$, there are no elliptical transfer orbits because there are no intersections between the circumferences of radii P_1F^*, P_2F^*. As the orbital energy is proportional to the semi-major axis, the transfer orbit with $a = a_{\rm m}$ is the *minimum-energy* transfer orbit. Transfers with $a < a_{\rm m}$ are only feasible if they are hyperbolic. The minimum value of the semi-major axis can be easily found by considering that

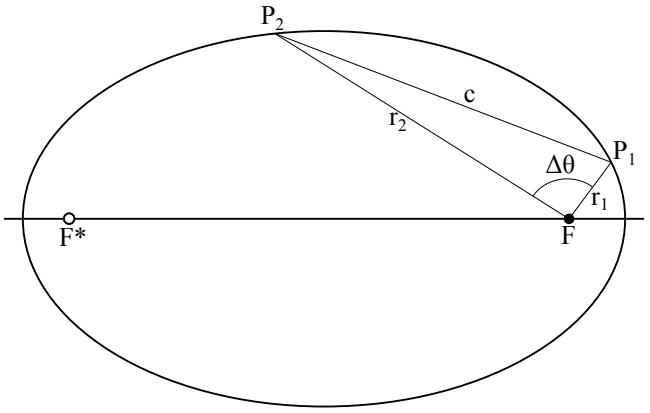
$$c = 2a_{\rm m} - r_1 + 2a_{\rm m} - r_2 \Rightarrow a_{\rm m} = \frac{r_1 + r_2 + c}{4} = \frac{s}{2}, \tag{64}$$

where $s \equiv (r_1 + r_2 + c)/2$ is the *semiperimeter* of the space triangle.

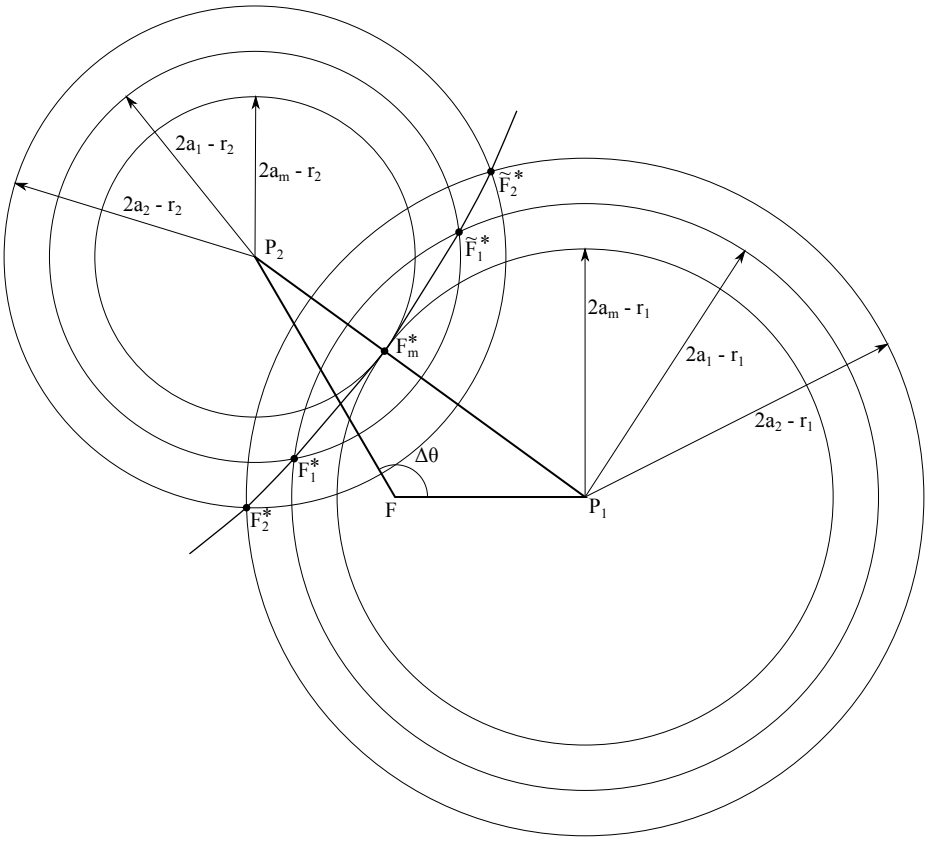
For $a > a_{\rm m}$, there are two intersections that correspond to two feasible transfer orbits with the same semi-major axis and different eccentricities. The difference of the distances between a vacant focus F^* and the points P_1, P_2 is

$$P_1F^* - P_2F^* = r_2 - r_1,$$

which is constant. Therefore, the locus of the vacant transfer orbit foci is a hyperbola with foci P_1, P_2. As the distance between the foci of each transfer orbit is $FF^* = 2ae$, transfer orbits with vacant foci on the side of the hyperbola closer to F have smaller eccentricity than those on the other side. In the figure, we denote transfer orbits with larger eccentricity with $\tilde{F}^*$.



Geometry of Lambert's problem.



Locus of the vacant foci of transfer orbits.

It can be shown that the locus of the eccentricity vectors is a straight line that is perpendicular to the chord. As a consequence, there exists a transfer ellipse with minimum eccentricity

$$e = \frac{r_2 - r_1}{c}.$$

Lambert's theorem and transfer time

Lambert's problem is named after the Swiss-German mathematician Johann Heinrich Lambert, who in the 18th century derived the following theorem through a geometrical proof:

Lambert's theorem: *The transfer time for an arc of ellipse connecting points P_1, P_2 at distances r_1, r_2 only depends on the semi-major axis a, the chord length c, and the sum $r_1 + r_2$,*

$$\Delta t = t_2 - t_1 = f(a, c, r_1 + r_2). \tag{65}$$

Note that the transfer time does not depend on the eccentricity of the ellipse.

By virtue of Kepler's equation, the transfer time is

$$\sqrt{\mu}(t_2 - t_1) = a^{3/2} \left[E_2 - E_1 - e(\sin E_2 - \sin E_1) \right]. \tag{66}$$

To prove Lambert's theorem, we would like to put this equation in the form of Equation (65). To do so, it is useful to express both the sum of radii $r_1 + r_2$ and the chord length c as a function of the quantities

$$E^+ = \frac{E_2 + E_1}{2}, \quad E^- = \frac{E_2 - E_1}{2}.$$

The orbit equation written in terms of the eccentric anomaly is

$$\boxed{r = a(1 - e\cos E)} \tag{67}$$

Therefore, the sum of the radii can be expressed as

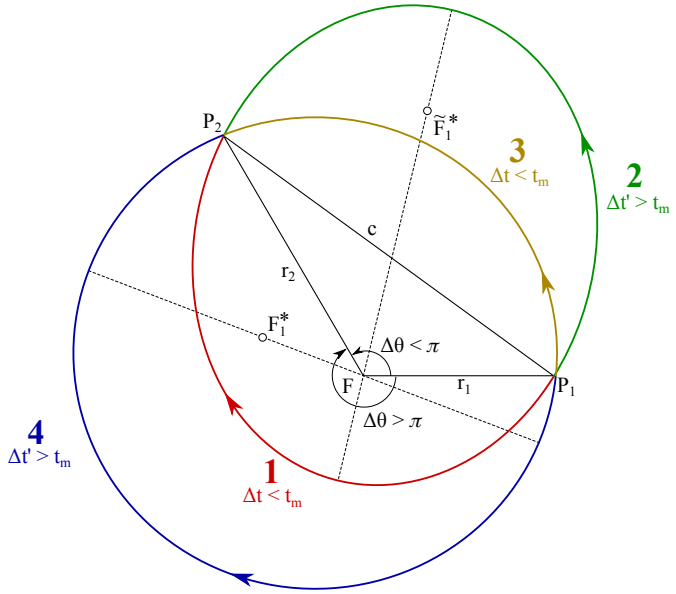
$$r_1 + r_2 = a \left[2 - e \left(\cos E_1 + \cos E_2 \right) \right] = 2a \left[1 - e \cos E^+ \cos E^- \right]. \tag{68}$$

We now look at the coordinates x, y of a point on the ellipse in Cartesian axes with origin in the center (and not in the focus!). These can be expressed as a function of the eccentric anomaly as

$$x = a\cos E, \quad y = b\sin E,$$

where $b = a\sqrt{1 - e^2}$ is the semi-minor axis. The chord length is

$$\begin{aligned} c = (x_2 - x_1)^2 + (y_2 - y_1)^2 = \\ a^2 \left[\left(\cos^2 E_2 - \cos^2 E_1 \right) + \left(1 - e^2 \right) \left(\sin^2 E_2 - \sin^2 E_1 \right) \right]. \end{aligned} \tag{69}$$



Transfer arcs corresponding to a given semi-major axis.

With the help of trigonometric identities, the term in the brackets can be expressed as a function of E^+, E^-,

$$\left(\cos^2 E_2 - \cos^2 E_1\right) + \left(1 - e^2\right)\left(\sin^2 E_2 - \sin^2 E_1\right) = \sin^2 E^- \left(1 - e^2 \cos^2 E^+\right).$$

The term on the right-hand side is "almost" the square of a sine! If we define

$$\cos\xi = e \cos E^+ \tag{70}$$

then the chord length and sum of radii become

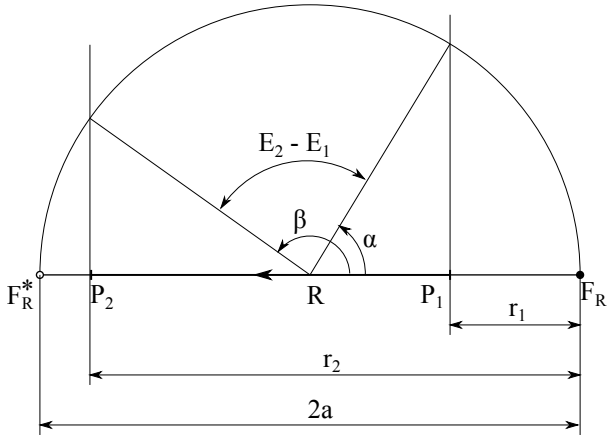
$$c = 2a \sin E^- \sin\xi$$
$$r_1 + r_2 = 2a\left(1 - \cos E^- \cos\xi\right).$$

Now we can combine these two equations to write

$$r_1 + r_2 + c = 2a\left(1 - \cos E^- \cos\xi + \sin E^- \sin\xi\right) = 2a\left(1 - \cos\alpha\right) = 4a \sin^2\left(\frac{\alpha}{2}\right)$$
$$r_1 + r_2 - c = 2a\left(1 - \cos E^- \cos\xi - \sin E^- \sin\xi\right) = 2a\left(1 - \cos\beta\right) = 4a \sin^2\left(\frac{\beta}{2}\right),$$

where we have defined the new angles

$$\alpha = \xi + E^-, \quad \beta = \xi - E^-. \tag{71}$$



Rectilinear transfer ellipse.

Finally, we can rewrite Equation (66) for the transfer time as

$$\begin{aligned} \sqrt{\mu}\Delta t = a^{3/2}\left[E_2 - E_1 - e\left(\sin E_2 - \sin E_1\right)\right] = \\ 2a^{3/2}[E^- - e\sin E^-\cos E^+] = \\ \boxed{\sqrt{\mu}\Delta t = a^{3/2}\left[\alpha - \beta - \left(\sin\alpha - \sin\beta\right)\right].} \end{aligned} \tag{72}$$

This form of Kepler's equation is also called *Lambert's equation*, and it gives the transfer time for all arcs of ellipse short than one complete revolution, $0 \leq \Delta\theta < 2\pi$. The angles α, β depend only on the semiperimeter s and the chord length c through the relations

$$\begin{aligned} \sin^2\left(\frac{\alpha}{2}\right) &= \frac{s}{2a} \\ \sin^2\left(\frac{\beta}{2}\right) &= \frac{s-c}{2a}, \end{aligned} \tag{73}$$

therefore Lambert's theorem is proved.

For a given $a > a_{\mathrm{m}}$, there are four feasible transfer arcs: looking at the figure, if $\Delta\theta \leq \pi$ we can follow arcs 2 and 3, whereas if $\Delta\theta > \pi$ we can follow arcs 1 and 4.[2] Now note that inverting Equation (73) yields principal values $0 \leq \alpha_0 \leq \beta_0 \leq \pi$; however, the quadrants of α and β are ambiguous! Different choices of quadrants for α, β correspond to different transfer arcs. To resolve this quadrant ambiguity, we make a geometrical argument: we move the foci to find a *rectilinear ellipse* with the same a, c, r_1+r_2 as the transfer ellipse and with eccentricity $e = 1$. We can trace the auxiliary circle enclosing the rectilinear ellipse (as done in the derivation of Kepler's equation) to find the points P_1, P_2 corresponding to

[2]Remember that here we're always assuming that $r_2 \geq r_1$! In the case in which $r_1 < r_2$ we can follow exactly the same reasoning, and just run our solution backwards in time, because gravity is a conservative force.

the eccentric anomalies E_1, E_2. In virtue of Lambert's theorem, the transfer time between P_1 and P_2 on the rectilinear ellipse does not depend on the eccentricity and therefore is the same as on the original transfer ellipse. Now we can particularise the definitions of α, β to the rectilinear ellipse for which $e = 1$. Making use of Equations (70), (71), we see that

$$\alpha|_{e=1} = E_2, \quad \beta|_{e=1} = E_1, \quad \alpha - \beta = E_2 - E_1$$

Therefore, *α and β are the eccentric anomalies on the rectilinear ellipse with the same transfer time as the original transfer ellipse.* It follows that arcs 2 and 3, with $\Delta\theta \leq \pi$, correspond to $\beta = \beta_0, 0 \leq \beta < \pi$; conversely arcs 1 and 4 with $\Delta\theta > \pi$ correspond to $\beta = -\beta_0, -\pi < \beta < 0$. The quadrant of α is determined by the transfer time: for the transfer with shorter transfer time $\alpha = \alpha_0, 0 \leq \alpha\pi$, whereas for the longer transfer time $\alpha = 2\pi - \alpha_0, \pi \leq \alpha < 2\pi$. The choice of quadrant is summarized in the table below. Finally, let's examine what happens to the transfer time as a function of the semi-major axis. For $0 < a < a_\mathrm{m}$, no elliptical transfers exist. For the minimum-energy transfer time, Equation (64) gives us the semi-major axis as $a_\mathrm{m} = s/2$. Plugging this into Equation (73) and inverting the, we obtain the α, β angles as

$$\alpha_\mathrm{m} = \pi, \quad \beta_{\mathrm{m},0} = 2 \arcsin \sqrt{\frac{s-c}{s}},$$

and the transfer time is

$$\Delta t_m = \sqrt{\frac{s^3}{8\mu}} \left(\pi - \beta_\mathrm{m} + \sin \beta_\mathrm{m}\right), \tag{74}$$

where β_m is derived from $\beta_{\mathrm{m},0}$ according to the table above. For $a > a_\mathrm{m}$, there are two values for the transfer time: if α is in the first or second quadrant ($\alpha = \alpha_0$), the transfer time is $\Delta t < \Delta t_\mathrm{m}$. Conversely, if α is in the third or fourth quadrant ($\alpha = 2\pi - \alpha_0$), the transfer time is $\Delta t > \Delta t_\mathrm{m}$. For $a \to \infty$, the transfer time tends asymptotically to the parabolic transfer time Δt_p, which is computed by taking the limit for $a \to \infty$ of Lambert's equation:

$$\sqrt{\mu}\Delta t_\mathrm{p} = \frac{\sqrt{2}}{3}\left[s^{3/2} - \mathrm{sign}(\sin\Delta\theta)\,(s-c)^{3/2}\right], \tag{75}$$

where $\mathrm{sign}(x) = 1$ if $x > 0$ and $\mathrm{sign}(x) = -1$ if $x < 0$.

$\Delta\theta < \pi, \Delta t \leq t_\mathrm{m}$	$\alpha = \alpha_0$	$\beta = \beta_0$
$\Delta\theta < \pi, \Delta t > t_\mathrm{m}$	$\alpha = 2\pi - \alpha_0$	$\beta = \beta_0$
$\pi \leq \Delta\theta < 2\pi, \Delta t \leq t_\mathrm{m}$	$\alpha = \alpha_0$	$\beta = -\beta_0$
$\pi \leq \Delta\theta < 2\pi, \Delta t > t_\mathrm{m}$	$\alpha = 2\pi - \alpha_0$	$\beta = -\beta_0$

Solving Lambert's problem

Let's assume to have $\mathbf{r}_1, \mathbf{r}_2, \Delta t = t_2 - t_1$. For instance, these vectors could correspond to ground radar measurements of a satellite at two distinct times t_2, t_1. Alternatively, we might be interested in calculating a transfer orbit from Earth at time t_1 to Mars at time t_2. Because the transfer time only depends on $r_1 + r_2, c = \|\mathbf{r}_2 - \mathbf{r}_1\|, a$, the transfer orbit is unique. However, we do need to fix the direction of travel to assign the correct quadrant to $\Delta\theta$. We can obtain the transfer orbit through the following steps:

1. First, compute the parabolic transfer time through Equation (75). If $\Delta t \leq \Delta t_{\mathrm{p}}$, the transfer orbit is parabolic or hyperbolic.
2. Compute the minimum-energy transfer time through Equation (74).
3. Solve Lambert's equation, Equation (72) for the semi-major axis a. You can do this through either classical iterative methods such as the Newton-Raphson method, or by using an efficient iterative scheme such as that described in Chapter 7 of Battin's book. The quadrant of α must be chosen depending on whether $\Delta t < \Delta t_{\mathrm{m}}$ ($\alpha = \alpha_0$) or $\Delta t \geq \Delta t_{\mathrm{m}}$ ($\alpha = 2\pi - \alpha_0$).
4. The terminal velocity vectors can be conveniently obtained by considering a set of skewed unit vectors

$$\mathbf{u}_1 \equiv \frac{\mathbf{r}_1}{r_1}, \quad \mathbf{u}_2 \equiv \frac{\mathbf{r}_2}{r_2}, \quad \mathbf{u}_c \equiv \frac{\mathbf{r}_2 - \mathbf{r}_1}{c}.$$

   In these coordinates, the terminal velocity vectors are expressed as

$$\begin{aligned}
\mathbf{v}_1 &= (B + A)\mathbf{u}_c + (B - A)\mathbf{u}_1 \\
\mathbf{v}_2 &= (B + A)\mathbf{u}_c - (B - A)\mathbf{u}_2 \\
A &= \sqrt{\frac{\mu}{4a}} \cot\frac{\alpha}{2} \\
B &= \sqrt{\frac{\mu}{4a}} \cot\frac{\beta}{2}.
\end{aligned}$$

Non-Keplerian Orbits

Thus far we have focussed on the classical two-body problem, where the dynamics were described by

$$\frac{d^2\mathbf{r}}{dt^2} + \mu\frac{\mathbf{r}}{r^3} = 0. \tag{76}$$

Although it is not possible to find a closed-form solution to this problem, we were able to find a number of constants of integration, and to thereby show that solutions to this equation result in the classical conic orbits described (in part) by Kepler.

We now address the problem of finding solutions to problems of orbital motion that contain a disturbing (or perturbing) acceleration $\mathbf{a}_d$, so that the above *equations of motion* become

$$\frac{d^2\mathbf{r}}{dt^2} + \mu\frac{\mathbf{r}}{r^3} = \mathbf{a}_d. \tag{77}$$

In the above, the disturbance acceleration term can result from gravitational forces from third bodies (e.g. the Sun, Moon or other planets for an Earth-orbiting satellite), disturbances due to solar pressure, tides, non-sphericity of the Earth, atmospheric drag etc. The term $\mathbf{a}_d$ can also be treated as an engine thrust term in problems of orbit targeting, spacecraft station-keeping, or general orbit transfers.

In any of the above cases, the problem of finding a solution to the above ODE becomes much more complicated compared to the unperturbed (or *Keplerian*) case. By considering a limited set of perturbations under simplified assumptions, it is still possible to derive analytical solutions. This is called the *general perturbations* approach, and this process of solution relies on taking the average of the rates of change of the orbital elements over one orbital period. The averaged equations can sometimes be solved for the evolution of a set of *mean* orbital elements that approximately describe the state of the spacecraft. General perturbations methods give solutions that are fast to evaluate and give insight into the dynamics of the problem. However, their accuracy is limited by the simplifying assumptions made in the averaging process and in the method of solution, especially over long time spans. Furthermore, analytical general perturbations solutions can quickly get very complicated (with hundreds of terms) when adding perturbations for different orbital regimes.

The accuracy of general perturbations solutions can be improved by solving the averaged equations of motion numerically, rather than analytically. This leads to so-called *semi-analytical* solutions, which offer a compromise between speed and accuracy.

The most accurate solutions can be obtained through *special perturbations methods*, which generate solutions by integrating the equations of motion numerically. The most straightforward special perturbations method, called *Cowell's method*, consists in the direct integration of equation (77) through a numerical integration method such as a Runge-Kutta or multistep integration scheme. Special perturbations methods give very accurate solutions,

but they are also the methods that are most computationally expensive. This means that they require a lot of computational resources to arrive at a solution. Therefore, we are interested in making as much use as possible of the information that we have from the solution to the unperturbed Keplerian motion to aid in this numerical solution. The procedure we use to do this is known as *Encke's method*.

Encke's Method

Partition the position $\mathbf{r}$ and velocity $\mathbf{v}$ into reference terms $(\mathbf{r}_o,\mathbf{v}_o)$ and perturbation terms $(\boldsymbol{\delta}, \boldsymbol{\nu})$:

$$\mathbf{r} = \mathbf{r}_o + \boldsymbol{\delta}$$
$$\mathbf{v} = \mathbf{v}_o + \boldsymbol{\nu}$$

We assume that the reference trajectory $\mathbf{r}_o(t)$ represents the solution to the unperturbed problem

$$\frac{d^2\mathbf{r}_o}{dt^2} + \mu\frac{\mathbf{r}_o}{r_o^3} = 0.$$

If we subtract the dynamics of the reference trajectory from the dynamics of the true trajectory $\mathbf{r}(t)$ and collect all terms in $\boldsymbol{\delta}$ on the left-hand side, we are left with

$$\frac{d^2\boldsymbol{\delta}}{dt^2} + \frac{\mu}{r_o^3}\boldsymbol{\delta} = \mathbf{a}_d + \frac{\mu}{r_o^3}\left(1 - \frac{r_o^3}{r^3}\right)\mathbf{r}.$$

The first term on the right hand side represents an acceleration due to external forcing. The second term accounts for incremental changes to the gravitational force when $\mathbf{r} \neq \mathbf{r}_o$. If we integrate this ODE, we get an *exact* value for the difference $\boldsymbol{\delta}$ between the nominal trajectory $\mathbf{r}_o$ and the true trajectory $\mathbf{r}$. Encke's method for orbit propagation uses this idea in the following algorithm:

Encke's Method

1. Calculate a reference trajectory $\mathbf{r}_0(t)$ over some interval $t \in [0, T]$ using Kepler's equation / Lagrange coefficients.

2. Integrate $\boldsymbol{\delta}(t)$ over the interval $t \in [0, T]$ with initial condition $\delta(0) = 0$.

3. *Rectify* the two orbits by setting $\mathbf{r}_0(T) \leftarrow \mathbf{r}_o(T) + \boldsymbol{\delta}(T)$

4. use $\mathbf{r}_o(T)$ as a new initial condition and repeat.

Encke's method creates a sequence of diverging and re-converging trajectories as shown in the figure below. These trajectories are referred to as *osculating orbits* (from the Latin verb *osculari* – to kiss).

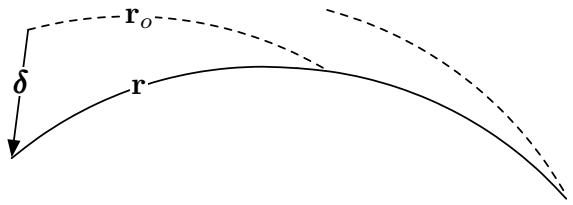


Figure : Osculating Orbits

The above procedure is an exact method and will work no matter how large $\boldsymbol{\delta}$ is allowed to get. However, in practice the idea is to employ this method only when $\mathbf{a}_d$ (and therefore $\boldsymbol{\delta}$) is relatively small. The main advantage of this method is that one can use a much larger step size in solving for $\boldsymbol{\delta}$ than would be possible if the original ODE in $(\mathbf{r}, \mathbf{a}_d)$ were to be solved directly.

Note that in the above algorithm, it is implied that the rectification step happens at fixed time intervals of size T. In practice one might instead choose to implement the rectification step when $\|\boldsymbol{\delta}\|$ reaches some threshold value relative to $\|\mathbf{r}\|$.

Improving Numerical Accuracy in Encke's Method

Although the method above is exact, the term

$$\left(1 - \frac{r_o^3}{r^3}\right)$$

can cause difficulties due to catastrophic cancellation when $\mathbf{r}_o \approx \mathbf{r}$. We can get around this difficulty by rewriting the awkward term above as

$$\frac{r_o^3}{r^3} = \left[\frac{(\mathbf{r}_o + \boldsymbol{\delta}) \cdot (\mathbf{r}_o + \boldsymbol{\delta})}{r_o^2}\right]^{-\frac{3}{2}} = \left[1 + \frac{\boldsymbol{\delta} \cdot (2\mathbf{r}_o + \boldsymbol{\delta})}{r_o^2}\right]^{-\frac{3}{2}}.$$

If we define a new variable $q = q(\boldsymbol{\delta}, \mathbf{r}_o)$ as

$$q := \frac{\boldsymbol{\delta} \cdot (2\mathbf{r}_o + \boldsymbol{\delta})}{r_o^2},$$

and define a new function Φ such that

$$\Phi(q) = 1 - (1+q)^{-\frac{3}{2}} = \left(1 - \frac{r_o^3}{r^3}\right).$$

It can be shown that

$$\Phi(q) = -q\frac{3+3q+q^2}{1+(1+q)^{3/2}}.$$

In this form, we avoid any catastrophic cancellation as there are no subtractions involved in the calculation of the term above. We can then rewrite our ODE for the perturbation term $\boldsymbol{\delta}$ as

$$\frac{d^2\boldsymbol{\delta}}{dt^2} + \frac{\mu}{r_o^3}\boldsymbol{\delta} = \mathbf{a}_d + \frac{\mu}{r_o^3}\Phi(q)(\mathbf{r}_o + \boldsymbol{\delta}).$$

with initial condition $\boldsymbol{\delta}(0) = 0$. The numerical conditioning of the above ODE is generally better than the original form, although it is still a nonlinear ODE that must be integrated numerically.

Perturbed motion in osculating elements

Trajectories in the presence of perturbations are not Keplerian conics any more. However, it is still possible to describe the state of the spacecraft by transforming position and velocity into orbital elements using the algorithms described in lecture 5. The resulting orbital elements describe a Keplerian conic that is tangent to the actual trajectory at each instant. In fact, if perturbations were "switched off" at a certain time τ, the spacecraft would follow the Keplerian conic described by its osculating orbital elements at time τ for all future times.

Because osculating elements are just another way of representing the state of the spacecraft, we can write their rates of change in the presence of perturbations. There are two main forms in which we can express the rates of change of the orbital elements mathematically: the *Lagrange planetary equations*, which are derived through the method of *variation of parameters*, and the *Gauss variational equations*, which are derived through vector calculus. Expressing the equations of motion in orbital elements allows us to integrate only the perturbations to obtain the evolution of the trajectory, as the Keplerian part of the dynamics will not result in a change in the orbital elements. In addition, they are needed to understand the qualitative effect on the orbit of perturbing accelerations and potentials.

Variation of parameters

The method of variation of parameters is a method of solution for general ordinary differential equations. In the previous lectures, we have seen that the unperturbed two-body problem,

$$\frac{d^2\mathbf{r}}{dt^2} + \mu\frac{\mathbf{r}}{r^3} = \mathbf{0}. \tag{76}$$

has a solution that can be expressed as a function of six independent constants of motion, which we can collect in the vector $\mathbf{c} = (a, e, i, \Omega, \omega, \theta)$. For instance, the constants can be

the classical orbital elements or the initial conditions in Cartesian coordinates. Therefore, we can write the position as

$$\mathbf{r} = \mathbf{f}\left(\mathbf{c}, t\right), \tag{78}$$

and taking its derivative

$$\mathbf{v} = \frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t} = \frac{\mathrm{d}\mathbf{f}\left(\mathbf{c}, t\right)}{\mathrm{d}t} = \frac{\partial \mathbf{f}}{\partial t} + \sum_{\alpha=1}^{6} \frac{\partial}{\partial c_\alpha}\mathbf{f}\left(\mathbf{c}, t\right) \cancelto{0}{\frac{\mathrm{d}c_\alpha}{\mathrm{d}t}}$$

where $\mathrm{d}c_\alpha/\mathrm{d}t = 0$ for all α, because the vector $\mathbf{c}$ is a constant. Thus we have

$$\mathbf{v} = \frac{\partial \mathbf{f}}{\partial t} \tag{79}$$

$$\mathbf{a} = \frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = \frac{\partial^2 \mathbf{f}}{\partial t^2} \tag{80}$$

Replacing the above expressions in Equation (76) results in

$$\frac{\partial^2 \mathbf{f}}{\partial t^2} + \mu \frac{\mathbf{f}\left(\mathbf{c}, t\right)}{\left\|\mathbf{f}\left(\mathbf{c}, t\right)\right\|^3} = \mathbf{0}. \tag{81}$$

We're now ready to introduce the method of variation of parameters.

Variation of Parameters: *We look for a solution of the differential equations expressing the perturbed two-body problem,*

$$\frac{d^2\mathbf{r}_\mathrm{p}(t)}{dt^2} + \mu \frac{\mathbf{r}_\mathrm{p}(t)}{r_\mathrm{p}^3} = \mathbf{a}_\mathrm{d}\left(\mathbf{r}_\mathbf{p}, \mathbf{v}_\mathrm{p}, t\right). \tag{77}$$

The solution must have the same mathematical form as for the unperturbed two-body problem,

$$\mathbf{r}_\mathrm{p}(t) = \mathbf{f}\left(\mathbf{u}(t), t\right), \tag{82}$$

with the only difference being that the coefficients $\mathbf{u}(t)$ *are now functions of time.*

Differentiating equation (82) and using the chain rule, we find the perturbed velocity[3]

$$\mathbf{v}_\mathrm{p}(t) = \frac{\mathrm{d}\mathbf{r}_\mathrm{p}(t)}{\mathrm{d}t} = \frac{\partial}{\partial t}\mathbf{f}(\mathbf{u}(t), t) + \sum_{\beta=1}^{6} \frac{\partial \mathbf{f}}{\partial u_\beta}\frac{\mathrm{d}u_\beta}{\mathrm{d}t}. \tag{83}$$

Because we want $\mathbf{v}_\mathrm{p}$ and $\mathbf{v}$ to have the same mathematical form, we impose that the last term in the above equation be zero:

$$\sum_{\beta=1}^{6} \frac{\partial \mathbf{f}}{\partial u_\beta}\frac{\mathrm{d}u_\beta}{\mathrm{d}t} = \mathbf{0}.$$

[3]The perturbed position $\mathbf{r}_\mathrm{p}(t)$ and velocity $\mathbf{v}_\mathrm{p}(t)$ are the solution to equation (77). The unperturbed position $\mathbf{r}(t)$ and velocity $\mathbf{v}(t)$ are the solution to equation (76).

This is called the *condition of osculation*, because it states that the velocity along the perturbed trajectory $\mathbf{v}_\mathrm{p}$ is the same as that along the unperturbed trajectory. Therefore, the orbital elements $\mathbf{u}(t)$ describe a Keplerian conic that is tangent (or *osculates*) the perturbed trajectory. Differentiating equation (83) after imposing the condition of osculation, we get an expression for the acceleration

$$\frac{\mathrm{d}^2\mathbf{r}_\mathrm{p}}{\mathrm{d}t^2} = \frac{\mathrm{d}\mathbf{v}_\mathrm{p}}{\mathrm{d}t} = \frac{\partial^2}{\partial t^2}\mathbf{f}(\mathbf{u}(t),t) + \sum_{\beta=1}^{6}\frac{\partial^2\mathbf{f}}{\partial t\partial u_\beta}\frac{\mathrm{d}u_\beta}{\mathrm{d}t}. \tag{84}$$

We can now plug equations (82), (84) into the perturbed two-body problem equation (77), obtaining

$$\underbrace{\frac{\partial^2}{\partial t^2}\mathbf{f}(\mathbf{u}(t),t) + \mu\frac{\mathbf{f}\left(\mathbf{u}(t),t\right)}{\left\|\mathbf{f}\left(\mathbf{u}(t),t\right)\right\|^3}}_{\mathbf{0}} + \sum_{\beta=1}^{6}\frac{\partial^2\mathbf{f}}{\partial t\partial u_\beta}\frac{\mathrm{d}u_\beta}{\mathrm{d}t} = \mathbf{a}_\mathrm{d},$$

where the first two terms evaluate to zero because the coefficients $\mathbf{u}(t)$ at each instant of time describe unperturbed motion, for which equation (81) is valid. Considering that $\mathbf{r}_\mathrm{p} = \mathbf{f}(\mathbf{u},t), \mathbf{v}_\mathrm{p} = \partial\mathbf{f}(\mathbf{u},t)/\partial t$, we are left with six (scalar) equations for the coefficients

$$\sum_{\beta=1}^{6}\frac{\partial\mathbf{r}_\mathrm{p}}{\partial u_\beta}\frac{\mathrm{d}u_\beta}{\mathrm{d}t} = \mathbf{0} \tag{85}$$

$$\sum_{\beta=1}^{6}\frac{\partial\mathbf{v}_\mathrm{p}}{\partial u_\beta}\frac{\mathrm{d}u_\beta}{\mathrm{d}t} = \mathbf{a}_\mathrm{d}, \tag{86}$$

Defining the state vector $\mathbf{x} := (\mathbf{r}(t), \mathbf{v}(t))$, we can write these equations in matrix-vector form,

$$\mathbf{L}(\mathbf{x}(\mathbf{u},t))\cdot\dot{\mathbf{u}} = \mathbf{P}, \qquad \mathbf{P} := \begin{bmatrix}\mathbf{0}\\ \mathbf{a}_\mathrm{d}\end{bmatrix} \tag{87}$$

where the element (i,j) of the matrix $\mathbf{L}$ is $L_{i,j} = \partial x_i/\partial u_j$. By manipulating equations (85), (86), we can show that

$$L_{i,j} = \sum_{k=1}^{3}\frac{\partial r_k}{\partial u_i}\frac{\partial v_k}{\partial u_j} - \frac{\partial r_k}{\partial u_j}\frac{\partial v_k}{\partial u_i},$$

and that the *Lagrange matrix* $\mathbf{L}$ is skew-symmetric, $\mathbf{L}^\intercal = -\mathbf{L}$. The elements of the matrix $\mathbf{L}$ are called *Lagrangian brackets*.

Lagrange planetary equations

From now on, we assume that the vector of coefficients represents the set of orbital elements $\mathbf{u} = (a, e, i, \Omega, \omega, \tau)$, where the time of periapsis passage τ is simply defined as $\tau := t - M/n$.

In the derivation of the Lagrange planetary equations, we consider the perturbing acceleration to be a conservative force that can be written as the gradient of a *perturbing potential* R,

$$\mathbf{a}_\mathrm{d} = \frac{\partial R^\intercal}{\partial \mathbf{r}} = \left[\frac{\partial R}{\partial r_x}, \frac{\partial R}{\partial r_y}, \frac{\partial R}{\partial r_z}\right]^\intercal. \tag{88}$$

Note that we regard position, velocity, and acceleration vectors as column vectors and the gradient as a row vector, hence the need for transposing the right-hand side of the equation above. By applying the chain rule, we express the gradient of the perturbing potential with respect to the orbital element vector $\mathbf{u} = (a, e, i, \Omega, \omega, \tau)$ as

$$\frac{\partial R}{\partial \mathbf{u}} = \left[\frac{\partial R}{\partial a}, \frac{\partial R}{\partial e}, \frac{\partial R}{\partial i}, \frac{\partial R}{\partial \Omega}, \frac{\partial R}{\partial \omega}, \frac{\partial R}{\partial \tau}\right] = \frac{\partial R}{\partial \mathbf{r}} \frac{\partial \mathbf{r}}{\partial \mathbf{u}},$$

where $\partial \mathbf{r}/\partial \mathbf{u}$ is a 6×6 Jacobian matrix.

We can derive expressions for the rates of change of the orbital elements $\dot{\mathbf{u}}$ by pre-multiplying equation (87) by $\mathbf{L}^{-1}$,

$$\dot{\mathbf{u}} = \mathbf{L}^{-1} \cdot \mathbf{P}.$$

Computing the inverse of the Lagrange matrix requires applying a few properties of the Lagrangian brackets, and several of its elements turn out to be zero. Details on this process can be found in Section 10.2 of Battin's book. Finally, we arrive at the rates of change of the osculating orbital elements due to a perturbing potential R,

$$\frac{\mathrm{d}a}{\mathrm{d}t} = -\frac{2a^2}{\mu}\frac{\partial R}{\partial \tau} \tag{89}$$

$$\frac{\mathrm{d}e}{\mathrm{d}t} = -\frac{\sqrt{1-e^2}}{\sqrt{\mu a}e}\frac{\partial R}{\partial \omega} - \frac{a\,(1-e^2)}{\mu e}\frac{\partial R}{\partial \tau} \tag{90}$$

$$\frac{\mathrm{d}i}{\mathrm{d}t} = \frac{1}{\sqrt{\mu a(1-e^2)}}\left(\frac{1}{\tan i}\frac{\partial R}{\partial \omega} - \frac{1}{\sin i}\frac{\partial R}{\partial \Omega}\right) \tag{91}$$

$$\frac{\mathrm{d}\Omega}{\mathrm{d}t} = \frac{1}{\sqrt{\mu a(1-e^2)}\sin i}\frac{\partial R}{\partial i} \tag{92}$$

$$\frac{\mathrm{d}\omega}{\mathrm{d}t} = -\frac{1}{\sqrt{\mu a(1-e^2)}\tan i}\frac{\partial R}{\partial i} + \frac{\sqrt{1-e^2}}{\sqrt{\mu a}e}\frac{\partial R}{\partial e} \tag{93}$$

$$\frac{\mathrm{d}\tau}{\mathrm{d}t} = \frac{2a^2}{\mu}\frac{\partial R}{\partial a} + \frac{a(1-e^2)}{\mu e}\frac{\partial R}{\partial e}. \tag{94}$$

These are the *Lagrange planetary equations*, which describe the evolution of the osculating orbital elements in perturbed motion. They are valid for all types of orbits; however, they are singular for both $i = 0°$ and $e = 0$. These are geometrical singularities that are due to the definition of the classical orbital elements: if $i = 0°$ we cannot define the node line, and if $e = 0$ we cannot define the major axis (and therefore the location of periapsis). For

this reason, the Lagrange planetary equations (along with the Gauss variational equations, which we will see in the following section), are seldom used for the numerical integration of equatorial or circular orbits. It is possible to eliminate these singularities by using alternative sets of orbital elements (therefore a different choice of $\mathbf{u}$), such as the *equinoctial orbital elements*, which are closely related to the classical orbital elements.

Gauss variational equations

Any orbital element u is a function of the state $\mathbf{x}(t)$, analogously to equation (82),

$$u = u(\mathbf{x}(t)) = u(\mathbf{r}(t), \mathbf{v}(t)).$$

It follows that the rate of change of the orbital element u is

$$\frac{\mathrm{d}u}{\mathrm{d}t} = \frac{\partial u}{\partial \mathbf{r}} \cdot \frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t} + \frac{\partial u}{\partial \mathbf{v}} \cdot \frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t}, \tag{95}$$

Plugging the expression of $\mathrm{d}\mathbf{v}/\mathrm{d}t$ from equation (77) in equation (95) yields:

$$\frac{\mathrm{d}u}{\mathrm{d}t} = \frac{\partial u}{\partial \mathbf{r}} \cdot \frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t} + \frac{\partial u}{\partial \mathbf{v}} \cdot \left(-\mu \frac{\mathbf{r}}{r^3}\right) + \frac{\partial u}{\partial \mathbf{v}} \cdot \mathbf{p}.$$

We can split the rate of change of the orbital element u into two terms: one that is due to two-body motion alone, and another that is due to perturbations:

$$\frac{\mathrm{d}u}{\mathrm{d}t} = \left.\frac{\mathrm{d}u}{\mathrm{d}t}\right|_{\text{Two-body}} + \left.\frac{\mathrm{d}u}{\mathrm{d}t}\right|_{\text{Perturbed}}.$$

The two-body rate of change is zero for all elements, except for the anomalies θ, E, M tracking the location of the spacecraft along its orbit. The anomalies are usually called the *fast* variables because their rate of change is proportional to the mean motion, whereas for weak perturbations the other elements change more slowly. The rates of change of the anomalies due to two-body motion only are

$$\left.\frac{\mathrm{d}\theta}{\mathrm{d}t}\right|_{\text{Two-body}} = \frac{h}{r^2}$$
$$\left.\frac{\mathrm{d}E}{\mathrm{d}t}\right|_{\text{Two-body}} = \frac{na}{r}$$
$$\left.\frac{\mathrm{d}M}{\mathrm{d}t}\right|_{\text{Two-body}} = n.$$

The rates of change of the orbital elements due to a perturbing acceleration $\mathbf{a}_\mathrm{d}$ can be obtained by differentiating the expressions for the orbital elements, with the aid of two-body formulas and vector calculus. It is particularly convenient to express the perturbing acceleration along the RSW frame, which is defined as follows:

$\mathbf{i}_r$	vector in the direction of the radius vector, $\mathbf{i}_r = \mathbf{r}/r$.
$\mathbf{i}_s$	completes the right-handed frame, $\mathbf{i}_s = \mathbf{i}_w \times \mathbf{i}_r$
$\mathbf{i}_w$	vector in the direction of the angular momentum vector, $\mathbf{i}_w = \mathbf{h}/h$.

The RSW rotates with the spacecraft position along the orbit. Note that $\mathbf{i}_s$ points towards the direction of motion but is not collinear with the velocity vector, except for circular orbits. The RSW frame can be obtained through a counter-clockwise rotation of the perifocal frame along the third axis $\mathbf{i}_w$, which is in common between the two frames. Therefore, the rotation matrix from the ECI to the RSW frame is obtained as

$$\begin{aligned}\mathbf{A}' &= [\mathbf{R}_3(\omega+\theta)\mathbf{R}_1(i)\mathbf{R}_3(\Omega)]^\intercal \\ &= \begin{bmatrix} \cos\Omega\cos u - \sin\Omega\sin u\cos i & -\cos\Omega\sin u - \sin\Omega\cos u\cos i & \sin\Omega\sin i \\ \sin\Omega\cos u + \cos\Omega\sin u\cos i & -\sin\Omega\sin u + \cos\Omega\cos u\cos i & -\cos\Omega\sin i \\ \sin u\sin i & \cos u\sin i & \cos i \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{i}_x\cdot\mathbf{i}_r & \mathbf{i}_x\cdot\mathbf{i}_s & \mathbf{i}_x\cdot\mathbf{i}_w \\ \mathbf{i}_y\cdot\mathbf{i}_r & \mathbf{i}_y\cdot\mathbf{i}_s & \mathbf{i}_y\cdot\mathbf{i}_w \\ \mathbf{i}_z\cdot\mathbf{i}_r & \mathbf{i}_z\cdot\mathbf{i}_s & \mathbf{i}_z\cdot\mathbf{i}_w \end{bmatrix}.\end{aligned}$$

which is equivalent to substituting the argument of latitude $u = \omega + \theta$ instead of the argument of periapsis ω in the rotation matrix from the perifocal to the inertial frame $\mathbf{A}$ (cfr. lecture 5). The acceleration in the RSW frame, with components $\mathbf{a}_{\mathrm{d_{RSW}}} = (a_{\mathrm{d}_r}, a_{\mathrm{d}_s}, a_{\mathrm{d}_w})$ is obtained from that in the ECI frame through

$$\mathbf{a}_{\mathrm{d_{RSW}}} = \mathbf{A}'^{\intercal}\mathbf{a}_{\mathrm{d_{ECI}}}.$$

The *Gauss variational equations*, which give the rates of change of the orbital elements due to the perturbing acceleration expressed in the RSW frame, are as follows:

$$\frac{\mathrm{d}h}{\mathrm{d}t} = r\ a_{\mathrm{d}_s} \tag{96}$$

$$\frac{\mathrm{d}e}{\mathrm{d}t} = \frac{h}{\mu}\sin\theta\ a_{\mathrm{d}_r} + \frac{1}{\mu h}\left[\left(h^2+\mu r\right)\cos\theta + \mu e r\right] a_{\mathrm{d}_s} \tag{97}$$

$$\frac{\mathrm{d}i}{\mathrm{d}t} = \frac{r}{h}\cos u\ a_{\mathrm{d}_w} \tag{98}$$

$$\frac{\mathrm{d}\Omega}{\mathrm{d}t} = \frac{r\sin u}{h\sin i}\ a_{\mathrm{d}_w} \tag{99}$$

$$\frac{\mathrm{d}\omega}{\mathrm{d}t} = -\frac{1}{eh}\left[\frac{h^2}{\mu}\cos\theta\ a_{\mathrm{d}_r} - \left(r+\frac{h^2}{\mu}\right)\sin\theta\ a_{\mathrm{d}_s}\right] - \frac{r\sin u}{h\tan i}\ a_{\mathrm{d}_w} \tag{100}$$

$$\frac{\mathrm{d}\theta}{\mathrm{d}t} = \frac{h}{r^2} + \frac{1}{eh}\left[\frac{h^2}{\mu}\cos\theta\ a_{\mathrm{d}_r} - \left(r+\frac{h^2}{\mu}\right)\sin\theta\ a_{\mathrm{d}_s}\right]. \tag{101}$$

In addition, the rate of change of the semi-major axis (which can be used in place of that for the angular momentum) is

$$\frac{\mathrm{d}a}{\mathrm{d}t} = \frac{2a^2}{h}\left(e\sin\theta\ a_{\mathrm{d}_r} + \frac{h^2}{\mu r}\ a_{\mathrm{d}_s}\right). \tag{102}$$

The set of equations (96)-(102) is equivalent to the Lagrange planetary equations (89)-(94) in the case of conservative perturbations. The Gauss variational equations are more general because they are valid for any perturbing acceleration and not just stemming from a perturbing potential. It is possible to also consider the evolution of a spacecraft under the simultaneous influence of both a perturbing acceleration and a perturbing potential, in which case the total rate of change of the elements is simply obtained as the sum of the right-hand sides of the Lagrange and Gauss equations. The Gauss variational equation also suffer from the same singularities for circular and equatorial orbits as the Lagrange planetary equations, and the same considerations regarding non-singular sets of orbital elements apply.

Within the Gauss variational equations, it is possible to clearly see the impact of in-plane perturbing accelerations, given by the components $a_{\mathrm{d}_r}, a_{\mathrm{d}_s}$, and the out-of-plane perturbing acceleration component a_{d_w}. In particular, h, a, e, θ only change because of in-plane accelerations, whereas i, Ω only change because of out-of-plane accelerations. The argument of periapsis ω changes due to both in-plane and out-of-plane accelerations. In-plane accelerations cause an in-plane rotation of the apse line, which is captured by the first term of the right-hand side of equation (100) (note that this is the opposite of the second term in equation (101)). Out-of-plane accelerations cause a rotation of the node line in the equatorial plane, which is captured by the last term of the right-hand side of equation (100). The evolution of ω reflects both of these phenomena.

Main perturbations acting on Earth satellites

Earth-orbiting satellites are subject to many perturbing accelerations that complicate the accurate analysis of their motion.

First of all, the Earth is not a perfect sphere with uniform density; instead, the radius of the Earth is about $20\,\mathrm{km}$ larger at the equator than at the poles due to the centrifugal force imparted by the spin. A good approximation for the Earth's shape is an oblate ellipsoid, which generates a gravitational field that is not spherically symmetric. In the equations of motion, this is considered by adding a small perturbing acceleration that is proportional to a coefficient that is proportional to the oblateness, and is commonly denoted as the J_2 coefficient. The oblateness perturbation is one of the principal perturbing accelerations acting on Earth satellites from LEO to GEO, although its strength decays relatively rapidly with the orbital radius. Departures of the gravitational field from the spherical symmetry due to finer variations in the Earth's mass distribution are captured through a perturbing potential. Perturbations due to small longitudinal asymmetries (i.e., along the parallels) are particularly important for geostationary orbit, as they affect the stability of satellites on the geostationary ring.

Satellites also experience aerodynamic accelerations that are due to the presence of the atmosphere. Although the atmosphere is very thin at orbital altitudes, orbital velocities in the order of a few $\mathrm{km\,s^{-1}}$ generate an often non-negligible dynamic pressure. Unlike gravitational accelerations, aerodynamic forces also depend on the physical characteristics of the satellite, such as its ballistic coefficient (which is in turn affected by the satellite's orientation with respect to its velocity vector). Whereas the aerodynamic lift of a satellite is negligible except for special or high-accuracy applications, the drag component has a relevant effect on the orbital evolution. The drag force is dissipative and acts to decrease the orbital energy (and therefore the semi-major axis) along with the eccentricity. Ultimately, a satellite will deorbit and collide with the Earth once the decrease in orbital radius brings it through the denser parts of the atmosphere. The atmospheric density decays exponentially with the altitude, and for altitudes larger than $1000\,\mathrm{km}$ to $1200\,\mathrm{km}$ this perturbation can be often neglected.

The presence of other celestial bodies in the Solar System also imparts additional gravitational accelerations that perturb the two-body motion. Close to the Earth, the most important accelerations originate from the Moon and the Sun, which are generally comparable in magnitude (with the lunar accelerations being a little larger than the solar). These accelerations act both on the spacecraft and the Earth; because the centre of mass of the Earth is also the origin of our reference system the perturbing accelerations are split into an *indirect* component, which is the acceleration on the origin of the reference frame, and a *direct* component that acts directly on the spacecraft. These perturbations are conservative; however, the corresponding perturbing potential is time-dependent due to the motion of the perturbing body. Increasing the altitude, the lunisolar perturbations grow in magnitude

whereas the point-mass Earth gravitational acceleration decreases. Trajectories at very high altitudes (for instance, past the geostationary altitude) eventually enter an N-body regime, in which the gravitational acceleration from the Moon or the Sun is comparable to the Earth's. After the Sun and the Moon, perturbing accelerations from Venus and Jupiter can be relevant for high-altitude orbits.

Depending on the orbital regime, satellites are subject to a variety of illumination conditions. Although photons are massless, they carry linear momentum that is transferred to a spacecraft when the photon hits its surface. Therefore, sunlit surfaces of a satellite experience a small pressure that is due to the photons impacting them; this is the physical cause of the *solar radiation pressure* (SRP). As with drag, the corresponding acceleration also depends on the physical characteristics of the spacecraft, and in particular on its area-to-mass ratio and reflective coefficient, which depends on the surface material. Because the solar radiation pressure depends on the distance from the Sun (which is much greater than typical orbital altitudes), the magnitude of this perturbation changes little depending on the orbital altitude. Solar radiation pressure can be particularly relevant past Medium Earth Orbits (MEOs), and in particular for geostationary satellites.

Manoeuvrable satellites may also fire their thrusters due to orbit transfers or station-keeping manoeuvres. Although the resulting accelerations are due to controls rather than the natural dynamics, they can be modelled in the same way as other perturbations. High-thrust chemical rockets are often modelled with impulsive Δv due to their short firing duration. In contrast, low-thrust electric thrusters impart accelerations of long duration and small magnitude.

With the exception of thruster firings, what all these perturbations have in common is that they are *weak* for most satellites from LEO to GEO. The very nature of the term "perturbations" implies that it is still useful to consider the effect of the forces as a small, perturbative effect superimposed on the main gravitational acceleration arising from spherical gravity. This is important, because it means that the trajectory will have small deviations from the unperturbed Keplerian trajectory over reasonably long time spans. Therefore, the orbital elements will still be a good description of the trajectory.

In the following, we will explain perturbations due to the non-spherical Earth gravitational field in more detail. For a quantitative treatment of the other perturbations, see Chapters 8 and 9 of Vallado and McClain's "Fundamentals of Astrodynamics".

Non-spherical Earth gravity

The unperturbed two-body problem,

$$\frac{\mathrm{d}^2\mathbf{r}}{\mathrm{d}t^2} + \mu\frac{\mathbf{r}}{r^3} = 0 \tag{1}$$

is derived under the assumption that the mass distribution of the Earth has spherical symmetry. However, the figure of the Earth is not a perfectly symmetrical sphere, and its mass is distributed unevenly. Therefore, we have to consider the perturbations that arise from the fact that the Earth does not have spherical symmetry.

It is particularly convenient to see gravitational accelerations as arising from a potential. The gravitational acceleration exerted by a point mass m can be written as

$$\ddot{\boldsymbol{\rho}} = \nabla U = \frac{\partial}{\partial \boldsymbol{\rho}} \left(\frac{Gm}{\rho} \right) = -\frac{Gm}{\rho^3} \boldsymbol{\rho}.$$

The quantity U is called the *gravitational potential*, while $\boldsymbol{\rho}$ is the radius vector going from the point mass to the spacecraft (which is assumed massless). Therefore, the total gravitational acceleration from an ensemble of point masses m_i is just

$$\ddot{\mathbf{r}} = \nabla \sum_i U_i = \nabla \sum_i \frac{Gm_i}{\rho_i}. \tag{103}$$

We can consider the Earth as being composed by a set of infinitesimal masses $\mathrm{d}m$, each giving a contribution $\mathrm{d}U = \mathrm{d}m/r$ to the total gravitational potential. Therefore, the total potential is given by the volume integral

$$U = \int_{\text{Earth}} \frac{\mathrm{d}m}{\rho}. \tag{104}$$

To be able to use this formula to compute the gravitational acceleration from an asymmetrical mass distribution, we need to relate the potential to the position vector of the spacecraft with respect to the Earth's centre of mass $\mathbf{r}$. Let us call $\mathbf{r}_i$ he position vector of the infinitesimal mass $\mathrm{d}m_i$ with respect to the Earth's centre of mass. Then, we have that

$$\boldsymbol{\rho} = \mathbf{r} - \mathbf{r}_i$$

and we can use the law of cosines to express the magnitude of ρ as

$$\rho = \sqrt{r^2 + r_i^2 - 2rr_i \cos \Lambda}$$

where $\cos \Lambda = (\mathbf{r} \cdot \mathbf{r}_i)/(rr_i)$. Equation (104) becomes

$$U = \int_{\text{Earth}} \frac{\mathrm{d}m}{\sqrt{r^2 + r_i^2 - 2rr_i \cos \Lambda}}.$$

Solving this integral for an arbitrary mass distribution is not very easy. To build a solution, we expand the denominator in the equation above using the binomial theorem. The result is

$$\frac{1}{\sqrt{r^2 + r_i^2 - 2rr_i \cos \Lambda}} = \frac{1}{r\sqrt{1 + \alpha^2 - 2\alpha \cos \Lambda}} = \sum_{l=0}^{\infty} \alpha^l P_l(\cos \Lambda),$$

where $\alpha = r/r_i$ and $\alpha < 1$ because the spacecraft is outside of the Earth's surface. The function P_l is the *Legendre polynomial* of degree l. Through spherical trigonometry, we can express the angle Λ as a function of the latitude ϕ and the longitude λ of the spacecraft in the Earth-centred, Earth-fixed frame (ECEF) and is therefore measured from the Greenwich meridian. After several algebraic manipulations, we can express the potential above as an expansion in *spherical harmonics*,

$$U = \frac{\mu}{r} + \sum_{l=2}^{\infty} \sum_{m=0}^{l} U_{l,m} = \frac{\mu}{r} \left\{ 1 + \sum_{l=2}^{\infty} \sum_{m=0}^{l} \left(\frac{R_\oplus}{r} \right)^l P_{l,m}(\sin \phi) \left[C_{l,m} \cos(m\lambda) + S_{l,m} \sin(m\lambda) \right] \right\}. \tag{105}$$

Let's unpack the meaning of this equation. The quantity $\mu = GM_\oplus$ is the Earth's gravitational parameter, and $R_\oplus$ is the Earth's equatorial radius. The spacecraft position is fully specified by the spacecraft radius measured from the Earth's centre of mass r, the latitude

$$\phi = \arcsin \frac{Z}{r},$$

where Z is the Z-coordinate of the spacecraft, and the longitude

$$\lambda = \text{arctan2}(Y, X),$$

where X, Y are the coordinates on the equatorial plane in the ECEF frame. The functions $P_{l,m}$ are called *associated Legendre functions*, defined as

$$P_{l,m}(x) = \frac{1}{2^l l!} \left(1 - x^2\right)^{m/2} \frac{\mathrm{d}^{l+m}}{\mathrm{d}x^{l+m}} \left(x^2 - 1\right)^l.$$

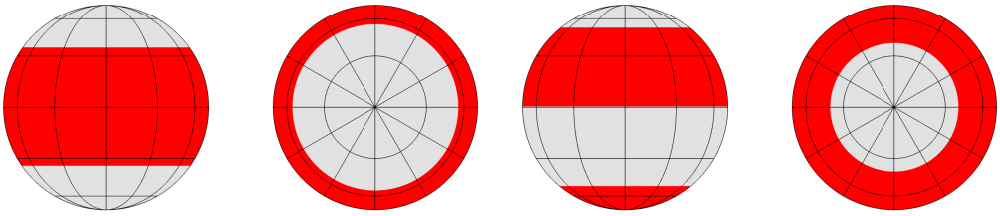
For $m = 0$, these functions are the Legendre polynomials of degree l.

Equation (105) represents the potential as a sum of harmonics $U_{l,m}$ of *degree* l and *order* m; each harmonic represents the contribution to the potential from a particular type of mass distribution. The harmonic coefficients $C_{l,m}, S_{l,m}$ represent the relative importance of these mass distributions. The zeroth harmonic is $U_{0,0} = \mu/r$, and it represents the two-body potential arising from a spherically symmetric mass distribution. The first harmonic is proportional to the distance between the origin of the reference frame and the center of mass of the Earth. We define our reference frame to coincide with the center of mass of the Earth, therefore $U_{1,0} = 0$.

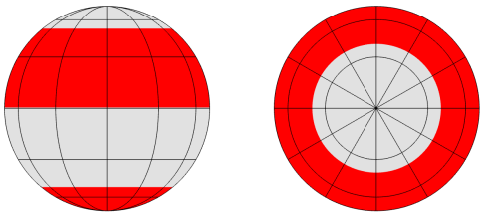
Harmonics of zeroth order, $m = 0$, do not depend on longitude and are called *zonal* harmonics. As a polynomial of order l has l roots, the zonal harmonics have zeros along l parallels and divide the sphere in $(l + 1)$ bands of latitude in which the potential changes

First few Legendre associated functions.

l	m	$P_{l,m}(x)$	$P_{l,m}(\sin\phi)$
1	0	x	$\sin\phi$
1	1	$\sqrt{1-x^2}$	$\cos\phi$
2	0	$\frac{1}{2}\left(3x^2-1\right)$	$\frac{1}{2}\left(3\sin\phi^2-1\right)$
2	1	$3x\sqrt{1-x^2}$	$3\sin\phi\cos\phi$
2	2	$-3(x^2-1)$	$3\cos^2\phi$
3	0	$\frac{1}{2}\left(5x^3-3x\right)$	$\frac{1}{2}\left(5\sin\phi^3-3\sin\phi\right)$
3	1	$\frac{3}{2}\sqrt{1-x^2}\left(5x^2-1\right)$	$\frac{3}{2}\cos\phi\left(5\sin^2\phi-1\right)$
3	2	$-15x\left(x^2-1\right)$	$15\sin\phi\cos^2\phi$
3	3	$15\left(1-x^2\right)^{3/2}$	$15\cos^3\phi$

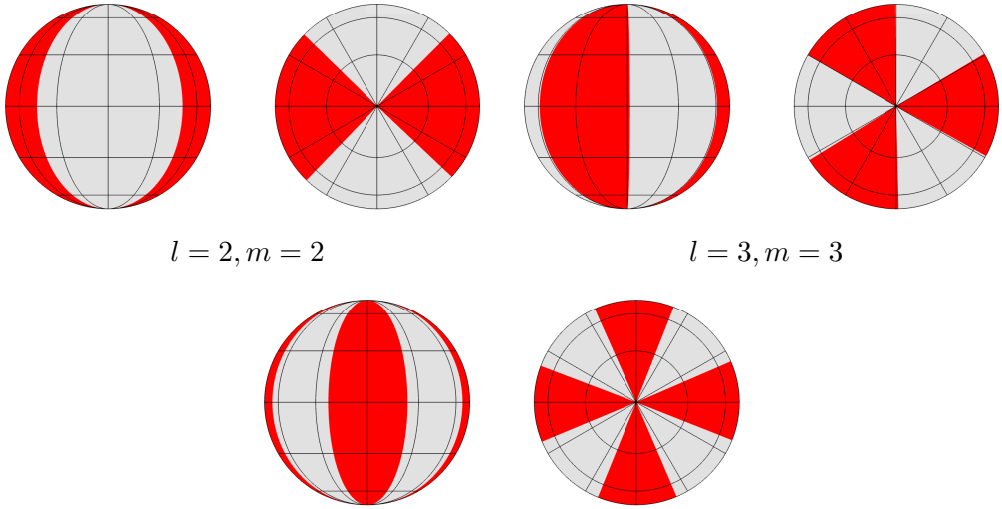


$l=2, m=0$ $l=3, m=0$



$l=4, m=0$

Top and side view of the perturbing potential due to zonal harmonics on the Earth's sphere. Red areas represent where $P_{l,0}<0$.



Top and side view of the perturbing potential due to sectorial harmonics on the Earth's sphere. Red areas represent where $P_{l,l} > 0$.

sign. The zonal harmonic coefficients are commonly defined in the literature with the notation

$$J_l = -C_{l,0}.$$

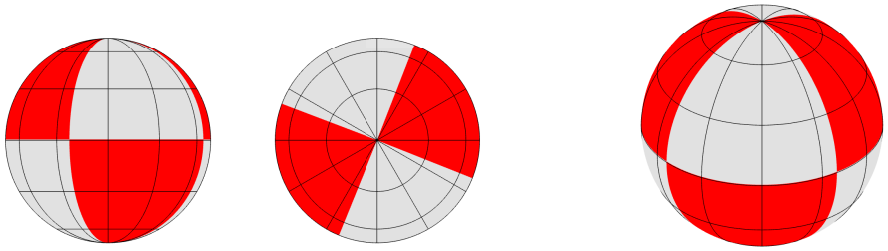
In the case of the Earth, by far the largest contribution to the non-spherical part of the gravitational potential is that of the second zonal harmonic. Because of the centrifugal force imparted by the Earth's rotation, the shape of the Earth can be described by a flattened ellipsoid. The equatorial bulge of the Earth implies that there's excess mass at the equator; this effect translates into a J_2 coefficient that is 1000 times larger than the next zonal harmonics.

When the order is equal to the degree, $m = l$, the Legendre associated functions $P_{l,l}(\sin\phi)$ only have zeros at the poles, $\phi = \pm 90°$. Therefore, the only changes in sign are due to the trigonometric functions $\cos(m\lambda), \sin(m\lambda)$. These harmonics are called *sectorial*, and they represent mass variations in bands of longitude. The remaining harmonics for $l \neq m, m > 0$ are called *tesserals*, and they represent mass variations in "tiles".

The magnitudes of the harmonic coefficients in the Earth's geopotential decrease with increasing order, following an approximate inverse square law that is known as *Kaula's rule*.

The perturbing acceleration due to the perturbing part of the geopotential is obtained by taking the gradient,

$$\ddot{\mathbf{r}} = \nabla U(r, \phi, \lambda) = -\frac{\mu}{r^3}\mathbf{r} + \mathbf{R}_3^{\intercal}[\theta_{\text{GMST}}]\nabla R(r, \phi, \lambda).$$



Perturbing potential due to the $l = 3, m = 2$ tesseral harmonic on the Earth's sphere. Red areas represent where $P_{l,m}(\sin\phi) \cdot [\cos(m\lambda) + \sin(m\lambda)] > 0$.

In the above equation, we have split the total gravitational gradient in the term of zeroth degree $U_0 = \mu/r$, which gives rise to the main gravitational acceleration, plus the perturbing term

$$R = \frac{\mu}{r} \sum_{l=2}^{\infty} \sum_{m=0}^{l} \left(\frac{R_\oplus}{r} \right)^l P_{l,m}(\sin\phi) \left[C_{l,m} \cos(m\lambda) + S_{l,m} \sin(m\lambda) \right].$$

The perturbing term is premultiplied by the rotation matrix $\mathbf{R}_3^\intercal[\theta_{\text{GMST}}]$ to rotate the acceleration components from the ECEF to the ECI frame. Note that perturbations to Keplerian motion (and thus variations in the orbital elements through the Lagrange planetary equations) arise only if the perturbing potential is non-zero.

Because the spherical harmonics expansion is written in spherical coordinates, to obtain the acceleration in Cartesian coordinates we need to use the chain rule,

$$\ddot{\mathbf{r}} = \nabla U(r, \phi, \lambda) = -\frac{\mu}{r^3}\mathbf{r} + \mathbf{a}_{\text{d}} = -\frac{\mu}{r^3}\mathbf{r} + \mathbf{R}_3^\intercal[\theta_{\text{GMST}}] \left(\frac{\partial R}{\partial r} \cdot \frac{\partial r}{\partial \mathbf{r}} + \frac{\partial R}{\partial \phi} \cdot \frac{\partial \phi}{\partial \mathbf{r}} + \frac{\partial R}{\partial \lambda} \cdot \frac{\partial \lambda}{\partial \mathbf{r}} \right).$$

Explicit expressions for the acceleration components in the ECEF frame are available in several references (for instance, in Vallado and McClain's "Fundamentals of Astrodynamics" book). The resulting expressions can get quite involved, so care must be taken in their implementation in code. The evaluation of the perturbing potential at high orders and degrees can get computationally expensive; therefore, the required trigonometric and Legendre associated functions are usually computed through recursive relations.

In practical computations, the geopotential expansion in equation (105) is truncated at a maximum degree and order that depend on the required accuracy and on the orbit that is being modelled. Because the harmonics are proportional to $1/r^{l+1}$, we truncate the geopotential at orders that are larger the smaller the semi-major axis of the orbit. For LEOs, the potential must be truncated at a high order and degree for high-fidelity simulations. As the magnitude of the Legendre associated functions grows quickly with the order, numerical issues affecting the accuracy of the geopotential calculations can arise. This can be solved by normalizing both the Legendre associated functions and the potential coefficients

according to the following formula:

$$\Pi_{l,m} = \sqrt{\frac{(l+m)!}{(l-m)!k(2l+1)}}$$

$$\overline{C}_{l,m} = \Pi_{l,m} C_{l,m} \qquad \overline{S}_{l,m} = \Pi_{l,m} S_{l,m} \qquad \overline{P}_{l,m} = \Pi_{l,m} P_{l,m}$$

,

where $\overline{C}_{l,m}, \overline{S}_{l,m}$ are the normalized spherical harmonics coefficients and $\overline{P}_{l,m}$ are normalized Legendre associated functions. Also, it is better to sum the terms in (105) from smaller to larger harmonics to reduce the accumulation of round-off error.

Earth oblateness effects

The J_2 coefficient in the gravitational potential expansion is larger than all the others by at least three orders of magnitude, and dominates geopotential perturbations. The corresponding perturbing potential term is

$$R_{2,0} = -\frac{\mu J_2}{r}\left(\frac{R_\oplus}{r}\right)^2 \frac{3}{2}\left(\sin^2\phi - \frac{1}{3}\right). \tag{106}$$

Taking the gradient results in the following acceleration components in the ECI frame:[4]

$$\begin{bmatrix} a_{\mathrm{d}_x} \\ a_{\mathrm{d}_y} \\ a_{\mathrm{d}_z} \end{bmatrix} = \frac{3}{2}\frac{J_2 \mu R_\oplus^2}{r^4}\begin{bmatrix} \frac{x}{r}\left(5\frac{z^2}{r^2} - 1\right) \\ \frac{y}{r}\left(5\frac{z^2}{r^2} - 1\right) \\ \frac{z}{r}\left(5\frac{z^2}{r^2} - 3\right) \end{bmatrix}. \tag{107}$$

Rotating the acceleration to the RSW frame results in the components

$$\begin{bmatrix} a_{\mathrm{d}_r} \\ a_{\mathrm{d}_s} \\ a_{\mathrm{d}_w} \end{bmatrix} = -\frac{3}{2}\frac{J_2 \mu R_\oplus^2}{r^4}\begin{bmatrix} 1 - 3\sin^2 i \sin^2(\omega+\theta) \\ \sin^2 i \sin 2(\omega+\theta) \\ \sin 2i \sin(\omega+\theta) \end{bmatrix}. \tag{108}$$

To understand the effect of the perturbing acceleration on the orbital elements, we plug Equation (108) into the Gauss variational equations (96)-(101). The resulting set of differential equations is not solvable analytically, but we can integrate it numerically. Alternatively, we can also numerically integrate the trajectory by using the acceleration in the ECI frame, Equation (107), together with the Cowell or Encke methods. In the latter case, we can transform the Cartesian coordinates to osculating elements to better visualize the qualitative evolution of the orbit.

In general, the evolution of the osculating elements for almost all perturbations acting on Earth satellites can be seen as the superposition of

[4] Perturbing accelerations from zonal harmonics are symmetric about the z-axis, and they don't change after the rotation from ECEF to ECI.

- a *secular* trend, that is a monotonous variation along time,
- *long-periodic* oscillations with period larger than the orbital period,
- a *short-periodic* oscillations with period equal to or smaller than the orbital period.

We can isolate the secular trends in the orbital elements due to the J_2 perturbation through the *method of averaging*. The idea is to average the right-hand side of either the Gauss or Lagrange planetary equations over one orbital period to "smooth out" the short-periodic perturbations. The averaged rate of change for the orbital element u is

$$\frac{\mathrm{d}\overline{u}}{\mathrm{d}t} = \frac{1}{2\pi}\int_0^{2\pi} \frac{\mathrm{d}u}{\mathrm{d}t}\mathrm{d}\theta. \tag{109}$$

Note that the averaged (or *mean*) element $\overline{u}$ will follow a different trajectory from that of the osculating element u, without short-periodic oscillations. We can express $\mathrm{d}u/\mathrm{d}t$ due to the J_2 perturbation by plugging equation (108) into the Gauss planetary equations, and then performing the averaging operation in Equation (109). The average rates of change for the mean orbital elements are then

$$\begin{aligned}
\overline{\dot{h}} &= \overline{\dot{a}} = \overline{\dot{e}} = \overline{\dot{i}} = 0 \\
\overline{\dot{\Omega}} &= -\frac{3}{2}\frac{nJ_2\mu^2R_\oplus^2}{h^4}\cos i \\
\overline{\dot{\omega}} &= \frac{3}{2}\frac{J_2\mu^2R_\oplus^2}{h^4}\frac{n}{2}\left(5\cos^2 i - 1\right),
\end{aligned}$$

where we highlight that the elements on the right-hand side are also *mean* elements, and we omit the overline for simplicity. This simplified set of equations tells us that the mean angular momentum, semi-major axis, eccentricity, and inclination are all constant under the J_2 perturbation. This is a specific instance of a more general result, as it can be shown that geopotential perturbations do not change the mean semi-major axis. Both the right ascension of ascending node and the argument of perigee precess because of J_2. Physically, this can be explained by observing that the equatorial bulge of the Earth results in an "extra force" when getting close to the equator. This extra force results in a non-zero average torque acting on the orbit, which results in a gyroscopic precession of the node and apse lines. This perturbation can be exploited to achieve two particular types of orbits:

- *Sun-synchronous orbits (SSOs):* these are orbits for which the mean rate of change of the right ascension of ascending node matches the apparent variation of the right ascension of the Sun, $\overline{\dot{\Omega}} = \dot{\Omega}_{\text{Sun}}$. In this way, the orbit sees constant lighting conditions during the year, which is convenient for Earth observation payloads and onboard power systems.
- *Frozen orbits:* these orbits correspond to the critical inclination for direct motion $i = 63.4°$, for which $\overline{\dot{\omega}} = 0$. Keeping the mean apse line fixed in inertial space

is useful for high-latitude telecommunications satellites that are placed on highly elliptical orbits. The apse line is oriented in such a way that the satellite spends most of its time over high latitudes. It can be shown that the condition $\overline{\dot{\omega}} = 0$ also minimizes the short-periodic variations in orbital radius, which can be also exploited for certain payloads such as laser altimeters.

The Gravity Gradient

In the previous lectures, we used Encke's method to integrate exactly the ODE

$$\frac{d^2\mathbf{r}}{dt^2} + \mu\frac{\mathbf{r}}{r^3} = \mathbf{a}_d.$$

This was done by partitioning the position and velocity vectors as

$$\mathbf{r} = \mathbf{r}_o + \boldsymbol{\delta}$$
$$\mathbf{v} = \mathbf{v}_o + \boldsymbol{\nu}$$

and integrating the disturbance terms separately from the nominal components $(\mathbf{r}_o, \mathbf{v}_o)$ that would be found in Keplerian motion.

All of the difficulties that arise in integrating this equation result from the fact that the gravity term

$$g(\mathbf{r}) = -\frac{\mu\mathbf{r}}{r^3}$$

is nonlinear. In this lecture we develop an *approximate* method to calculate perturbations to the position $\mathbf{r}$ and velocity $\mathbf{v}$ when a disturbing acceleration is present. Our approach will be based on linearization of the gravity term.

The Gravity Gradient

We can approximate the gravitational acceleration at a point $\mathbf{r} = \mathbf{r}_o + \boldsymbol{\delta}$ as

$$g(\mathbf{r}) \approx g(\mathbf{r}_o) + G(\mathbf{r}_o)\boldsymbol{\delta} + \mathcal{O}(\delta^2),$$

where the term

$$G(\mathbf{r}) = \nabla g(\mathbf{r}) = \frac{\partial g(\mathbf{r})}{\partial \mathbf{r}}$$

is the *gravity gradient*, and the term $\mathcal{O}(\delta^2)$ represents all of the higher order terms. If $\boldsymbol{\delta}$ is small, then the dynamics of our system can be written as

$$\frac{d^2\mathbf{r}_o}{dt^2} + \frac{d^2\boldsymbol{\delta}}{dt^2} = g(\mathbf{r}_o) + G(\mathbf{r}_o)\boldsymbol{\delta} + \mathbf{a}_d.$$

If we assume that the reference trajectory $\mathbf{r}_o$ is the solution to

$$\frac{d^2\mathbf{r}_o}{dt^2} = g(\mathbf{r}_0),$$

then these terms can be cancelled, leaving

$$\frac{d^2\boldsymbol{\delta}}{dt^2} = G(\mathbf{r}_o)\boldsymbol{\delta} + \mathbf{a}_d.$$

The critical difference between this ODE from δ and the one used in Encke's method is that the above, although approximate, is *linear*, and therefore much easier to solve.

State-Space Form of the Perturbation Errors

An important application of the above linearization is for spacecraft control. If we rewrite the dynamics of the position error $\boldsymbol{\delta}$ in terms of the velocity error term $\boldsymbol{\nu}$, we get

$$\begin{aligned}\frac{d\boldsymbol{\delta}}{dt} &= \boldsymbol{\nu} \\ \frac{d\boldsymbol{\nu}}{dt} &= G(\mathbf{r}_o)\boldsymbol{\delta} + \mathbf{a}_d.\end{aligned}$$

The above can be rewritten as

$$\frac{d}{dt}\begin{bmatrix}\boldsymbol{\delta} \\ \boldsymbol{\nu}\end{bmatrix} = \begin{bmatrix}0 & I \\ G(\mathbf{r}_o) & 0\end{bmatrix}\begin{bmatrix}\boldsymbol{\delta} \\ \boldsymbol{\nu}\end{bmatrix} + \begin{bmatrix}0 \\ I\end{bmatrix}\mathbf{a}_d,$$

which is in standard state-space form. In the above the disturbing acceleration $\mathbf{a}_d$ is best viewed as a thrust that can be used to force the position error $\boldsymbol{\delta}$ to zero for either trajectory following (e.g. a moon landing problem) or stationkeeping (e.g. to keep a geostationary satellite at its desired location).

Properties of the Gravity Gradient

We now work out in detail some of the properties of the gravity gradient term $G(\mathbf{r}_0)$:

$$\begin{aligned}G(\mathbf{r}) &= \frac{\partial g(\mathbf{r})}{\partial \mathbf{r}} \\ &= \frac{\partial}{\partial \mathbf{r}}\left(-\mu\frac{\mathbf{r}}{r^3}\right) \\ &= \frac{1}{r^3}\mu\left(\frac{3\mathbf{r}\mathbf{r}^\top}{r^2} - I\right),\end{aligned}$$

where the term $\mathbf{r}\mathbf{r}^\top$ is the *outer product* of the vector $\mathbf{r}$ with itself, i.e. a 3×3 matrix formed by the multiplication of column by a row.

Note that if the nominal position vector $\mathbf{r}_o$ is written as

$$\mathbf{r}_o = \begin{bmatrix}r_x \\ r_y \\ r_z\end{bmatrix},$$

then the above is equivalent to

$$G(\mathbf{r}_o) = \frac{\mu}{r_o^5}\begin{bmatrix}3r_x^2 - r_o^2 & 3r_xr_y & 3r_xr_z \\ 3r_xr_y & 3r_y^2 - r_o^2 & 3r_yr_z \\ 3r_xr_z & 3r_yr_z & 3r_z^2 - r_o^2\end{bmatrix}.$$

The Clohessy-Wiltshire Equations

In this lecture we consider the *relative* motion of two objects (e.g. two satellites) orbiting a central body. We assume that the mass of both objects is negligible relative to the mass of the central body.

We designate one of the two satellites as the reference object, and denote its position as $\mathbf{r}_o$. We assume that the other object is in a nearby orbit with position $\mathbf{r} = \boldsymbol{\delta} + \mathbf{r}_o$, and assume that $\delta \ll r_o$.

Recall from the previous lecture that, using a linear approximation, the dynamics of the relative position vector $\boldsymbol{\delta}$ can be written as

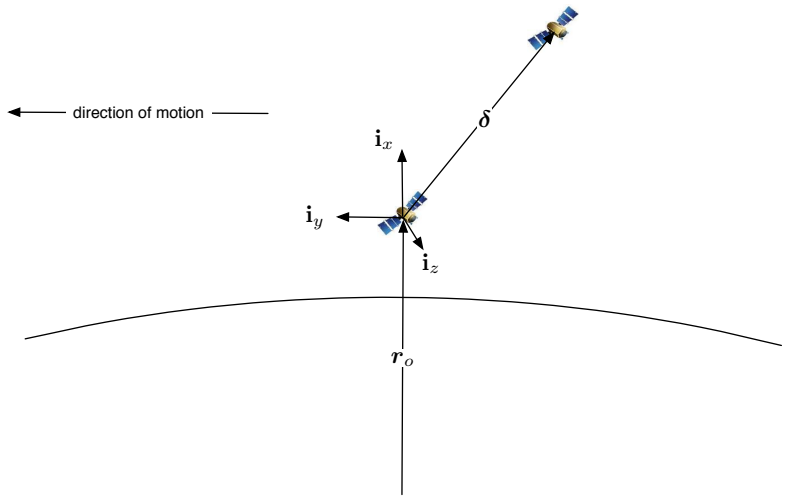
$$\frac{d^2\boldsymbol{\delta}}{dt^2} = G(\mathbf{r}_o)\boldsymbol{\delta}.$$

The matrix term $G(\mathbf{r}_o)$ is the gravity gradient evaluated at the reference position $\mathbf{r}_o$ and can be written as

$$G(\mathbf{r}_o) = \frac{\mu}{r_o^3}\left(\frac{3\mathbf{r}_o\mathbf{r}_o^\top}{r_o^2} - I\right).$$

Relative Position in a Rotation Reference Frame

In order to describe the relative position of the two bodies, we will find it convenient to work in frame of reference that is *fixed* with respect to the reference object, as shown in the figure below:



The vector $\mathbf{i}_x$ represents the local vertical direction. The vectors $(\mathbf{i}_y, \mathbf{i}_z)$ are locally horizontal, with $\mathbf{i}_y$ facing in the direction of orbital motion and $\mathbf{i}_z$ facing in the direction of the angular momentum vector of the reference orbit.

We assume that the reference orbit is circular, so that $\mathbf{r}_o$ is constant and the period of the reference orbit is

$$P = 2\pi\sqrt{\frac{r_o^3}{\mu}}.$$

The angular velocity ω of the rotating reference frame therefore satisfies

$$\omega = \sqrt{\frac{\mu}{r_o^3}}.$$

We note that the acceleration of the relative position term $\boldsymbol{\delta}$ due to the gravity gradient has thus far always been in an inertial frame. To emphasise this point, we write it henceforward as $\boldsymbol{\delta}_I$.

We denote the relative position in the rotating frame fixed to the reference orbit as $\boldsymbol{\delta}_B$. A standard result is

$$\underbrace{\ddot{\boldsymbol{\delta}}_I}_{\text{gravity}} = \underbrace{\ddot{\boldsymbol{\delta}}_B}_{\text{observed}} + \underbrace{2(\boldsymbol{\omega} \times \dot{\boldsymbol{\delta}}_B)}_{\text{Coriolis}} + \underbrace{(\dot{\boldsymbol{\omega}} \times \boldsymbol{\delta}_B)}_{\text{Euler}} + \underbrace{\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \boldsymbol{\delta}_B)}_{\text{centripetal}}.$$

Note that the Euler term above is zero in this case since $\dot{\omega} = 0$.

If we write

$$\boldsymbol{\omega} = \omega \mathbf{i}_z, \quad \mathbf{r}_o = r_o \mathbf{i}_x, \quad \boldsymbol{\delta}_B = x\mathbf{i}_x + y\mathbf{i}_y + z\mathbf{i}_z,$$

then the above can be simplified to

$$\underbrace{2\omega^2 x\mathbf{i}_x - \omega^2 y\mathbf{i}_y - \omega^2 z\mathbf{i}_z}_{\text{gravity}} = \underbrace{\ddot{x}\mathbf{i}_x + \ddot{y}\mathbf{i}_y + \ddot{z}\mathbf{i}_z}_{\text{observed}} + \underbrace{2\omega\dot{x}\mathbf{i}_y - 2\omega\dot{y}\mathbf{i}_x}_{\text{Coriolis}} - \underbrace{\omega^2 x\mathbf{i}_x - \omega^2 y\mathbf{i}_y}_{\text{centripetal}}.$$

The above vector differential equation can be separated into a set of three coupled linear constant-coefficient differential equations, known as the *Clohessy-Wiltshire equations*.

Clohessy-Wiltshire Equations
The position of a satellite relative to a circular reference orbit is modelled approximately by

$$\begin{aligned} \ddot{x} - 3\omega^2 x - 2\omega\dot{y} &= 0 \\ \ddot{y} + 2\omega\dot{x} &= 0 \\ \ddot{z} + \omega^2 z &= 0 \end{aligned}$$

An important point to notice is that equation in (x, y) are coupled to each other, but not to z. This indicates that local vertical and horizontal motions are coupled to each other and independent of motion normal to the reference plane.

Note that if we wish to add disturbing acceleration terms (e.g. due to an engine on board the satellite), then we simply add these to the right-hand side of the C-W equations, i.e.

$$\begin{aligned} \ddot{x} - 3\omega^2 x - 2\omega\dot{y} &= a_x \\ \ddot{y} + 2\omega\dot{x} &= a_y \\ \ddot{z} + \omega^2 z &= a_z. \end{aligned}$$

In this case a_x represents an acceleration in the local vertical direction, a_y in the local horizontal direction, and a_z out of the reference plane.

Solution to the Clohessy-Wiltshire Equation

The C-W equations are linear constant-coefficient differential equations, and can be solved exactly. If initial values are available for the relative position (x_0, y_0, z_0) and relative velocity $(\dot{x}_0, \dot{y}_0, \dot{z}_0)$, then the solution to the C-W equations is

$$\begin{bmatrix} x(t) \\ y(t) \\ \dot{x}(t) \\ \dot{y}(t) \end{bmatrix} = \begin{bmatrix} 4 - 3\cos\omega t & 0 & \frac{1}{\omega}\sin\omega t & \frac{2}{\omega}(1 - \cos\omega t) \\ 6(\sin\omega t - \omega t) & 1 & \frac{2}{\omega}(\cos\omega t - 1) & \frac{1}{\omega}(4\sin\omega t - 3\omega t) \\ 3\omega\sin\omega t & 0 & \cos\omega t & 2\sin\omega t \\ 6\omega(\cos\omega t - 1) & 0 & -2\sin\omega t & 4\cos\omega t - 3 \end{bmatrix} \begin{bmatrix} x_0 \\ y_0 \\ \dot{x}_0 \\ \dot{y}_0 \end{bmatrix}$$

and

$$\begin{bmatrix} z(t) \\ \dot{z}(t) \end{bmatrix} = \begin{bmatrix} \cos\omega t & \frac{1}{\omega}\sin\omega t \\ -\omega\sin\omega t & \cos\omega t \end{bmatrix} \begin{bmatrix} z_0 \\ \dot{z}_0 \end{bmatrix}$$

Gravity Gradient Torque

Recall that the gravity gradient at a point $\mathbf{r}_o$ can be written as[5]:

$$G(\mathbf{r}_o) = \frac{\mu}{r_0^3}\left[\frac{3}{r_o^2}\mathbf{r}_o\mathbf{r}_o^\top - E_3\right].$$

This allows us to approximate the acceleration due to gravity $g(\mathbf{r})$ at a point $\mathbf{r} = \mathbf{r}_o + \boldsymbol{\delta}$ as

$$g(\mathbf{r}) \approx g(\mathbf{r}_o) + G(\mathbf{r}_o)\boldsymbol{\delta}.$$

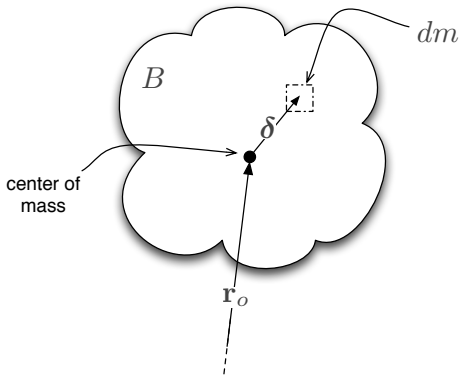
Until now, we have treated orbiting bodies as points, whose motion is determined by the location of their centers of mass. For small bodies (e.g. for satellites orbiting the earth), this gives very good results. However, no matter how small the satellite, the gravity gradient term above will still result in a net moment on the satellite in most cases. In this lecture we derive a method for calculating this moment.

Gravity Gradient Torque

We can write the total moment on a body due to gravity as

$$\mathbf{m} = \int_B \mathbf{r} \times g(\mathbf{r})\, dm,$$

where the integral is taken over the entire volume of the body B, dm is a differential element of the body's mass, and $\mathbf{r}$ is the location of the element dm.



We assume that the centre of mass of the body is at $\mathbf{r}_o$. If we describe the location of the mass element dm according to its relative position $\boldsymbol{\delta}$ to this center, i.e. by setting $\mathbf{r} = \mathbf{r}_o + \boldsymbol{\delta}$, then the total moment is approximately

$$\mathbf{m} = \int_B (\mathbf{r}_o + \boldsymbol{\delta}) \times \left(g(\mathbf{r}_o) + \frac{\mu}{r_o^3}\left[\frac{3}{r_o^2}\mathbf{r}_o\mathbf{r}_o^\top - E_3\right]\boldsymbol{\delta}\right) dm.$$

[5]We use the notation E_3 for the 3×3 identity matrix here, in order to avoid conflicts with the inertia tensor I to be introduced later on.

Recalling that $g(\mathbf{r}_o) = -(\mu/r_o^3)\mathbf{r}_o$, the above can be rewritten as:

$$\begin{aligned}\mathbf{m} &= \int_B (\mathbf{r}_o + \boldsymbol{\delta}) \times \left(\frac{\mu}{r_o^3} \left[-(\mathbf{r}_o + \boldsymbol{\delta}) + \frac{3}{r_o^2} \mathbf{r}_o \mathbf{r}_o^\top \boldsymbol{\delta} \right] \right) dm \\ &= \int_B (\mathbf{r}_o + \boldsymbol{\delta}) \times \left(\frac{\mu}{r_o^5} 3\mathbf{r}_o (\mathbf{r}_o^\top \boldsymbol{\delta}) \right) dm \\ &= \int_B \boldsymbol{\delta} \times \left(\frac{\mu}{r_o^5} 3\mathbf{r}_o (\mathbf{r}_o^\top \boldsymbol{\delta}) \right) dm.\end{aligned}$$

Note that the last step comes from considering the right hand term in the cross product as a vector proportional to $(\mathbf{r}_o^\top \boldsymbol{\delta}) \cdot \mathbf{r}_o$, where the term $\mathbf{r}_o^\top \boldsymbol{\delta}$ is a scalar.

We can also move this scalar to multiply the left hand term of the cross product instead, and thereby rewrite the expression for the moment due to gravity as

$$\begin{aligned}\mathbf{m} &= \frac{3\mu}{r_o^5} \int_B \boldsymbol{\delta} (\mathbf{r}_o^\top \boldsymbol{\delta}) \times \mathbf{r}_o \, dm. \\ &= \frac{3\mu}{r_o^5} \int_B \boldsymbol{\delta} (\boldsymbol{\delta}^\top \mathbf{r}_o) \times \mathbf{r}_o \, dm. \\ &= \frac{3\mu}{r_o^5} \int_B (\boldsymbol{\delta}\boldsymbol{\delta}^\top) \mathbf{r}_o \times \mathbf{r}_o \, dm.\end{aligned}$$

We next recall the definition of the moment of inertia tensor I, which can be written in vector form as

$$I = \int_B \left(\delta^2 E_3 - \boldsymbol{\delta}\boldsymbol{\delta}^\top \right) dm$$

Using this definition, we can rewrite the moment due to gravity gradient as

$$\begin{aligned}\mathbf{m} &= -\frac{3\mu}{r_o^5} \int_B (-\boldsymbol{\delta}\boldsymbol{\delta}^\top + \boldsymbol{\delta}^2 E_3) \mathbf{r}_o \times \mathbf{r}_o - \delta^2 E_3 \mathbf{r}_o \times \mathbf{r}_o \, dm. \\ &= -\frac{3\mu}{r_o^5} \int_B (-\boldsymbol{\delta}\boldsymbol{\delta}^\top + \boldsymbol{\delta}^2 E_3) \mathbf{r}_o \times \mathbf{r}_o \, dm. \\ &= -\frac{3\mu}{r_o^5} I\mathbf{r}_o \times \mathbf{r}_o = \frac{3\mu}{r_o^5} \mathbf{r}_o \times I\mathbf{r}_o\end{aligned}$$

Note that in the above, the position vector $\mathbf{r}_o$ can be treated as a vector centered on the satellite, pointing towards the body about which the satellite is orbiting (i.e. a change in sign for $\mathbf{r}_o$ does not change the result).

Gravity Gradient Torque *A satellite with inertia tensor I experiences a moment due to gravity gradient*

$$\mathbf{m} = \frac{3\mu}{r_o^5} \cdot \mathbf{r}_o \times I\mathbf{r}_o,$$

where $\mathbf{r}_o$ is the position of the central body with respect to the satellite.

Rotational Motion and Gravity Gradient Torque

Recall that Euler's equations of motion for or a rotating rigid body are

$$\boxed{\mathbf{m} = I\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times I\boldsymbol{\omega},}$$

where all vectors are in a body-fixed reference frame.

Assume that the only moment acting on the body is the gravity gradient torque, i.e.

$$\frac{3\mu}{r_o^5} \cdot \mathbf{r}_o \times I\mathbf{r}_o = I\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times I\boldsymbol{\omega}.$$

We can then try to determine whether there are any equilibrium conditions for the satellite. Assume that all vectors are in a body fixed reference frame aligned with the principal axes of the satellite, so that

$$I = \begin{bmatrix} A & 0 & 0 \\ 0 & B & 0 \\ 0 & 0 & C. \end{bmatrix}.$$

If we write the position vector $\mathbf{r}_o$ and angular rate $\boldsymbol{\omega}$ as

$$\mathbf{r}_o = \begin{bmatrix} r_x \\ r_y \\ r_z \end{bmatrix}, \quad \boldsymbol{\omega} = \begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix},$$

then Euler's equations become

$$\begin{aligned} A\dot{\omega}_x + \omega_y\omega_z(C - B) &= \frac{3\mu}{r_o^5} \cdot r_y r_z(C - B) \\ B\dot{\omega}_y + \omega_x\omega_z(A - C) &= \frac{3\mu}{r_o^5} \cdot r_x r_z(A - C) \\ C\dot{\omega}_z + \omega_x\omega_y(B - A) &= \frac{3\mu}{r_o^5} \cdot r_x r_y(B - A). \end{aligned}$$

Stable conditions for satellites under gravity gradient torques
If a satellite is in a circular orbit, then it rotational motion is in a stable equilibrium if:

- ω_z *matches the angular rate of the orbit,* and
- $\omega_x = \omega_y = 0$, and
- *The position of the central body in the body-fixed frame is* $r_x = \|\mathbf{r}_o\|$, and
- $C > B > A$.

Note that the first three conditions above are sufficient to describe an equilibrium point. The final condition ensures that this point is stable. One can of course define similar conditions for other other orientations, e.g. any ω_i matching the angular rate with $r_i = \|\mathbf{r}_o\|$, with all other rate and position components zero, will also be an equilibrium point.

Spacecraft Stability

A number of different mechanisms effect and can be used to control/maintain a satellite's stability. These range from the role of atmospheric drag on the spacecraft to the the use of spin to provide stability. As we have already derived the equations for the gravity gradient and the torques derived from the gravity gradients, we will look at gravity gradient derived stability.

Gravity Gradient Derived Stability

We will start to consider the stability of a satellite subject to gravitational gradients by using the equation below, and examining small perturbations about the axes (and reordering to give roll, pitch and yaw).

$$\begin{aligned} B\dot{\omega}_y + \omega_x\omega_z(A-C) &= \frac{3\mu}{r_o^5}\cdot r_x r_z(A-C) \\ C\dot{\omega}_z + \omega_x\omega_y(B-A) &= \frac{3\mu}{r_o^5}\cdot r_x r_y(B-A) \\ A\dot{\omega}_x + \omega_y\omega_z(C-B) &= \frac{3\mu}{r_o^5}\cdot r_y r_z(C-B). \end{aligned} \tag{110}$$

To consider the orbital frame's angular velocity along the body from the following transformation is required.

$$\big[Q\big] = \begin{bmatrix} 1 & \phi_{\text{yaw}} & -\theta_{\text{pitch}} \\ -\phi_{\text{yaw}} & 1 & \psi_{\text{roll}} \\ \theta_{\text{pitch}} & -\psi_{\text{roll}} & 1 \end{bmatrix}$$

Which can be used to calculate the satellite's inertial angular velocity along the body frame axes,

$$\begin{aligned} \omega_y &= \omega_{y\,\text{rel}} + \Omega_y \\ \omega_z &= \omega_{z\,\text{rel}} + \Omega_z \\ \omega_x &= \omega_{x\,\text{rel}} + \Omega_x \end{aligned}$$

Considering perturbations about the principal axes,

$$\begin{bmatrix} \Omega_y \\ \Omega_z \\ \Omega_x \end{bmatrix} = \begin{bmatrix} 1 & \phi_{\text{yaw}} & -\theta_{\text{pitch}} \\ -\phi_{\text{yaw}} & 1 & \psi_{\text{roll}} \\ \theta_{\text{pitch}} & -\psi_{\text{roll}} & 1 \end{bmatrix} \begin{bmatrix} 0 \\ n \\ 0 \end{bmatrix} = \begin{bmatrix} n\phi_{\text{yaw}} \\ n \\ -n\phi_{\text{roll}} \end{bmatrix}$$

Therefore,

$$\begin{aligned}\omega_y &= \omega_{y\,\text{rel}} + n\phi_{\text{yaw}} &&= \dot{\psi}_{\text{roll}} + n\Phi_{\text{yaw}}\\ \omega_z &= \omega_{z\,\text{rel}} + n &&= \dot{\theta}_{\text{pitch}} + n\\ \omega_x &= \omega_{x\,\text{rel}} - n\psi_{\text{roll}} &&= \dot{\phi}_{\text{yaw}} - n\psi_{\text{roll}}\end{aligned}$$

which can be differentiated with respect to time to give,

$$\begin{aligned}\dot{\omega}_y =& \ddot{\psi}_{\text{roll}} + n\Phi_{\text{yaw}}\\ \dot{\omega}_z =& \ddot{\theta}_{\text{pitch}}\\ \dot{\omega}_x =& \ddot{\phi}_{\text{yaw}} - n\psi_{\text{roll}}\end{aligned} \tag{111}$$

Substituting equations 111 into equations 110 along with,

$$\begin{bmatrix} r_y \\ r_z \\ r_x \end{bmatrix} = \begin{bmatrix} 1 & \phi_{\text{yaw}} & -\theta_{\text{pitch}} \\ -\phi_{\text{yaw}} & 1 & \psi_{\text{roll}} \\ \theta_{\text{pitch}} & -\psi_{\text{roll}} & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ r_0 \end{bmatrix} = \begin{bmatrix} -r_o\theta_{\text{pitch}} \\ r_0\psi_{\text{roll}} \\ r_0 \end{bmatrix},$$

and

$$A = I_{\text{yaw}} \quad B = I_{\text{roll}} \quad C = I_{\text{pitch}}$$

results in,

$$\begin{aligned} I_{\text{roll}}\left(\ddot{\psi}_{\text{roll}} + n\dot{\phi}_{\text{yaw}}\right) + \left(I_{\text{yaw}} - I_{\text{pitch}}\right)\left(\dot{\theta}_{\text{pitch}} + n\right)\left(\dot{\phi}_{\text{yaw}} - n\psi_{\text{roll}}\right) &= 3\left(I_{\text{yaw}} - I_{\text{pitch}}\right)n^2\psi_{\text{roll}}\\ I_{\text{pitch}}\ddot{\theta}_{\text{pitch}} + \left(I_{\text{roll}} - I_{\text{yaw}}\right)\left(\dot{\psi}_{\text{roll}} + n\phi_{\text{yaw}}\right)\left(\phi_{\text{yaw}} - n\psi_{\text{roll}}\right) &= -3\left(I_{\text{roll}} - I_{\text{yaw}}\right)n^2\theta_{\text{pitch}}\\ I_{\text{yaw}}\left(p\ddot{h}i_{\text{yaw}} + n\dot{\psi}_{\text{roll}}\right) + \left(I_{\text{pitch}} - I_{\text{roll}}\right)\left(\dot{\theta}_{\text{pitch}} + n\right)\left(\dot{\psi}_{\text{roll}} - n\phi_{\text{yaw}}\right) &= -3\left(I_{\text{pitch}} - I_{\text{roll}}\right)n^2\theta_{\text{pitch}}\psi_{\text{roll}}. \end{aligned}$$

Which can be simplified if only linear terms are considered,

$$\begin{aligned} I_{\text{roll}}\ddot{\psi}_{\text{roll}} + \left(I_{\text{roll}} - I_{\text{pitch}} + I_{\text{yaw}}\right)n\dot{\phi}_{\text{yaw}} + 4\left(I_{\text{pitch}} - I_{\text{yaw}}\right)n^2\psi_{\text{roll}} =& 0\\ I_{\text{pitch}}\ddot{\theta}_{\text{pitch}} + 3\left(I_{\text{roll}} - I_{\text{yaw}}\right)n^2\theta_{\text{pitch}} =& 0\\ I_{\text{yaw}}p\ddot{h}i_{\text{yaw}} + \left(I_{\text{pitch}} - I_{\text{roll}}\right)n^2\phi_{\text{yaw}} + \left(I_{\text{pitch}} - I_{\text{roll}} - I_{\text{yaw}}\right)n\dot{\psi}_{\text{roll}} =& 0. \end{aligned} \tag{112}$$

These differential equations govern the spacecraft's angular motion under the influence of gravity-gradient derived torques for small angles and rates. These equations can be solved for, and the stability of the equations can be considered. Let us first consider the pitch

equation, it can be seen that it is not coupled to the other equations so can be solved independently of them. If we assume the solution is of the form (removing the *pitch* subscript from θ).

$$\theta = Pe^{pt}$$

yields the solution, $I_{\text{pitch}}p^2 + 3\left(I_{\text{roll}} = I_{\text{yaw}}\right)n^2 = 0$, which can be solved for the p to give the two roots,

$$p_{1,2} = \pm ß\sqrt{3\frac{\left(I_{\text{roll}} - I_{\text{yaw}}\right)n^2}{I_{\text{pitch}}}}$$

To avoid an unstable root p must be imaginary, otherwise there will be one positive and one negative root, resulting in $\theta \to \infty$. Therefore for a steady state oscillatory solution,

$$I_{\text{roll}} > I_{\text{yaw}} \tag{113}$$

resulting in a stable frequency of

$$\omega_{f\text{pitch}} = n\sqrt{\frac{\left(I_{\text{roll}} - I_{\text{yaw}}\right)}{I_{\text{pitch}}}}$$

Turning to the yaw and roll equations, it can be seen that there is coupling between the roll and yaw motion. Assuming the following solutions,

$$\phi = Ye^{qt} \quad \psi = Re^{qt} \tag{114}$$

and factoring our I_{yaw}, the following equations are obtained,

$$\begin{aligned}\left[\frac{I_{\text{pitch}} - I_{\text{roll}}}{I_{\text{yaw}}}n^2 + q^2\right]Y + \left[\frac{I_{\text{pitch}} - I_{\text{roll}}}{I_{\text{yaw}}} - 1\right]nqR &= 0 \\ \left[1 - \frac{I_{\text{pitch}} - I_{\text{yaw}}}{I_{\text{roll}}}\right]nqY + \left[4\frac{I_{\text{pitch}} - I_{\text{yaw}}}{I_{\text{roll}}}n^2 + q^2\right]R &= 0\end{aligned} \tag{115}$$

If we let

$$k_Y = \frac{I_{\text{pitch}} - I_{\text{roll}}}{I_{\text{yaw}}} \quad \text{and} \quad k_R = \frac{I_{\text{pitch}} - I_{\text{yaw}}}{I_{\text{roll}}}, \tag{116}$$

from the definitions of I,

$$I_{\text{pitch}} = \int_m x^2 + y^2 dm, \quad I_{\text{roll}} = \int_m x^2 + z^2 dm, \quad I_{\text{yaw}} = \int_m y^2 + z^2 dm$$

it can be shown that $|k_Y| < 1$ and $|k_R| < 1$ (have a go at showing this yourself, substitute in the values of I into the definitions of k_Y and k_R and simplify it noting that squared terms have to be positive).

Equation 115 can therefore be written in the following matrix form,

$$\begin{bmatrix} k_Y n^2 + q^2 & (k_Y - 1)\, nq \\ (1 - k_R)\, nq & 4k_R n^2 + q^2 \end{bmatrix} \begin{bmatrix} Y \\ R \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

which yields the characteristic equation for q below.

$$q^4 + bn^2 q^2 + cn^4 = 0 \tag{117}$$

where

$$b = 3k_R + k_Y k_R + 1 \quad c = 4k_Y k_R.$$

This results in a quartic equation with four roots. To give a stable solution for ϕ and ψ (equation 114) the roots $q_{1,2,3,4}$ must be negative or imaginary, otherwise the perturbation will get larger. Identifying that only even powers are present for q in equation 117 the following substitution can be made,

$$q = \pm n\sqrt{\lambda}$$

allowing the the characteristic equation to be written as,

$$\lambda^2 + b\lambda + c = 0.$$

Which has the roots,

$$\lambda_{1,2} = -\frac{1}{2}\left(b \pm \sqrt{b^2 - 4c}\right).$$

To ensure that q is not positive we require that λ must be real and negative, which results in $b > 2\sqrt{c}$, or

$$3k_R + k_Y k_R + 1 > 4\sqrt{k_R k_Y}. \tag{118}$$

For λ to be negative $b^2 > b^2 - 4c$, which means $c > 0$, or,

$$k_Y k_R > 0. \tag{119}$$

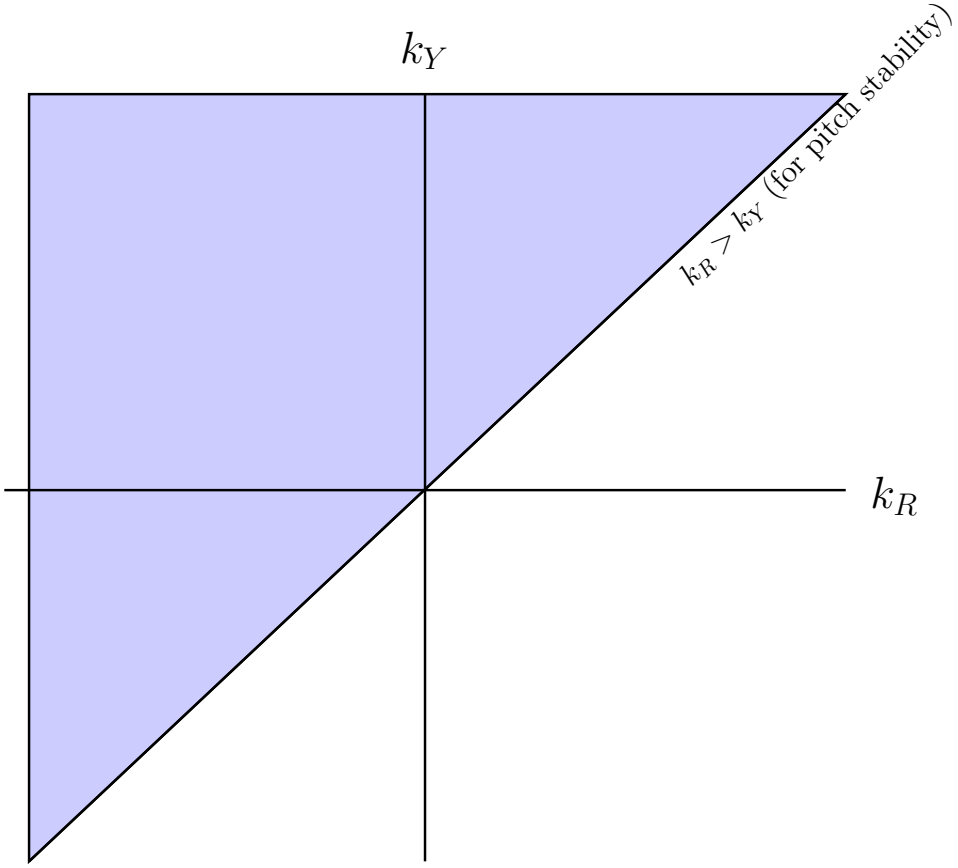
Equations 113, 118 and 119 are therefore the conditions for stability. Equations 116 can be rearranged to give,

$$I_{\text{yaw}} = \frac{1 - k_R}{1 - k_Y kR} I_{\text{pitch}}, \quad I_{\text{roll}} = \frac{1 - k_Y}{1 - k_Y kR} I_{\text{pitch}}, \tag{120}$$

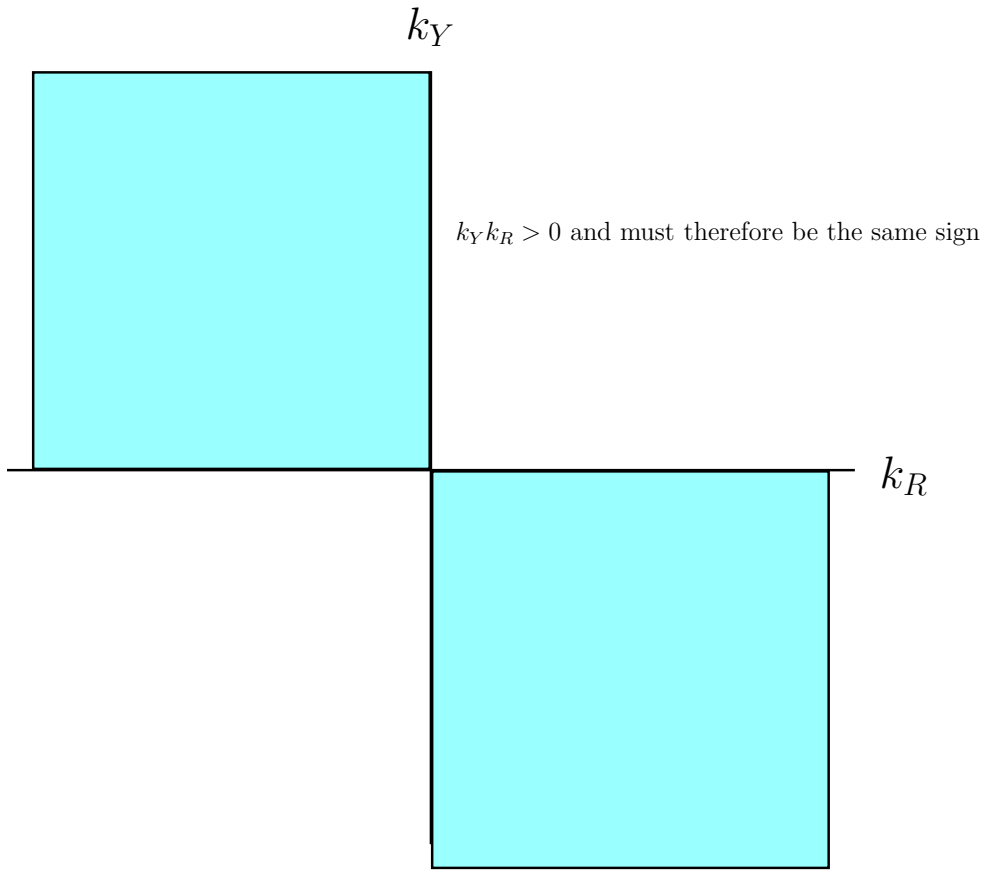
which when substituted into condition 113, gives,

$$\frac{1 - k_Y}{1 - k_R} > 1$$

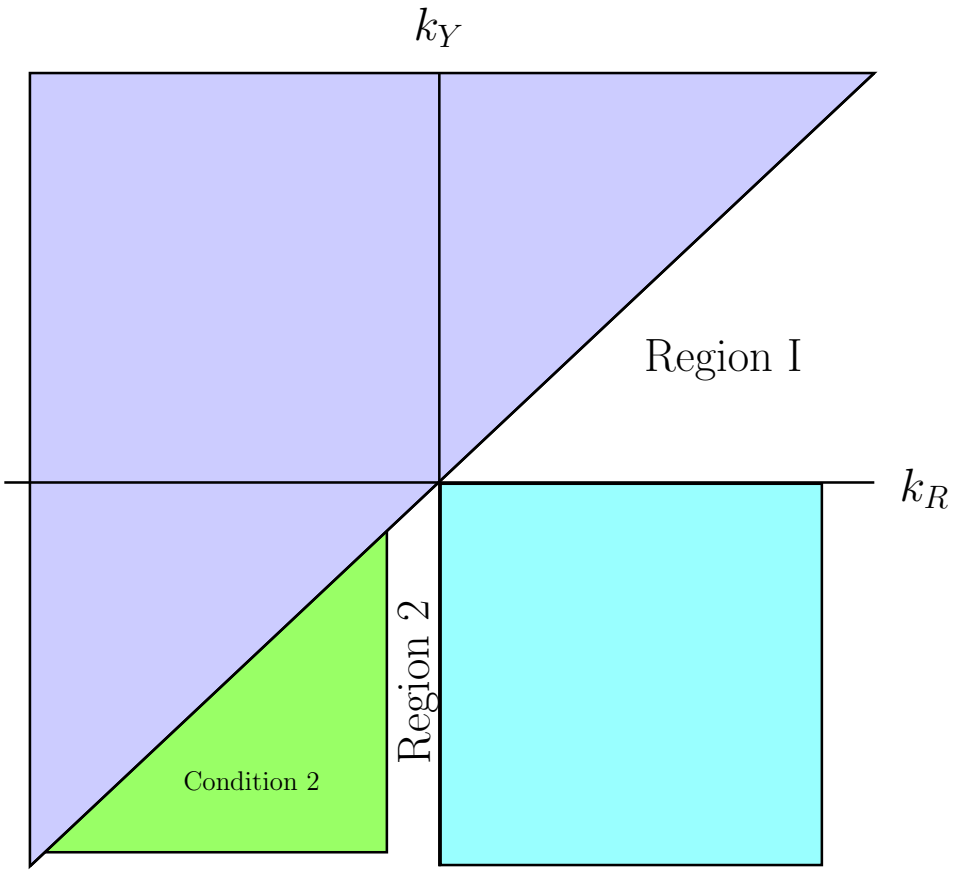
and, as previously shown $|k_R| < 1$, this means that $k_Y < k_R$. This is essentially equation 113 rewritten in terms of k_Y and k_R. We can not plot the stable regions where all three



Stability condition 1 - Pitch Stability Requirement ($k_Y < k_R$, equation 113)



Stability condition 3



All stability conditions, note the addition of condition 2.

conditions are met in terms of k_Y and k_R, along with the knowledge that all three moments of inertia must be positive. Three conditions can all be plotted and related to the moments of inertia.

From the figures it can be seen that there are two stable regions. In region one (as shown in figure 15), both k_R and k_Y are positive, and, because they are less than one (as shown above), and from equations 120,

$$I_{\text{pitch}} > I_{\text{roll}} \quad \text{and} \quad I_{\text{pitch}} > I_{\text{yaw}}$$

which combined with the pitch stability requirement ($I_{\text{roll}} > I_{\text{yaw}}$ means that,

$$I_{\text{pitch}} > I_{\text{roll}} > I_{\text{yaw}}.$$

Region II is more complex, and is determined by equation refeq:GGStabCond2. In this region both k_R and k_Y are negative, which from 120 results in,

$$I_{\text{yaw}} > I_{\text{pitch}} \quad \text{and} \quad I_{\text{roll}} > I_{\text{pitch}}$$

which combined with the pitch stability requirement ($I_{\text{roll}} > I_{\text{yaw}}$ means that,

$$I_{\text{roll}} > I_{\text{yaw}} > I_{\text{pitch}}.$$

While this configuration is stable in the short term, long term effects resulting in energy dissipation makes this configuration unstable at long timescales.